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Modelling the food web in the Irish Sea in the context of a depleted commercial fish community

Part 2:

ICES Ecopath with Ecosim Key Run



SAMS Marine Institute Foras na Mara ICES CIEM

Modelling the food web in the Irish Sea in the context of a depelted commercial fish community

Part 2: ICES Ecopath with Ecosim Key Run

Project partners

Scottish Association for Marine Science (SAMS): Jacob Bentley, Dr Natalia Serpetti, Dr Clive Fox

Marine Institute Ireland (MI): Professor David Reid

European Marine Board: Professor Sheila JJ Heymans

Additional contributions from

International Council for the Exploration of the Sea (ICES): WKIrish, WGSAM, WKEWIEA

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Preface

This work was undertaken as part of the PhD titled '*Modelling the food web in the Irish Sea in the context of a depleted commercial fish community',* funded via the Marine Institutes Cullen Fellowship and hosted by the Scottish Association for Marine Science (SAMS; accredited by the University of the Highlands and Islands). The project was designed under the remit of the first ICES Integrated Benchmark Assessment, WKIrish. WKIrish is a multi-year process focussing on improving single-species stock assessments (principally cod, haddock, whiting, plaice, and herring), incorporating a mixed fisheries model, and developing the integration of ecosystem aspects and working towards an integrated assessment and advice. For complex ecosystem models to provide practical management advice it is preferable that they attain 'key run' status, wherein a group of experts assess the quality and performance of a model. This report describes the development of a key run Ecopath with Ecosim model of the Irish Sea (Figure 0.1), allowing the interested reader to understand the methodology and data used to construct the model and provide guidance for attaining an EwE key run. Importantly, the report is intended to provide transparency to the model construction process and highlight the limitations of the data and thus the caveats attached to model outputs.

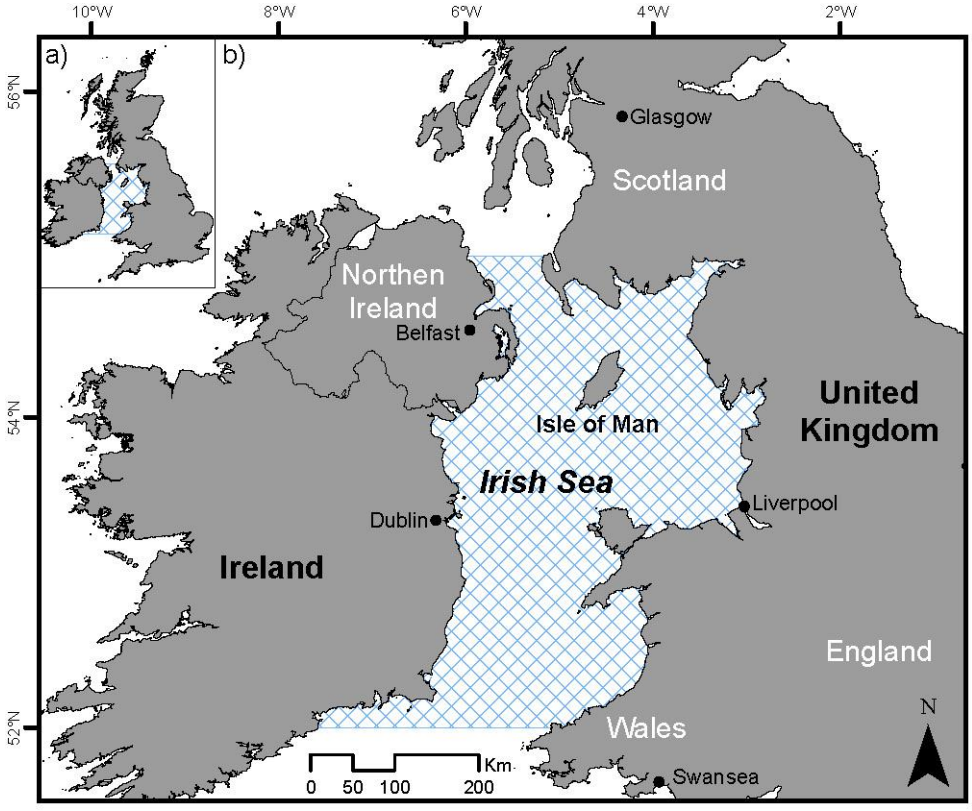
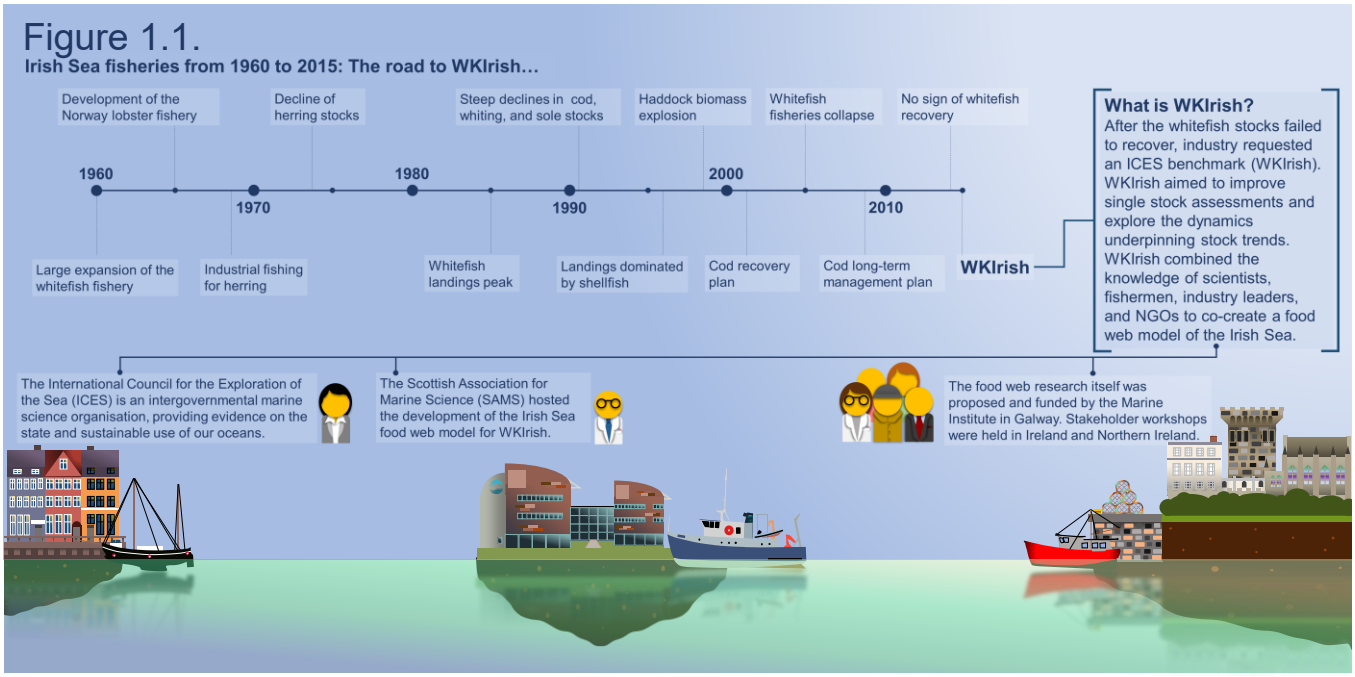


Figure 0.1. *Map of (a) Ireland and the British Isles showing (b) the extent of the model area for the Irish Sea EwE key run (hatched area- ICES division VIIa).*

Chapter 1: Introduction

1.1 The road to WKIrish

Following declines in commercial finfish stocks, the fisheries landings from the Irish Sea shifted in the 1990's from being dominated by finfish such as cod (*Gadus morhua*), whiting (*Merlangius merlangus*), herring (*Clupea harengus*), sole (*Solea solea*), and plaice (*Pleuronectes platessa*) to invertebrates such as Norway lobster (*Nephrops norvegicus*), scallops (*Aequipecten opercularis* and *Pecten maximus*), blue mussels (*Mytilus edulis*), and whelks (*Buccinum undatum*). In the early 2000's measures were established by the European Commission (EC) in an effort to reduce the fishing mortality of cod and aid its recovery (EC, 2000, ICES, 2001). This was followed by the implementation a total allowable catch (TAC) in 2004 (EC, 2004) and a long-term plan for cod in 2008 (EC, 2008). Despite tighter regulations on fishing activities there was limited evidence of stock recovery from 2008 to 2016 according to International Council for the Exploration of the Sea (ICES) stock assessments (ICES, 2010, ICES, 2016c). It has been suggested that this failure stemmed from the plans inability to deal with systemic uncertainties, such as mixed fisheries, ecosystem, or environmental factors (Kelly et al., 2006). In 2015 the first ICES Integrated Benchmark was established for the Irish Sea (WKIrish) (Figure 1.1). WKIrish included a series of workshops to improve single species stock assessments and conducted a holistic exploration of ecosystem and ecological aspects (ICES, 2016b, ICES, 2017c, ICES, 2015, ICES, 2018e, ICES, 2018d).



1.2 The Irish Sea Ecopath with Ecosim model

One of the aims of WKIrish was to combine scientific and stakeholder knowledge to build multi-species food web models that may ultimately be used to provide advice in an ecosystem-based context. One of the potential food web modelling approaches identified by attendees of the initial WKIrish workshop (ICES 2015) was Ecopath with Ecosim (EwE) (Christensen et al., 2008). The Ecopath approach was initially presented by Polovina (1984b, 1984a) to estimate the biomass and food consumption of functional groups within an ecosystem and was subsequently combined with approaches from theoretical ecology for the analysis of energy flows between functional groups (Ulanowicz and Kay, 1991). The system has since been optimised for use in fisheries assessment and ecological investigations (Christensen and Pauly, 1992, Walters et al., 1997, Walters et al., 1999, Christensen et al., 2008).

The Irish Sea EwE model uses the data gathered by WKIrish and the updated stock assessments to parameterise the biological characteristics of 41 functional groups (Figure 1.2). The model also uses information from literature, and open access databases (i.e. DAPSTOM: DAtabase and Portal for STOMach records, Pinnegar (2014)) and knowledge co-created by scientists and stakeholders.

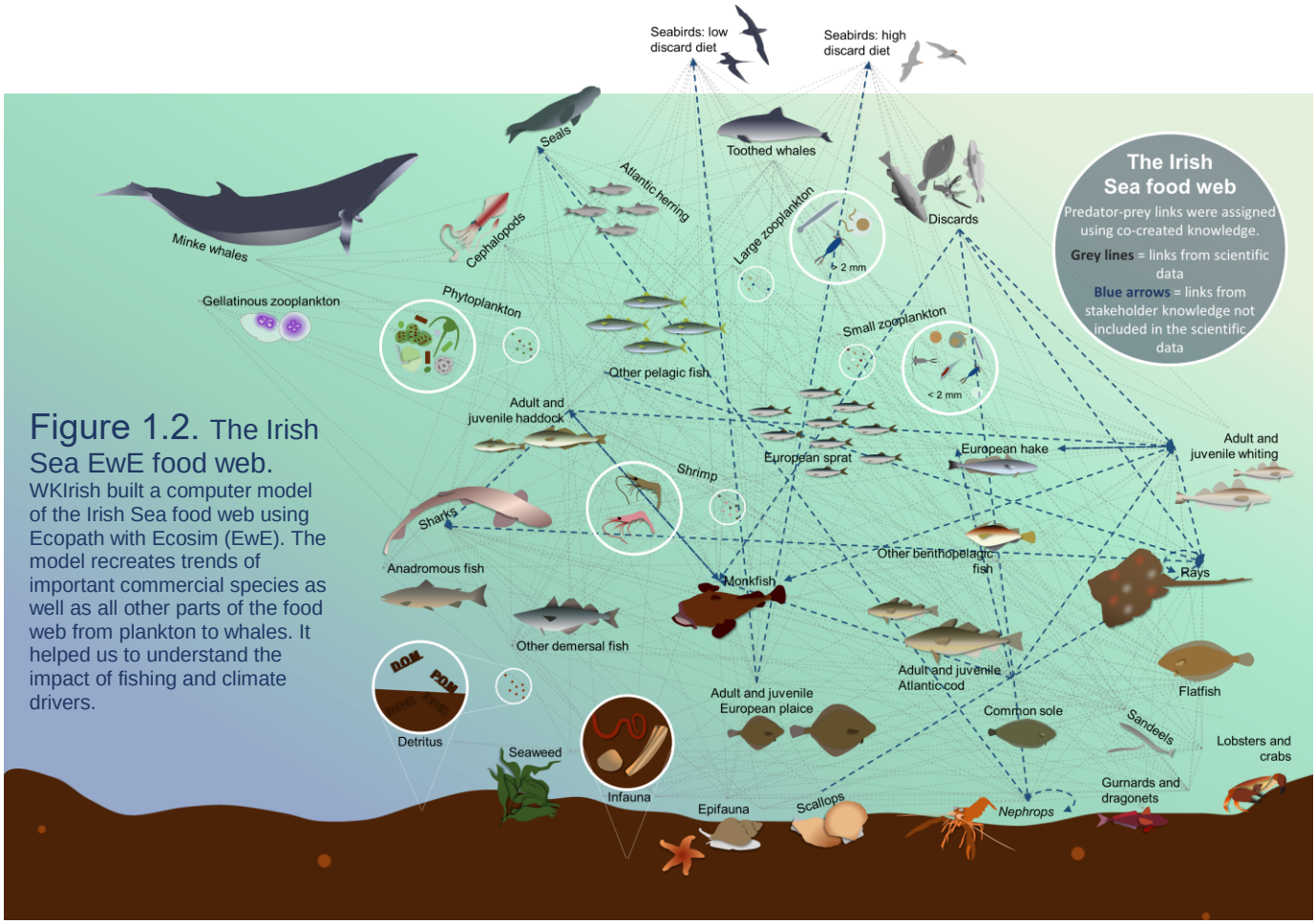


Figure 1.2. The Irish Sea EwE food web.
WKIrish built a computer model of the Irish Sea food web using Ecopath with Ecosim (EwE). The model recreates trends of important commercial species as well as all other parts of the food web from plankton to whales. It helped us to understand the impact of fishing and climate drivers.

1.3 The benefit of co-created knowledge

Fishers' knowledge holds considerable value as a source of information to support and inform fisheries research and management (Johannes et al., 2000, Haggan et al., 2007, Moreno-Báez et al., 2010, Macdonald et al., 2014, Raymond-Yakoubian et al., 2017, Bentley et al., 2019c). The unique experiences and perspectives of fishers provide not only fishery information, but also the ecological, institutional, social, and economic knowledge often passed from generation to generation (Berkes et al., 2000, Haggan et al., 2007, Silvano and Valbo-Jorgensen, 2008, Martins et al., 2018).

During WKIrish workshops in Dun Laoghaire, Ireland, on the 23-27 October 2017 and in Kilkeel, Northern Ireland, on the 8 December 2017 (ICES, 2018e), fishers provided information on the diets of commercial species, which enhanced the long term stomach records, and provided historic fishing effort trends to fill gaps in the available data (Figure 1.3). Combining scientific and stakeholder knowledge improved the EwE models ability to simulate the biomass and catch trends of commercial stocks (Bentley et al., 2019c) and led to the recognition of additional predator-prey interactions which influence the advice and indicators derived from the model (Bentley et al., 2019b). By disseminating these results back to stakeholders, we were able to demonstrate the visible impact and value of their knowledge, increasing their buy-in and engagement. Stakeholders continue to guide the direction of future collaboration and research objectives, recommending that researchers prioritise ecosystem modelling in the future, with the desire to better understand: (i) **the roles of food web components not currently under management**, (ii) **the impact of area closures on commercial stocks and the ecosystem**, and (iii) **how best to maximise the sustainability of fisheries and socio-economic benefits.**

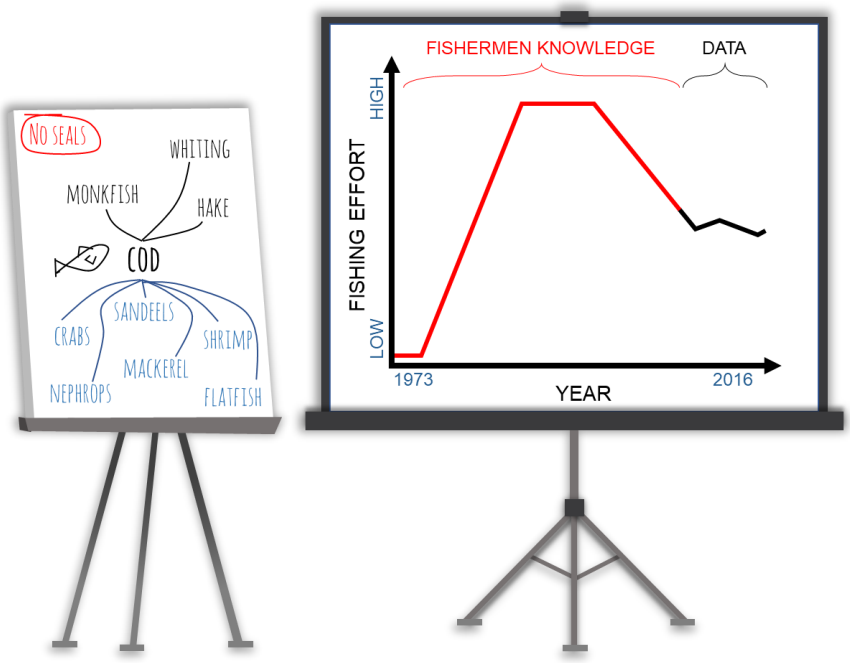
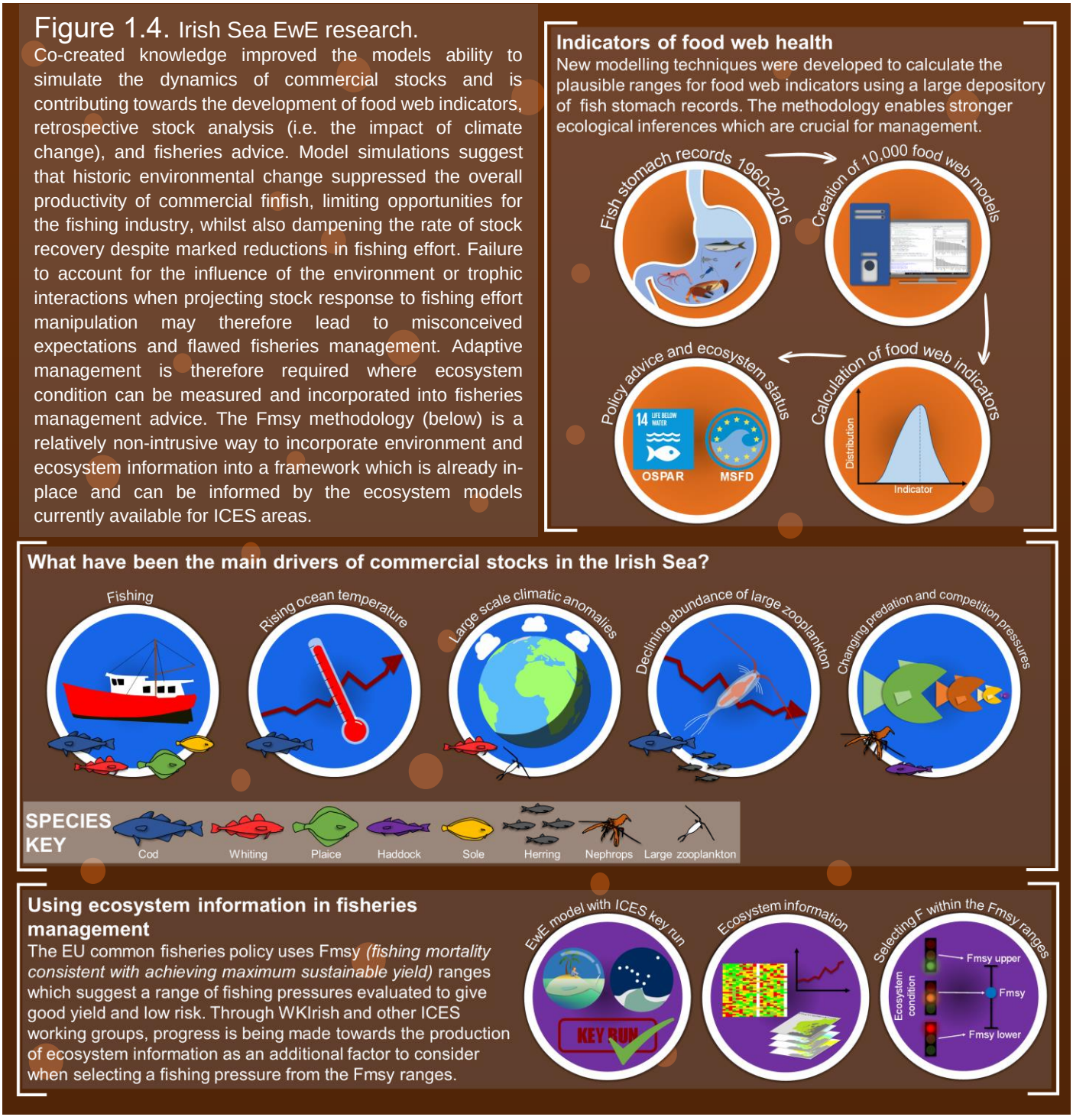


Figure 1.3. Fishers' knowledge. During WKIrish workshops fishers shared their knowledge of commercial species' diets and historic fishing effort, to run parallel to and fill gaps in scientific knowledge.

1.4 Research themes

The Irish Sea EwE model has thus far been used to develop food web indicators for the region, explore the retrospective drivers (environmental, trophic, fishing) of commercial stocks, and provide ecosystem information to advise fisheries advice (Figure 1.4).

Figure 1.4. Irish Sea EwE research.
Co-created knowledge improved the models ability to simulate the dynamics of commercial stocks and is contributing towards the development of food web indicators, retrospective stock analysis (i.e. the impact of climate change), and fisheries advice. Model simulations suggest that historic environmental change suppressed the overall productivity of commercial finfish, limiting opportunities for the fishing industry, whilst also dampening the rate of stock recovery despite marked reductions in fishing effort. Failure to account for the influence of the environment or trophic interactions when projecting stock response to fishing effort manipulation may therefore lead to misconceived expectations and flawed fisheries management. Adaptive management is therefore required where ecosystem condition can be measured and incorporated into fisheries management advice. The Fmsy methodology (below) is a relatively non-intrusive way to incorporate environment and ecosystem information into a framework which is already in-place and can be informed by the ecosystem models currently available for ICES areas.



Chapter 2: Ecopath

In this chapter we include a description of the Irish Sea Ecopath model.

2.1 Functional groups

The Irish Sea model comprises 41 functional groups including two detrital groups (detritus, discards), two primary producers (phytoplankton, seaweed), ten invertebrate groups, 22 finfish groups, two seabird groups (low discard diet and high discard diet), and three marine mammal groups (Figure 2.1). A detailed description of functional group design can be found in Part 1 of this report series (Bentley et al., 2018). Table 2.1 summarises the basic input parameters for functional groups. See Annex 1.1 for an extensive list of references and assumptions for each input parameter.

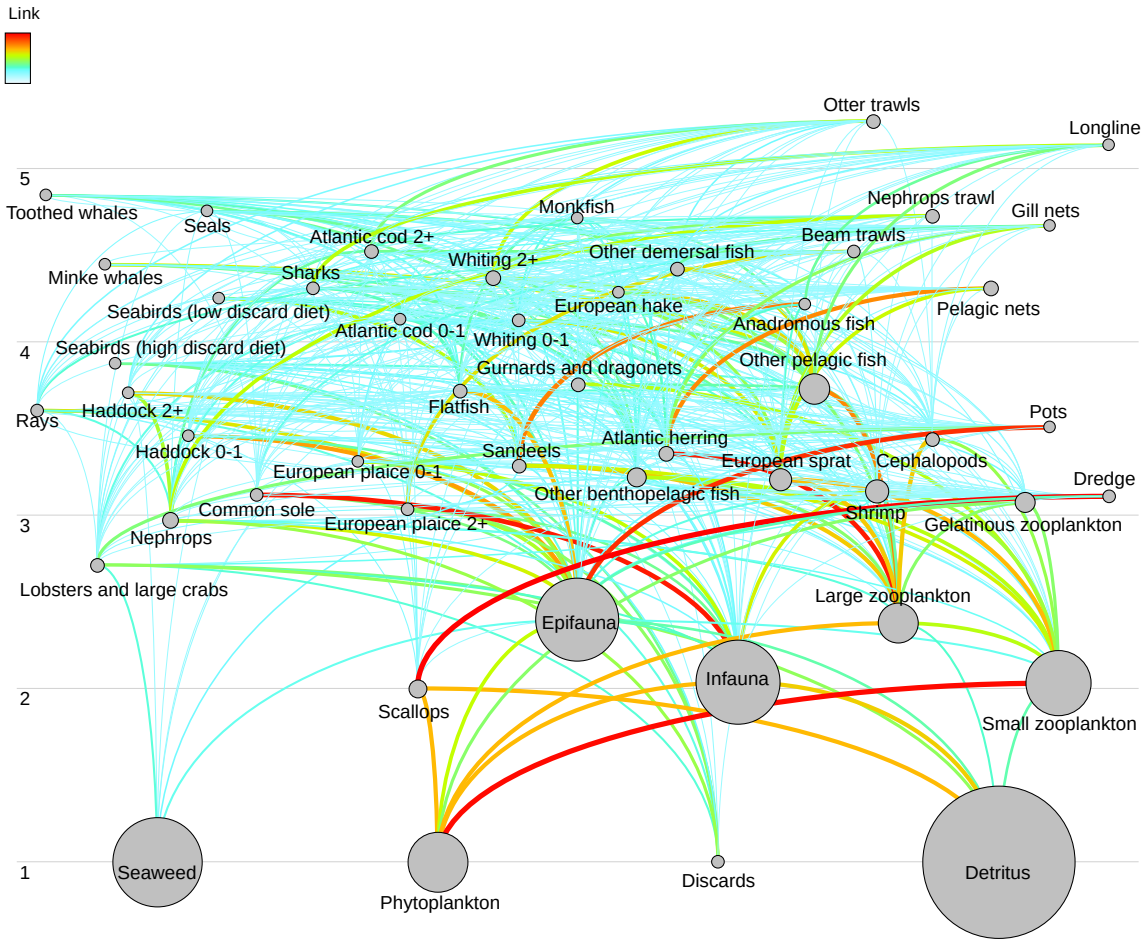


Figure 2.1. *Energy flow and biomass diagram for the Irish Sea Ecopath food web model. Functional groups and fleets are represented by nodes. The relative size of nodes for functional groups denotes their biomass in the ecosystem, whereas the relative size of nodes for fishing fleets represents catch volume. Lines represent the flow of energy and the y-axis denotes group trophic level (1-5). The colour and width of lines denote the relative size of energy flows between functional groups.*

Table 2.1. *Input parameters from the Irish Sea Ecopath key run. Parameters include trophic level (TL), biomass t.km^{-2} (B), production/biomass year^{-1} (PB), consumption/biomass year^{-1}, ecotrophic efficiency (EE), and production/consumption (PQ). Orange parameters have been calculated by Ecopath.*

	Functional group	TL	B	PB	QB	EE	PQ
1	Toothed whales	4.845	0.005	0.020	15.520	0.642	0.001
2	Minke whales	4.445	0.066	0.020	6.580	0.136	0.003
3	Seals	4.753	0.002	0.060	14.430	0.117	0.004
4	Seabirds (high discard diet)	3.874	0.002	0.400	66.020	0.000	0.006
5	Seabirds (low discard diet)	4.249	0.002	0.400	69.210	0.000	0.006
6	Sharks	4.309	0.290	0.400	4.010	0.344	0.100
7	Rays	3.603	0.254	0.845	3.280	0.500	0.258
8	Atlantic cod 2+	4.524	0.339	0.820	3.120	0.920	0.263
9	Atlantic cod 0-1	4.128	0.066	1.010	7.641	0.916	0.132
10	Whiting 2+	4.368	0.560	0.762	5.390	0.973	0.141
11	Whiting 0-1	4.124	0.236	1.524	11.883	1.000	0.128
12	Haddock 2+	3.703	0.086	0.890	4.420	0.962	0.201
13	Haddock 0-1	3.455	0.048	1.780	10.285	0.979	0.173
14	European plaice 2+	3.034	0.155	0.790	3.750	0.997	0.211
15	European plaice 0-1	3.307	0.031	1.380	10.298	0.934	0.134
16	Common sole	3.117	0.113	0.810	5.610	0.958	0.144
17	Flatfish	3.719	0.470	1.518	6.070	0.956	0.250
18	Monkfish	4.712	0.033	0.490	2.746	0.687	0.178
19	European hake	4.284	0.093	0.981	3.924	0.950	0.250
20	Sandeels	3.286	0.354	2.053	8.210	0.950	0.250
21	Gurnards and dragonets	3.755	0.340	1.403	5.610	0.997	0.250
22	Other demersal fish	4.423	0.359	0.960	4.320	0.950	0.222
23	Other benthopelagic fish	3.218	1.569	1.845	7.380	0.950	0.250
24	Atlantic herring	3.354	0.720	1.350	6.590	0.951	0.205
25	European sprat	3.207	2.304	2.425	9.700	0.950	0.250
26	Other pelagic fish	3.731	4.180	1.278	5.110	0.950	0.250
27	Anadromous fish	4.215	0.030	0.644	3.170	0.336	0.203
28	Lobsters and large crabs	2.713	0.507	0.620	4.133	0.950	0.150
29	Nephrops	2.967	0.924	1.270	8.467	0.950	0.150
30	Shrimp	3.137	2.316	2.670	17.800	0.950	0.150
31	Cephalopods	3.439	0.319	1.981	15.000	0.950	0.132
32	Scallops	2.000	1.248	1.146	7.643	0.950	0.150
33	Epifauna	2.394	25.546	1.007	4.028	0.950	0.250
34	Infauna	2.036	25.817	1.700	6.800	0.950	0.250
35	Gelatinous zooplankton	3.070	1.700	0.762	2.540	0.690	0.300
36	Large zooplankton	2.377	6.810	10.000	33.333	0.682	0.300
37	Small zooplankton	2.031	16.180	18.000	60.000	0.521	0.300
38	Seaweed	1.000	29.300	1.000	-	0.393	-
39	Phytoplankton	1.000	13.830	226.300	-	0.327	-
40	Discards	1.000	0.219	-	-	0.127	-
41	Detritus	1.000	100.000	-	-	0.115	-

2.2 Multi-stanzas

Cod, whiting, haddock, and plaice functional groups were represented with two life stages, adult and juvenile. Multi-stanza representation of life stages enables the model to account for ontogenetic changes in diet preference and fishing mortality. When multi-stanza groups are included in Ecopath, the 'usual' Ecosim differential equations to model biomass change are replaced by a set of difference equations that track monthly changes in the number and mean body weight of animals of all monthly cohorts. Production of juveniles does not occur in a yearly pulse, as in reality, but spread evenly across the annual cycle. Thus, juvenile biomass estimated by EwE models is considerably lower (approximately $1/12^{th}$) of real biomass and can only be considered a proxy variable (Walters et al., 2010, ICES, 2011, ICES, 2016d).

Table 2.2. *Multi-stanza parameters based on reported values for the Irish Sea.*

Multi-stanza	Parameter	Value	Source
Cod	VBGF K	0.16	(Froese and Pauly, 2017)
	Recruit power	1	-
	BA/B	0	-
	Adult stanza start month	24	(ICES, 2016b)
	Wmat/Winf	0.137	(Bentley et al., 2018)
Whiting	VBGF K	0.34	(Froese and Pauly, 2017)
	Recruit power	1	-
	BA/B	0	-
	Adult stanza start month	24	(ICES, 2016b)
	Wmat/Winf	0.221	(Bentley et al., 2018)
Haddock	VBGF K	0.29	(Froese and Pauly, 2017)
	Recruit power	1	-
	BA/B	0	-
	Adult stanza start month	24	(ICES, 2016b)
	Wmat/Winf	0.173	(Bentley et al., 2018)
Plaice	VBGF K	0.13	(Froese and Pauly, 2017)
	Recruit power	1	-
	BA/B	0	-
	Adult stanza start month	24	(ICES, 2016b)
	Wmat/Winf	0.165	(Bentley et al., 2018)

It is difficult to compare the stock-recruitment (SR) relationships from the EwE model with single species predictions because in addition to fishing effects, the dynamics of the adult juvenile groups are affected by the environmental forcing function. Key run models for the North Sea (ICES, 2016e) and Baltic Sea (ICES, 2017e) suggest that the closest we can get to demonstrating these emergent patterns is to switch off environmental forcing functions and apply a 'V'-shaped fishing pattern to drive stock biomasses

through high and low values. The emergent SR trends are dependent upon the effects of the fishing pattern and the resulting multi-species interactions, providing an indication of how recruitment changes as adult biomass changes in the model. Even though this test has been applied before, it should be noted that it is not entirely robust as the emergent SR patterns are sensitive to how fishing mortality is applied. However, this check does help to reaffirm whether the links between adult biomass and recruitment are sensible by ensuring SR patterns are not unrealistically erratic. The parameterisation of the key run (under a 'V'-shaped fishing pattern) leads to stock trajectories for cod and plaice that are relatively flat over a large range of biomass, with steep recruitment declines below a low biomass threshold (Figure 2.2). Whiting and haddock SR trends are more dome-shaped (Figure 2.2)

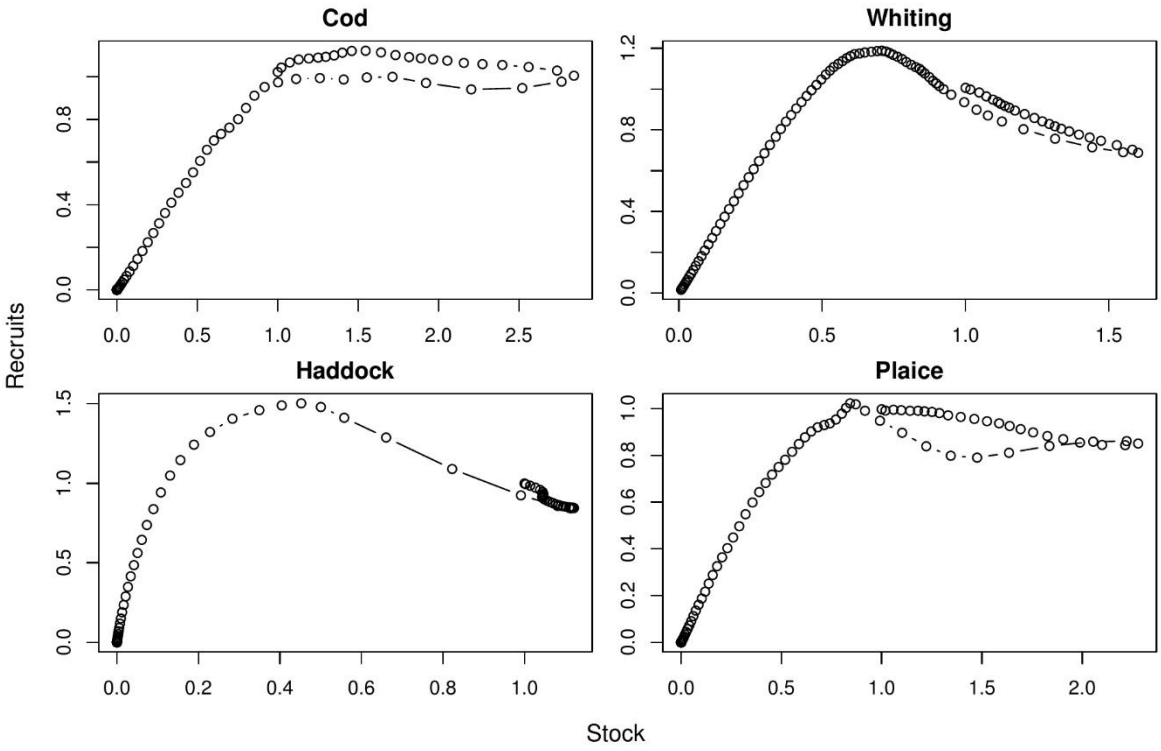


Figure 2.2. Stock recruitment trajectories under a 'V' shaped fishing regime for all multi-stanza groups.

2.3 Diet compositions

The models initial diet matrix for finfish was built using information held in the Cefas integrated DAtabase and Portal for STOMach records (DAPSTOM) (Pinnegar, 2014, Bentley et al., 2019a). DAPSTOM contains 26,765 records from 38,295 stomachs covering 96 predator species from the Irish Sea. Records date back as far as 1836 but for the purpose of the present model only data from 1960-2016 were used (23,331 records). The default format for data entry is as counts of prey item found per predator. In order to convert count data to weight (kg) data, weights were assigned to each prey item. For fish, these weights were obtained by converting length to weight. Length information was obtained from the BTS-VIIa in order to generate an average length and weight relative to the Irish Sea. For invertebrates, average weight data was acquired from SeaLifeBase (Palomares and Pauly, 2017). Weighted diets were then transformed into proportion diets, as is required by Ecopath. The diet extracted for each predator included the mean proportion of prey items consumed based on all the data available from 1960-present day. In addition, minimum, maximum, standard deviation and percentile values were extracted, providing a range of plausible diet proportions each prey may contribute towards a predator's diet (Figure 2.3). Diets of functional groups comprising multiple species were prorated by the biomass proportion each species contributes towards the functional group. When balancing Ecopath models, diet preferences are often the first biological parameters revisited and adjusted in order to fix model imbalances (Heymans et al., 2016). This is often done using 'ad hoc' tuning but with the diet preference ranges (min, max, percentiles) produced using DAPSTOM records we were able to take a systematic approach to diet tuning, reducing the need for 'ad hoc' adjustments.

Diets for mammals, seabirds and invertebrates were taken from literature sources as described in Bentley et al. (2018) and Annex 1.1 of this report.

Fishers' knowledge regarding the diets of commercially important species was shared during a WKIrish workshop (WKIrish4) held in Dun Laoghaire, Ireland, on the 23-27 October 2017 (ICES, 2018e). The aim of the workshop was to update the Irish Sea model so that it used both scientific knowledge and fishers' knowledge of predator-prey interactions for the species they commonly encountered in their operations, and where they would have observed stomach contents whilst processing catches. During the workshop, cod, whiting, haddock, plaice, rays (R*aja spp.*), and Norway lobster were identified as the species for which fishers' felt they had substantial knowledge.

Fishers' diet links were quantified in the model on a case-by-case basis. New interactions were added to the model diet matrix whilst ensuring that the combined predation and fishing mortality placed on each functional group did not exceed production, as for an Ecopath model to be mass balanced total

consumption cannot exceed the production of the species (Figure 2.3). To ensure diets remained balanced, additions of new prey proportions were counterbalanced by adjusting other prey proportions within the range of plausible values. Due to the inherent uncertainty in quantifying qualitative information, the diets of functional groups which were altered by fishers' knowledge were assigned large confidence intervals (±80%) in the models 'pedigree' routine (section 2.6) (Christensen and Walters, 2004). The large confidence intervals assigned to diets altered by fishers ensured, when later applying Monte Carlo simulations, that a large range of parameters could be tested to reflect data uncertainty in model outputs.

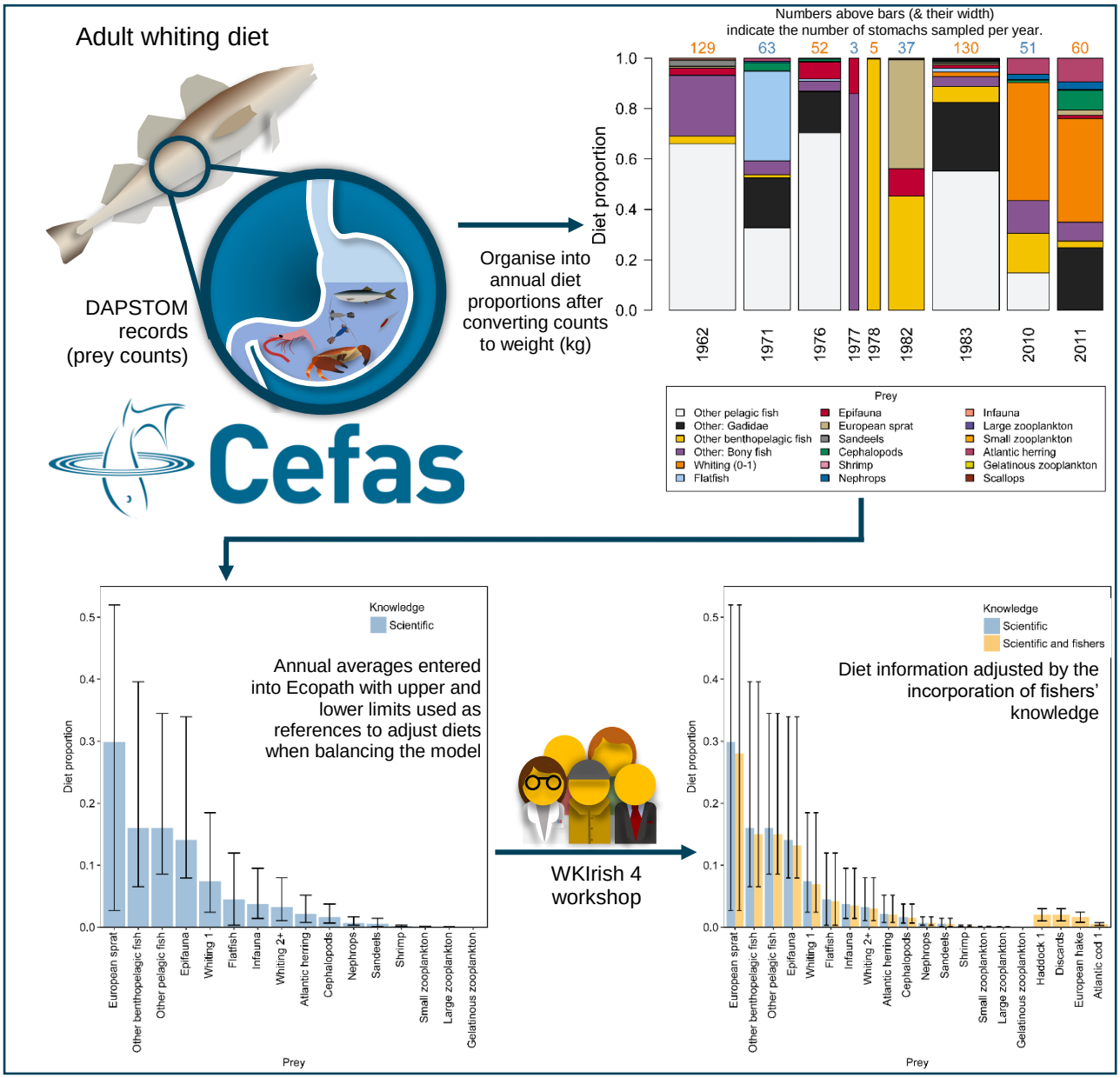


Figure 2.3. *Conceptual diagram for the assignment of diet information to functional groups, specifically commercial finfish (in this case adult whiting).*

2.4 Fishing fleet structure and parameterisation

The Irish Sea model contains 8 fleets which are based on the aggregation of gear categories used in the STECF (Scientific, Technical and Economic Committee for Fisheries) reports (STECF, 2018) (Table 2.3).

Table 2.3. *Irish Sea Ecopath model fleet groups and their corresponding equivalents in STECF data.*

Ecopath fleet	STECF gears	STECF description
Beam trawl	BT2	Beam trawls > 80 mm and < 120 mm
	BEAM	Beam trawl
Otter trawl	TR1	Bottom trawls and seines > 100 mm
	TR3	Beam trawls > 16 mm and < 32 mm
	OTTER	Otter trawl
Nephrops trawl	TR2	Bottom trawls and seines > 70 mm and < 100 mm
Pelagic nets	PEL_TRAWL	Pelagic trawl
	PEL_SEINE	Pelagic seine
Gill nets	GN1	Gillnets and entangling nets
Pots	POTS	Pots and creel
Dredge	DREDGE	Dredge
Longline	LL1	Longlines

Total landings for functional groups in 1973 were taken from ICES landing statistics (ICES, 2018a) and working group reports (ICES, 2017d, ICES, 2018c). In the absence of fleet based data prior to 2003, landings were proportioned amongst Ecopath fleets based on the proportions landed by STECF gears in 2003. Total discards for cod, whiting, haddock, plaice, whiting, and *Nephrops* were taken from ICES stock assessments (ICES, 2019) and distributed amongst gear types based on STECF proportions. For functional groups without ICES discards estimates, total discards were calculated using the discard/landings ratio from STECF using ICES landings from 1973. Bycatch of marine mammals were included in the parameterisation of discards based on literature and records from the Irish and Celtic Seas (Bentley et al., 2018). The total and proportional landing and discards of functional groups by fleet are shown in Figure 2.4. See Annex 1.3 for 1973 landings and discards data tables.

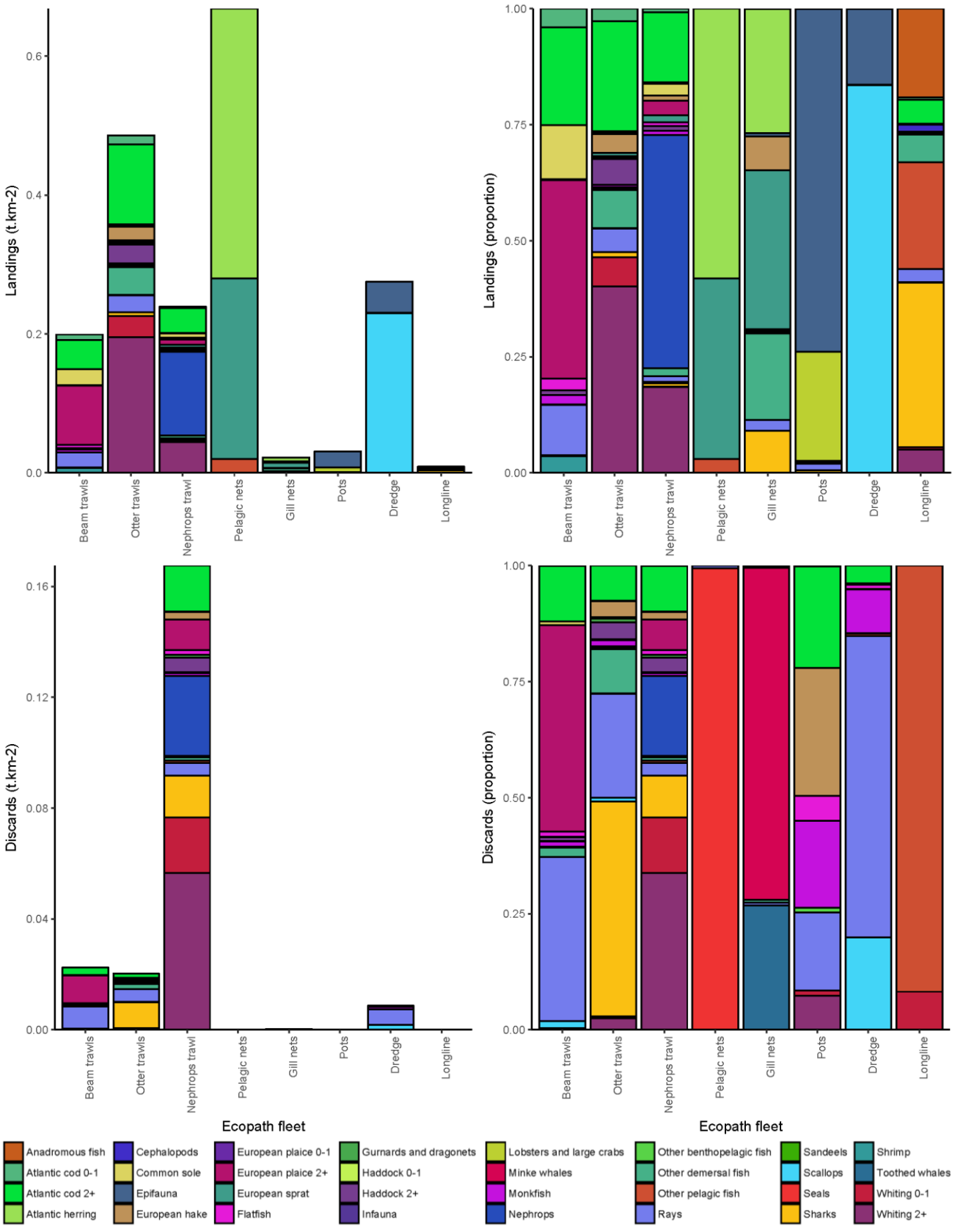


Figure 2.4. Landings and discards (t.km^{-2} and proportions) of functional groups by fishing fleet in the Irish Sea Ecopath model.

2.5 PreBal diagnostics

Pre-balance (PreBal) diagnositcs described by Link (2010) provide a means to judge the ecological quality of Ecopath models. Below, in Table 2.4, we list the PreBal criteria and our models performance against each.

Table 2.5. *PreBal diagnostics for the Irish Sea EwE key run.*

Criteria	Irish Sea model results	Comment
Biomasses should span 5-7 orders of magnitude.	Spans 5 orders of magnitude	-
Biomass slope (on log scale) around 5-10% decline with increasing TL.	8.9%	-
Are any functional group biomasses notably above/below the line?	See figure 2.5.	Seabirds, seals, whales, and discards below line: anthropogenic impacts. Epifauna, infauna, and other pelagic fish above line. These groups have high model estimated biomasses due to their importance as prey groups.
Compared across taxa, the ratio between predator and prey biomass should be < 1 and ~ 1-2 decimal places, depending on TL.	**66.14 %** > 0.009 & < 1 **18.86 %** < 0.009 **15 %** > 1 See table 2.5.	Ratios below 2 decimal places are prominent for the relationships between seals, seabirds, and their prey and for predators of epifana and infauna. Top predators have low relative biomasses whilst epifauna and infauna have high biomasses. Ratios greater than 1 are most commonly seen between high level trophic levels which predate upon each-other and also when comparing predator biomass to discard biomass.
PB should decline with increasing TL (excluding homeotherms).	Criteria met; see figure 2.7.	When excluding homeotherms, benthic invertebrates and seaweed fall below the line whilst small & large zooplankton and phytoplankton are above.
QB should decline with increasing TL (excluding homeotherms).	Criteria met; see figure 2.8.	When excluding homeotherms, benthic invertebrates, gellatinous zooplankton, adult plaice, and rays fall below the line whilst small & large zooplankton, shrimp, cephalopods, and juvenile whiting are above.
No taxa should have a PB greater than phytoplankton	Criteria met; see figure 2.7.	-
PQ should fall below 1 for all functional groups and between 0.1 and 0.3 for finfish.	Criteria met; see figure 2.9.	-
PR should fall below 1 for all functional groups	Criteria met; see figure 2.6.	-
EE should fall below 1 for all functional groups	Criteria met; see table 2.1.	-
Total production and consumption should decrease with increasing TL.	Criteria met; see figure 2.10.	-

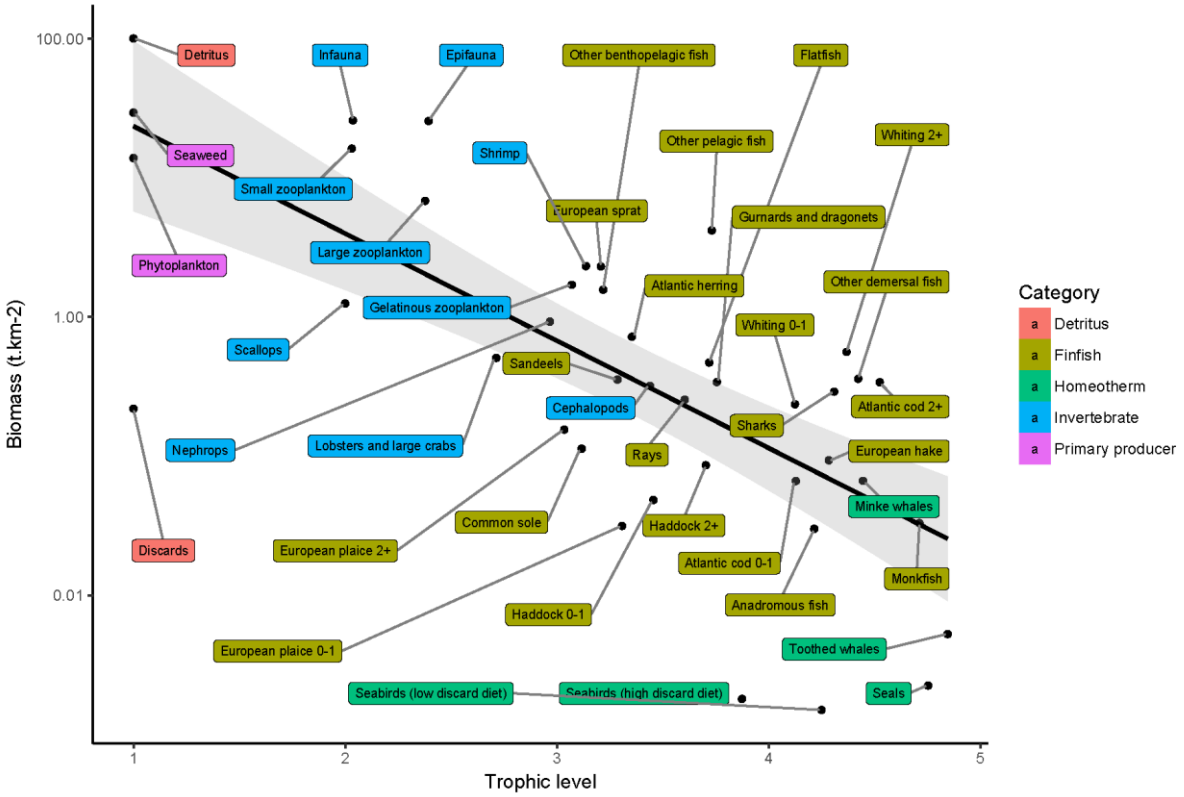


Figure 2.5. Declining biomass with increasing trophic level. Line: linear regression biomass ~ TL. Colours represent groupings used in further PreBal analysis. The grey bands around the line represent the standard error of the regression line.

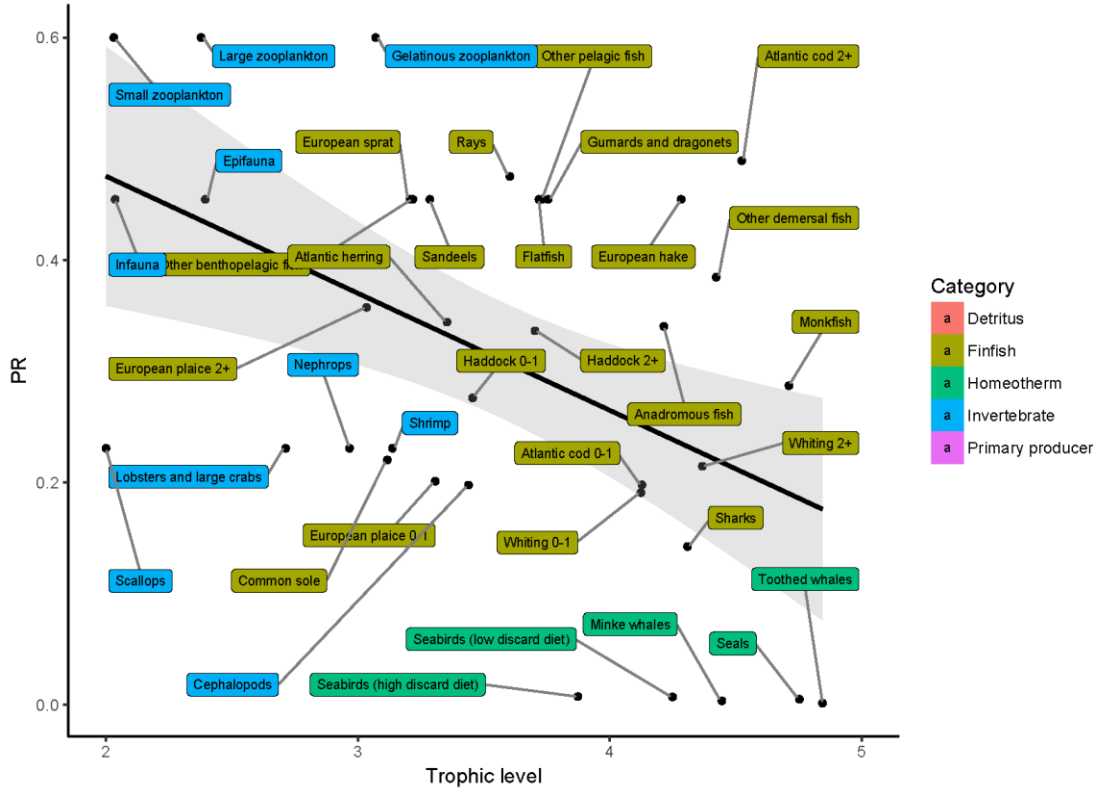


Figure 2.6. Declining Production/Respiration (PR) ratio with increasing trophic level. Line: linear regression PR ~ TL. Colours represent groupings used in further PreBal analysis. The grey bands around the line represent the standard error of the regression line.

Table 2.5. *Predator/prey biomass ratios. Green = 1< and >0.009; blue = cannibalism (1); orange = <0.009; red = > 1. Ratios have been rounded to their closest integer.*

Prey \ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
1 Toothed whales	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2 Minke whales	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Seals	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4 Seabirds (high discard diet)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5 Seabirds (low discard diet)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
6 Sharks	0	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7 Rays	-	-	0	-	-	1	1	-	-	-	-	-	-	-	-	-	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
8 Atlantic cod 2+	0	-	0	-	0	1	1	-	-	-	-	-	-	-	-	-	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9 Atlantic cod 0-1	0	-	0	0	-	4	4	-	-	8	-	-	-	-	-	-	-	0	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
10 Whiting 2+	0	-	0	-	-	1	-	1	0	1	-	-	-	-	-	-	1	0	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-
11 Whiting 0-1	0	-	-	0	-	1	1	1	0	2	-	-	-	-	-	-	2	0	0	-	-	2	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-
12 Haddock 2+	-	-	-	-	-	3	-	4	-	-	-	-	-	-	-	-	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
13 Haddock 0-1	-	-	-	-	-	6	5	7	-	12	-	-	-	-	-	-	-	1	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14 European plaice 2+	0	-	0	-	-	2	-	2	-	-	-	-	-	-	-	-	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
15 European plaice 0-1	-	-	-	-	0	9	-	11	-	-	-	-	-	-	-	-	15	1	-	-	11	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
16 Common sole	0	-	0	-	-	3	2	3	1	-	-	-	-	-	-	-	4	-	1	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
17 Flatfish	0	-	0	0	0	1	1	1	0	1	1	0	-	-	-	-	1	0	-	-	1	1	-	2	-	-	-	-	-	-	-	-	-	-	-	-	-
18 Monkfish	0	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
19 European hake	0	-	-	-	-	-	-	4	-	6	-	-	-	-	-	-	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 Sandeels	0	0	0	0	0	1	1	-	0	2	1	0	-	0	0	-	1	-	-	1	1	1	-	-	-	-	0	-	-	-	1	-	-	-	-	-	-
21 Gurnards and dragonets	-	-	0	0	0	1	1	1	0	-	1	-	-	-	0	-	1	0	-	-	1	1	5	2	-	-	-	-	-	-	1	-	-	-	-	-	-
22 Other demersal fish	0	-	0	0	0	1	-	1	0	-	-	-	-	-	-	-	-	0	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
23 Other benthopelagic fish	0	-	0	0	0	0	0	0	0	0	0	0	0	0	0	-	0	0	0	-	0	0	1	0	-	3	0	0	-	-	0	-	-	-	1	-	-
24 Atlantic herring	0	0	0	0	0	0	-	0	0	1	-	-	-	-	-	-	-	0	0	-	-	0	-	-	-	-	0	-	-	-	0	-	-	-	-	-	-
25 European sprat	0	0	0	0	0	0	-	0	0	0	0	0	0	-	0	-	0	0	0	-	0	0	-	-	-	2	0	-	-	-	-	-	-	-	-	-	-
26 Other pelagic fish	0	0	0	0	0	0	0	0	0	0	0	-	-	-	0	-	0	0	0	-	0	0	0	0	-	1	-	-	-	-	0	-	-	-	-	-	-
27 Anadromous fish	0	-	0	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
28 Lobsters and large crabs	-	-	0	-	0	1	1	1	0	-	-	0	0	0	-	-	1	-	-	-	1	-	3	-	-	-	-	1	-	-	1	-	-	-	-	-	-
29 Nephrops	-	-	0	-	-	0	0	0	0	1	0	0	0	-	-	-	1	0	-	-	0	0	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
30 Shrimp	-	0	-	-	-	0	0	0	0	0	0	0	0	0	0	-	0	0	0	-	0	0	1	0	-	2	0	0	0	-	-	-	-	-	1	-	-
31 Cephalopods	0	0	0	0	-	1	1	1	0	2	-	0	-	-	0	-	1	0	-	-	1	1	5	-	-	-	-	-	3	-	1	-	-	-	-	-	-
32 Scallops	-	-	-	-	-	0	0	-	0	-	-	-	-	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	20	-	-	-	-
33 Epifauna	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-	0	0	0	0	0	0	0	-	0	0	0	0	-	1	-	0	-	-
34 Infauna	-	-	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-	-	-	0	0	0	0	0	-	0	0	0	0	-	1	1	-	-	-	-
35 Gelatinous zooplankton	-	-	-	-	-	0	-	0	-	0	0	-	-	-	-	-	0	-	-	-	0	-	1	0	1	-	-	-	-	-	0	-	-	-	-	-	-
36 Large zooplankton	-	0	-	0	0	0	0	0	0	0	0	0	0	0	0	-	0	0	0	0	0	0	0	0	0	1	0	-	0	0	0	-	-	-	0	1	-
37 Small zooplankton	-	0	-	-	-	0	0	0	0	0	0	0	0	-	0	-	0	-	-	0	0	0	0	0	0	-	-	-	0	0	0	-	2	-	0	0	1
38 Seaweed	-	-	-	-	-	0	-	-	-	-	-	0	-	-	0	-	0	-	-	-	-	-	0	-	-	-	-	0	-	-	-	-	1	-	-	-	-
39 Phytoplankton	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	2	2	0	0	1
40 Discards	-	-	-	0	0	-	1	-	-	3	-	-	-	1	0	-	-	-	-	-	-	-	-	-	-	-	-	2	4	11	-	-	-	-	-	-	-
41 Detritus	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	-	-	-	0	0	0	-	0	0

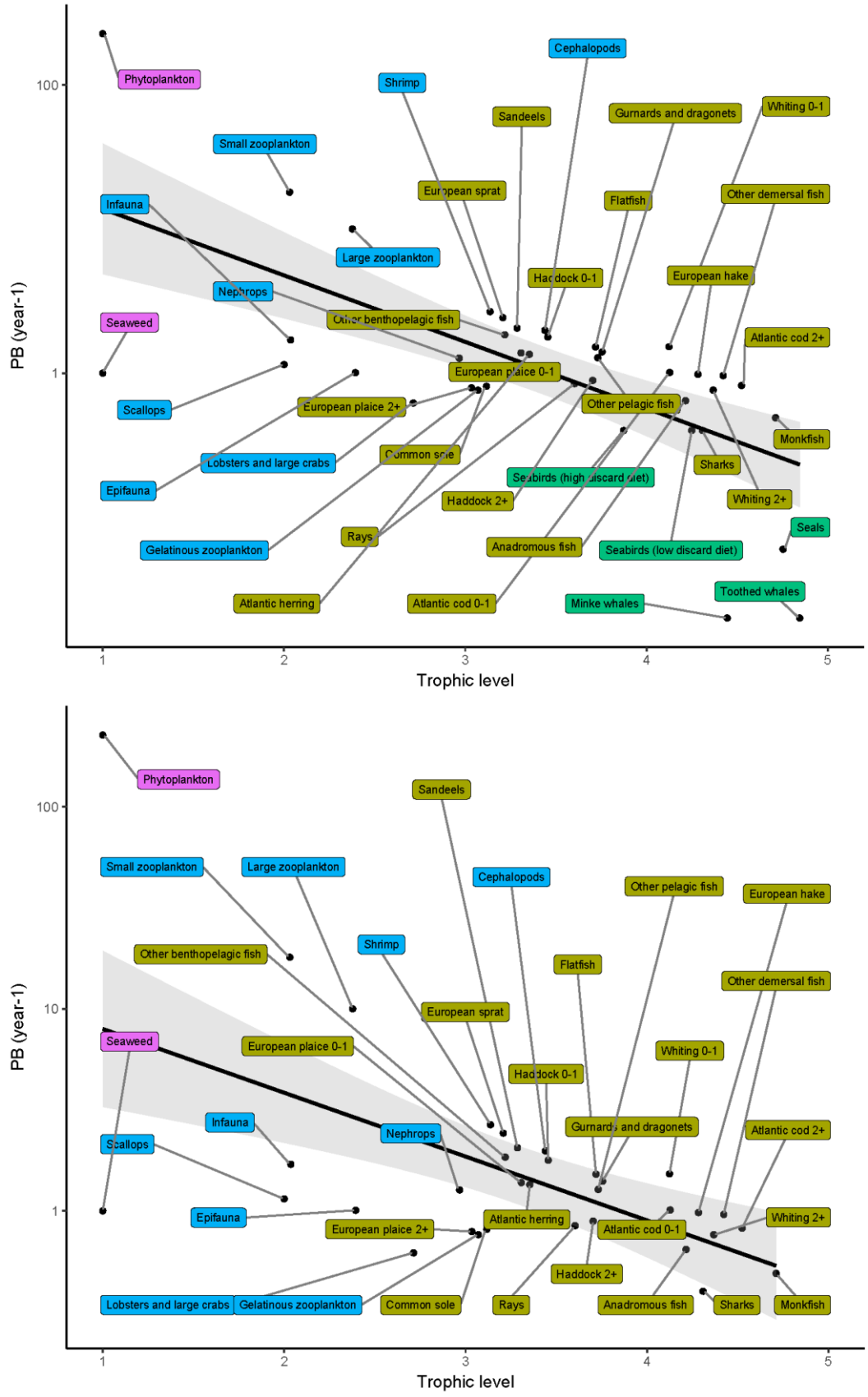


Figure 2.7. Declining PB ratios with increasing trophic level. Line: linear regression PB ~ TL. For the interpretation of the colours see figure 2.5. Analysis was conducted with and without homeotherms. The grey bands around the line represent the standard error of the regression line.

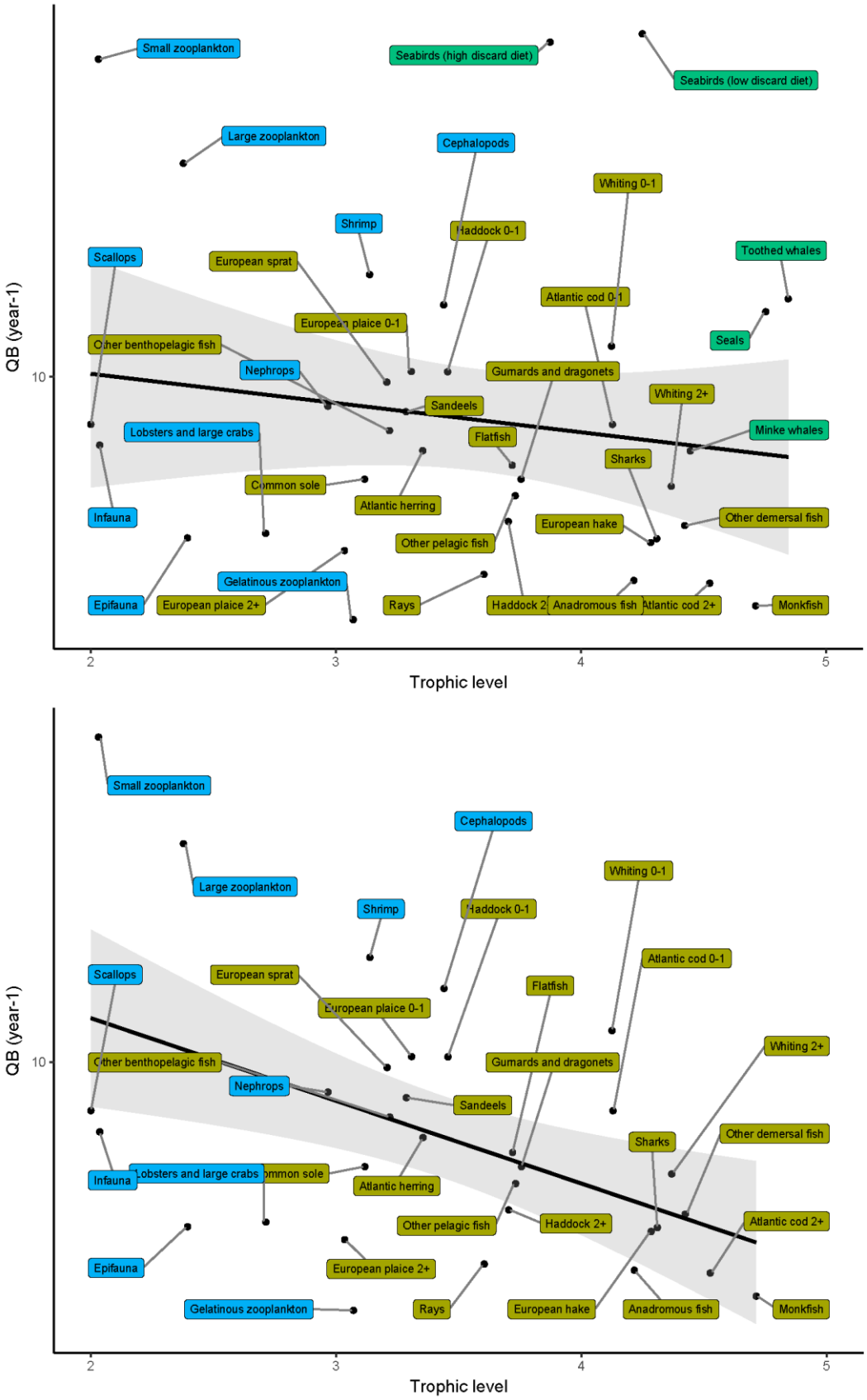


Figure 2.8. *Declining QB ratios with increasing trophic level. Line: linear regression QB ~ TL. For the interpretation of the colours see figure 2.5. Analysis was conducted with and without homeotherms. The grey bands around the line represent the standard error of the regression line.*

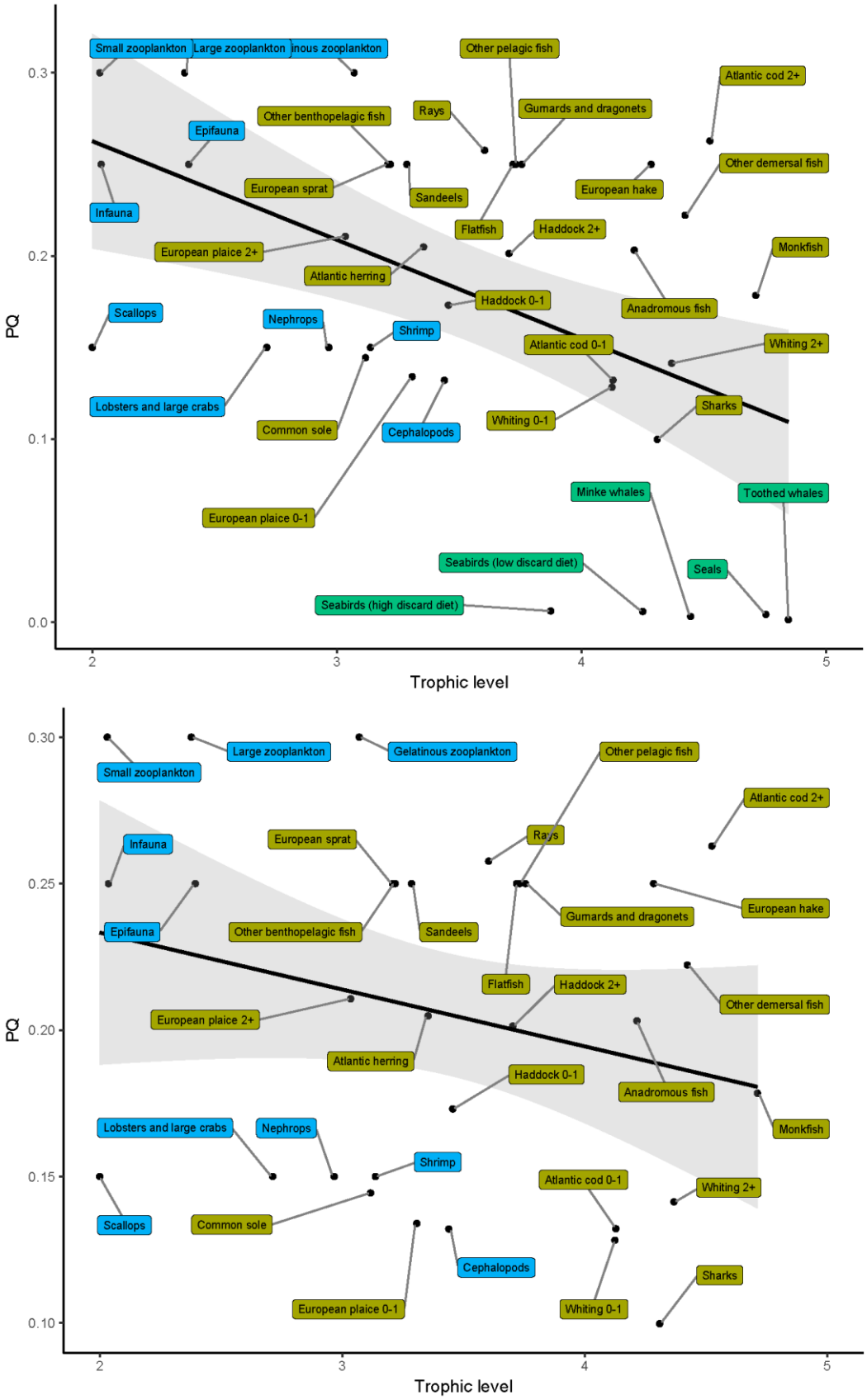


Figure 2.9. *PQ ratios with increasing trophic level. Line: linear regression PQ ~ TL. For the interpretation of the colours see figure 2.5. Analysis was conducted with and without homeotherms. The grey bands around the line represent the standard error of the regression line.*

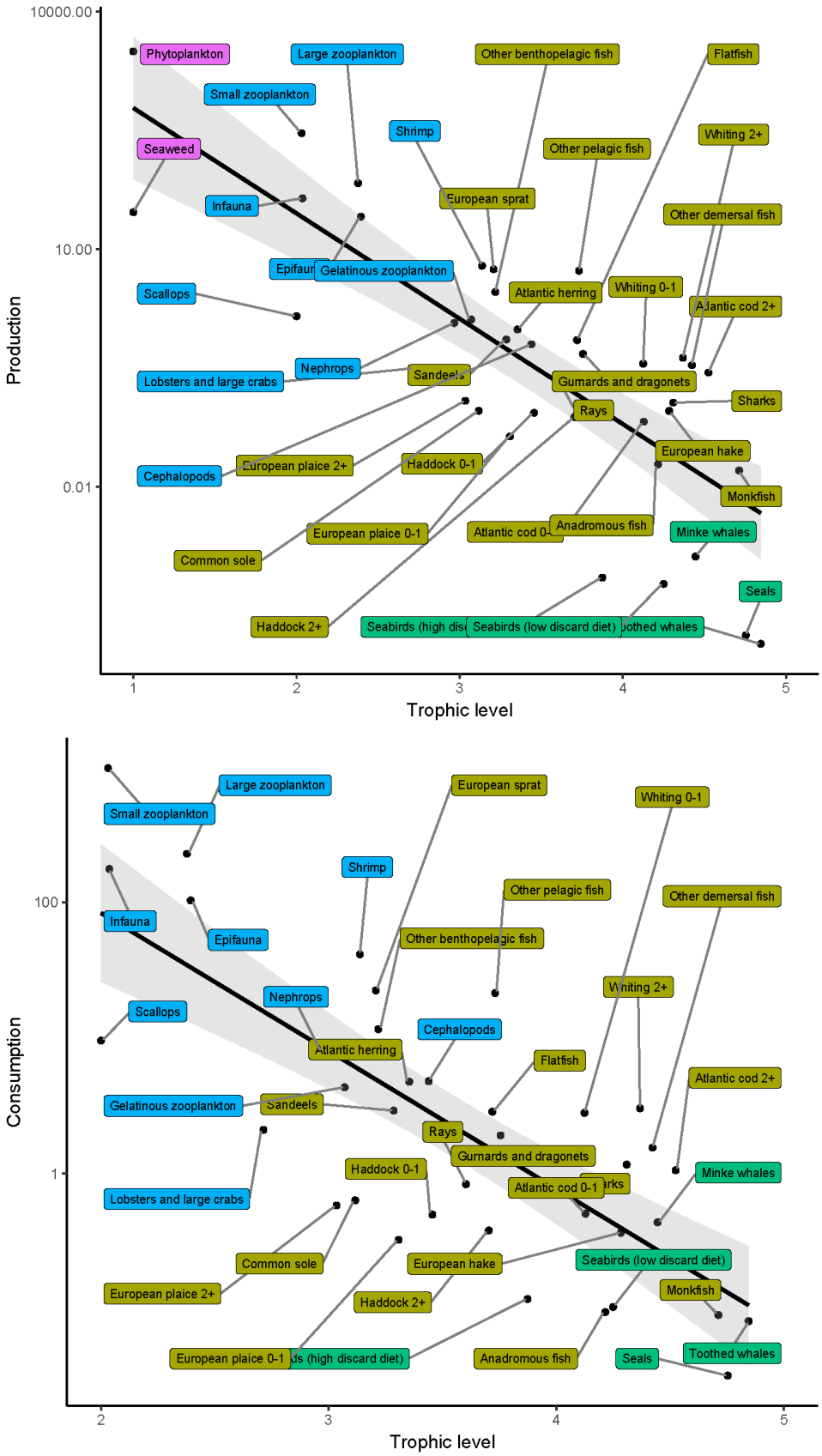


Figure 2.10. *Declining total production and consumption with increasing trophic level. Line: linear regression vital rate ~ TL. For the interpretation of the colours see figure 2.5. The grey bands around the line represent the standard error of the regression line.*

2.6 Basic input pedigree

Confidence intervals were assigned to the model's basic input parameters using the EwE pedigree routine (Table 2.6) (Christensen and Walters, 2004). Confidence intervals are assigned based on the quality of the data origin. These confidence intervals provide the basis for parameter uncertainty when using Monte Carlo simulations to generate 95% confidence intervals in model outptut.

Table 2.6. *Input data pedigree indicating data origin and confidence intervals.*

	Functional group	B	PB	QB	Diet	Catch
1	Toothed whales	0.5	0.5	0.5	0.5	0.5
2	Minke whales	0.5	0.5	0.5	0.5	0.5
3	Seals	0.5	0.5	0.5	0.8	0.3
4	Seabirds (high discard diet)	0.3	0.5	0.5	0.8	-
5	Seabirds (low discard diet)	0.3	0.5	0.5	0.8	-
6	Sharks	0.3	0.5	0.5	0.8	0.3
7	Rays	0.8	0.5	0.5	0.8	0.3
8	Atlantic cod 2+	0.1	0.5	0.5	0.1	0.3
9	Atlantic cod 0-1	0.8	0.7	0.8	0.1	0.3
10	Whiting 2+	0.3	0.5	0.5	0.8	0.3
11	Whiting 0-1	0.8	0.7	0.8	0.1	0.3
12	Haddock 2+	0.3	0.5	0.5	0.1	0.3
13	Haddock 0-1	0.8	0.7	0.8	0.1	0.3
14	European plaice 2+	0.1	0.5	0.5	0.8	0.3
15	European plaice 0-1	0.8	0.7	0.8	0.8	0.3
16	Common sole	0.1	0.5	0.5	0.1	0.3
17	Flatfish	0.3	0.8	0.5	0.1	0.3
18	Monkfish	0.3	0.5	0.5	0.8	0.3
19	European hake	0.8	0.8	0.5	0.8	0.3
20	Sandeels	0.8	0.8	0.5	0.1	0.3
21	Gurnards and dragonets	0.3	0.8	0.5	0.1	0.3
22	Other demersal fish	0.8	0.5	0.5	0.1	0.3
23	Other benthopelagic fish	0.8	0.8	0.5	0.1	0.3
24	Atlantic herring	0.1	0.5	0.5	0.1	0.3
25	European sprat	0.8	0.8	0.5	0.1	0.3
26	Other pelagic fish	0.8	0.8	0.5	0.1	0.3
27	Anadromous fish	0.8	0.5	0.5	0.1	0.3
28	Lobsters and large crabs	0.8	0.5	0.8	0.1	0.3
29	Nephrops	0.8	0.5	0.8	0.8	0.3
30	Shrimp	0.8	0.5	0.8	0.1	0.3
31	Cephalopods	0.8	0.5	0.6	0.1	0.3
32	Scallops	0.8	0.5	0.8	0.8	0.3
33	Epifauna	0.8	0.5	0.8	0.5	0.3
34	Infauna	0.8	0.5	0.8	0.5	0.3
35	Gelatinous zooplankton	0.3	0.8	0.5	0.5	-
36	Large zooplankton	0.5	0.5	0.8	0.8	-
37	Small zooplankton	0.5	0.5	0.8	0.8	-
38	Seaweed	0.3	0.5	-	-	-
39	Phytoplankton	0.3	0.5	-	-	-
40	Discards	0.5	-	-	-	-
41	Detritus	0.8	-	-	-	-

Confidence interval classifications

Biomass (B)
0.8: Estimated by Ecopath/from another model/guesstimate
0.5: Approximate or indirect method
0.3: Sampling/locally, low precision
0.1: sampling locally, high precision

Production/biomass (PB)
0.8: Estimated by Ecopath
0.7: Guesstimate
0.5: Empirical relationship

Consumption/biomass (QB)
0.8: Estimated by Ecopath
0.6: From another model
0.5: Empirical relationship

Diet
0.8: Altered by fishers' knowledge/from another model
0.5: General species information from literature
0.1: Stomach records (quantitative, detailed, from Irish Sea: DAPSTOM)

Catch
0.5: National statistics
0.3: Local study/ICES catch statistics for VIIa

Chapter 3: Ecosim

3.1 Stability tests

Prior to the application of forcing functions (fishing, environmental drivers), the stability of the model was examined by adjusting baseline fishing effort and examining functional group response and rate of recovery to equilibrium (Figure 3.1). The system was perturbed under three scenarios, (i) the complete cessation of fishing, (ii) a short period of fishing cessation (8 years), and (iii) a short period of increased fishing effort (2x) (8 years).

Before running stability simulations, Ecosim parameters were updated. In Ecosim it is possible to account for the satiation of functional groups and/or a functional groups capacity to alter its foraging behaviour. This is achieved by adjusting the 'Feeding time adjustment rate' of functional groups. Adjustment rates determine how quickly groups are able to adjust feeding times to stabilise consumption (Q/B) at the Ecopath base rate. Adjustment rates fall between 0 and 1, with larger values representing more rapid adjustments to foraging time. If a functional group has an adjustment rate of 0 it is incapable of altering the time it spends foraging and is assumed to be consistently active in the foraging arena. In general it is recommended that adjustment rates for all groups are initially set to 0, apart from marine mammals where a value of 0.5 is deemed reasonable (Christensen et al., 2008). Adjustment rates of 0.5 were therefore applied to marine mammals whilst the adjustment rates for all other groups were set to 0 (Figure 3.1a). When running the model without fishing we found instabilities in the biomass projections of multi-stanza groups (Figure 3.1a). To fix these instabilities feeding time adjustment rates of 0.2 were applied to all juvenile groups (Figure 3.1b).

The removal of fishing resulted in a relatively large increase in the biomass of Atlantic cod 2+ and monkfish. These groups experience large increases in their biomass as fishing mortality accounts for the majority of their explained mortality. This is reasonable due to their roles as top predators in the ecosystem. The removal of fishing effort for a short period led to large relative increases in the biomass of monkfish, Atlantic cod 2+, European plaice 2+, rays, Atlantic herring, whiting 2+, and common sole (Figure 3.1c). These species are the most heavily exploited by the fishing fleets in the base Ecopath model therefore this response is reasonable. When doubling fishing effort, Atlantic cod 2+, Atlantic cod 0-1, monkfish, European plaice 2+, European plaice 0-1, and rays experience the largest relative biomass declines (Figure 3.1d). The biomasses of haddock 2+ and flatfish increased. The rise in haddock biomass reflects the Irish Sea haddock explosion observed by surveys, fishers', and recent stock assessments (ICES, 2018c).

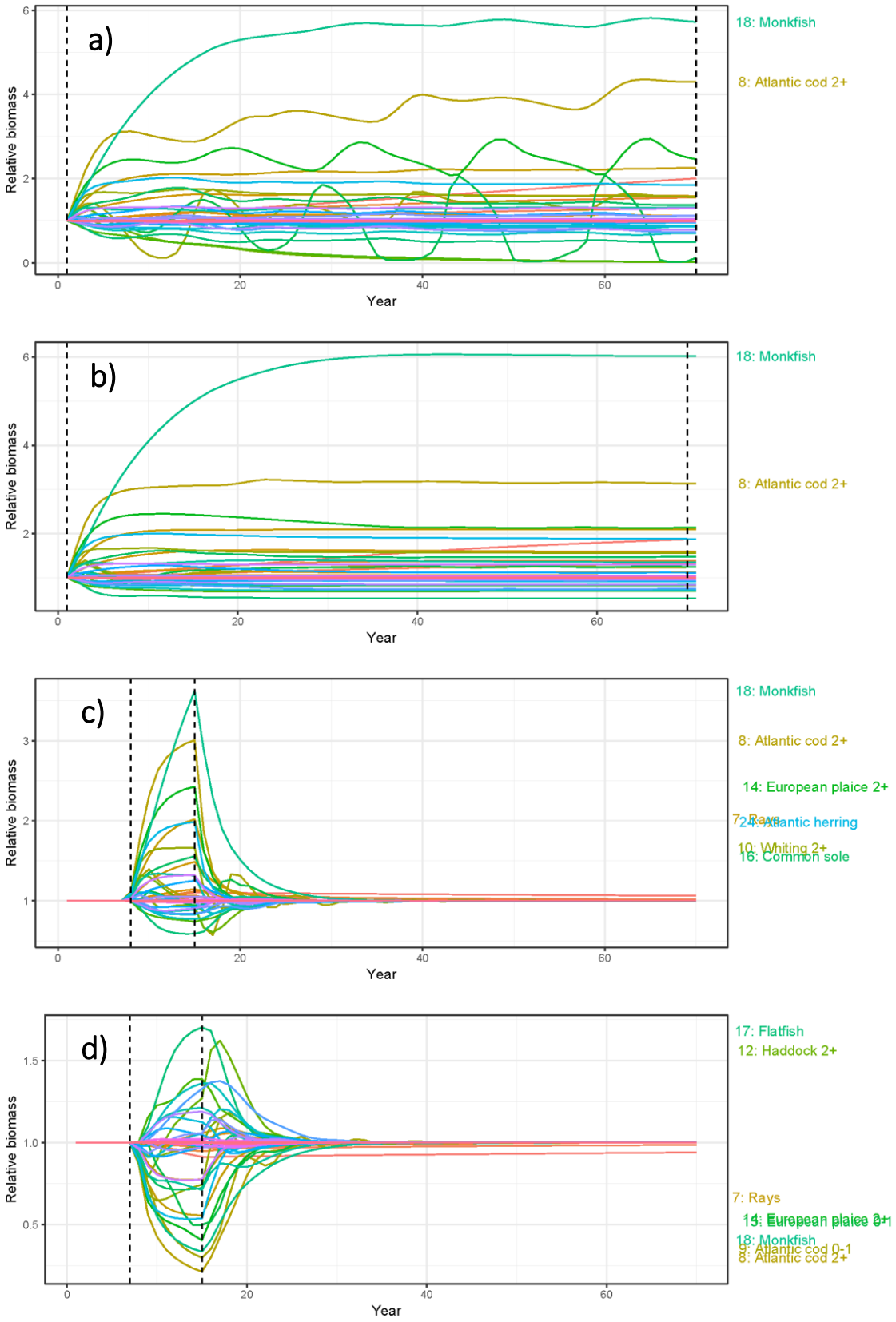


Figure 3.1. *Impacts of stopping fishing for all gears (a) without adjusted feeding rates and (b) with adjusted feeding rates. The models ability to recover from a period of fishing adjustment was tested by (c) stopping fishing for a short period and (d) doubling fishing for a short period. Dashed lines indicate the period within which fishing rates were altered. Species with the largest responses are noted on the right of each graph.*

3.2 Retrospective trend analysis

To identify which environmental time series data best accounted for observed changes in recruitment and production in the lower trophic levels of the Irish Sea model we collated environmental time series data on the Atlantic Multidecadal Oscillation (AMO), North Atlantic Oscillation winter index (NAOw), temperature, phytoplankton colour index, zooplankton abundance (Figure 3.2), and recruitment of multi-stanza functional groups.

Correlations between environmental variables, trends in zooplankton abundance, and fish recruitment were tested using Pearson's product-moment correlation after establishing data normality (p<0.05) using the Shapiro-Wilk test. Recruitment estimates were $\log_{10}$ transformed prior to analysis to stabilise the variance within and between time series. Recruitment and environmental data are often strongly autocorrelated as values in a given year are closely related to values in previous years (Pyper and Peterman, 1998). Therefore classical correlation tests tend to lead to a greater rate of Type 1 errors, where there is an increased chance of concluding that a correlation is statistically significant when in fact no correlation is present (Jenkins and Watts, 1968). We therefore adjust the degrees of freedom in the statistical tests to compensate for autocorrelation using the equation proposed by Chelton (1984) and modified by Pyper and Peterman (1998):

$$\frac{1}{N^*} = \frac{1}{N} + \frac{2}{N}\sum_j r_{XX}(j)\, r_{YY}(j) \qquad \text{Eq. 1}$$

where $N^*$is the number of independent joint observations on the time series X and Y, N is the sample size and $r_{XX}(j)$ and $r_{YY}(j)$ are the autocorrelation of X and Y at lag j. Estimates of r were calculated using the Box-Jenkins' equation (Box et al., 1976) modified by Chatfield (1989):

$$r_{XX}(j) = \frac{N}{N-j}\,\frac{\sum_{t=1}^{N-j}(X_{t+j}-\bar{X})(X_{t+j}-\bar{X})}{\sum_{t=1}^{N}(X_t-\bar{X})^2} \qquad \text{Eq. 2}$$

where $\bar{X}$ is the overall mean. In the present analysis, most time series comprised 43 common observations for which we applied nine lags (approximately $N/5$) following Pyper and Peterman (1998). Lags were adjusted for time series with fewer common observations.

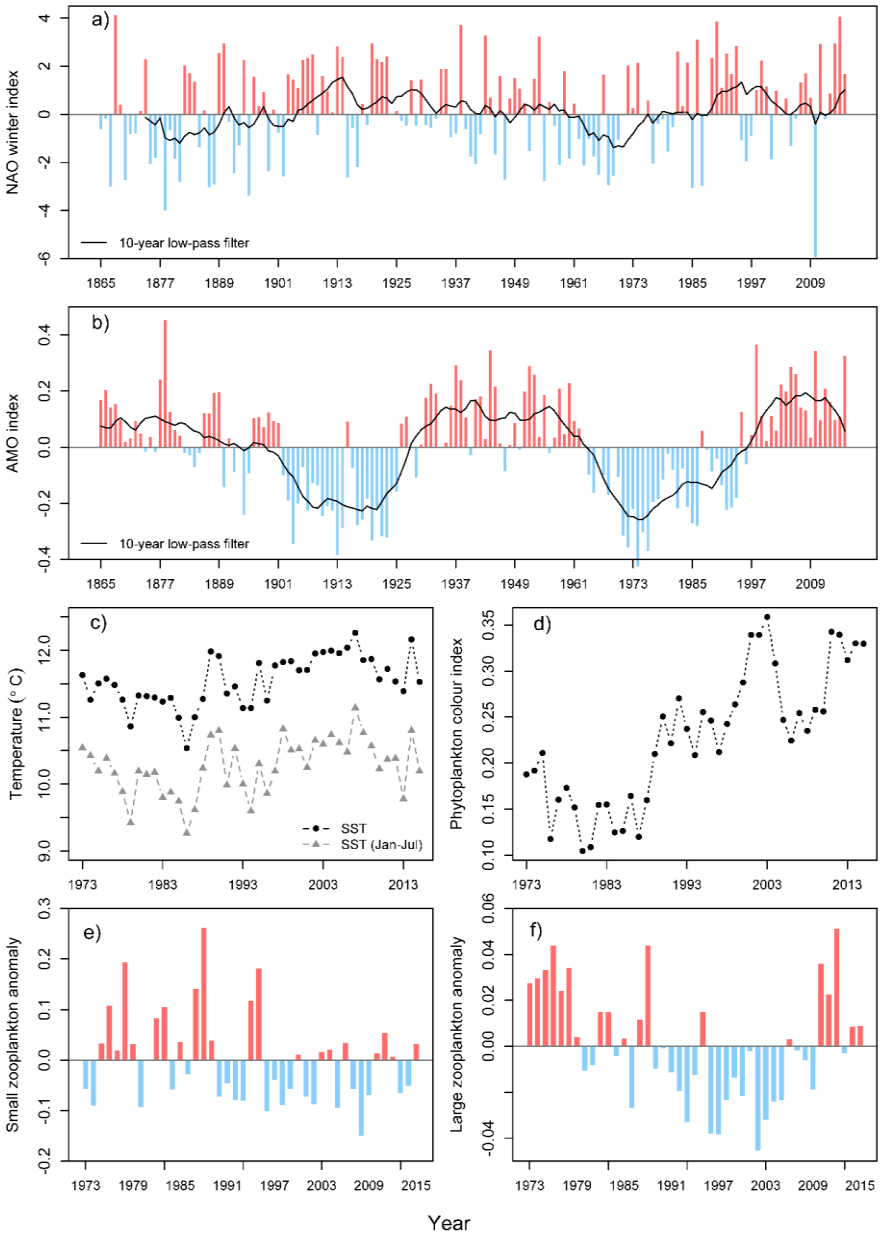


Figure 3.2. *Annual time series for environmental and plankton trends in the Irish; (a) Winter North Atlantic Oscillation (NAO) index, (b) Atlantic Multidecadal Oscillation (AMO) index, (c) annual averaged sea surface temperature (°C) (SST) and winter-spring (Jan-Jul) SST, (d) phytoplankton colour index, (e) small zooplankton (< 2mm) abundance, and (f) large zooplankton (> 2mm) abundance.*

The time series trend for large zooplankton was best explained by a 10 year running average of the NAOw. Recruitment of cod and whiting were best explained by a 10 year running average of the AMO. The smoothed NAOw was incorporated as a driver of the 'other mortality' of large zooplankton to replicate a direct negative effect. The inverse AMO trend was used to drive cod and whiting recruitment in the model. The EwE model uses a Deriso-Schnute delay-difference model (Deriso, 1980, Schnute, 1987)to keep track of the number of juveniles which recruit to the adult stage, and the number at age/size in the adult groups. Within this model, the number of age 0 juveniles produced each month is

calculated as a function of adult biomass, numbers, and food consumption. To have the AMO as a driver of recruitment success, the recruitment function at time= *t* was multiplied by the AMO anomaly at time= *t*, where the AMO anomaly multiplier at t_1 is 1. Figure 3.3 shows the results of the correlation analysis supporting the choice of environmental time series used in fitting the model.

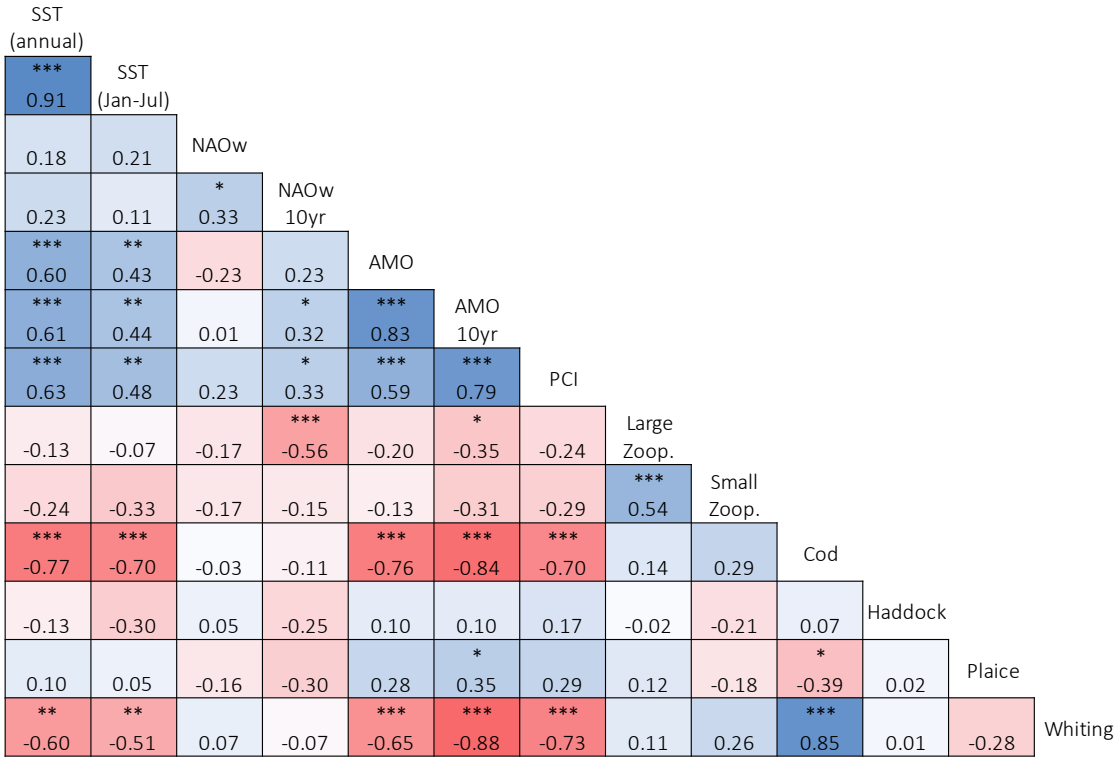


Figure 3.3. *Correlation matrix for environmental variables, plankton trends, and fish recruitment in the Irish Sea using Pearson's cross product-moment correlation. Variables include sea surface temperature (SST; °C), phytoplankton colour index (PCI), North Atlantic Oscillation winter index with a 10-year low-pass filter (NAO), Atlantic Multidecadal Oscillation with a 10-year low-pass filter (AMO), large zooplankton abundance (L.zoop.), and small zooplankton abundance (S.zoop.). Fish recruitment time series were taken from ICES stock assessments for cod, haddock, plaice, and whiting. The correlation matrix is shaded to signify the strength of positive (blue) and negative (red) correlations in relation to their r values. Statistically significant correlations are denoted: *p<0.05; **p<0.01; ***p<0.001.*

3.3 Functional responses to temperature

Previous Ecosim studies which investigated the impact of ocean warming on ecosystem structure and fisheries yield (Bentley et al., 2017, Serpetti et al., 2017, Corrales et al., 2018) established a methodology for the incorporation of temperature response functions for functional groups in Ecosim. Temperature functional responses affect the consumption rates of consumers. The functional response imposes a multiplier on the consumption equation, which can range from zero to one. Temperature tolerance ranges have previously been applied as Gaussian response functions (Bentley et al., 2017,

Serpetti et al., 2017) however Corrales et al. (2018) used trapezoid response functions which are likely more appropriate. When using Gaussian response functions, functional groups are very sensitive to temperature deviation from their optimum. Using a trapezoid response function allows us to assign an optimum temperature range using 10th and 90th preferable temperature quantiles, making groups less responsive to slight deviations from their optimum. The Irish Sea key run therefore uses trapezoid response functions. At a group's optimum temperature range, consumption is multiplied by one, therefore temperature has no effect on consumption. As temperature moves away from a group's optimum range the multiplier moves towards zero, therefore reducing the consumption rate. At a group's upper and lower tolerance levels the multiplier is equal to zero and groups are unable to ingest energy and thus survive.

Optimum temperature and tolerance ranges for Irish Sea functional groups were taken from AquaMaps (Kaschner et al., 2016) (Figure 3.4). Temperature ranges for functional groups with multiple species were prorated by species contribution to the group's biomass. A depth integrated temperature (DIT) time series was constructed for the Irish Sea from 1973- 2016 using data spanning 11 depth layers from 0 m (surface temperature) to 150 m (max depth temperature) and was derived from two datasets:

- Dataset 1: Atlantic-European North West Shelf-Ocean Physics Reanalysis from Met Office (1985-2016)
- Dataset 2: Hadley Centre Sea Ice and Sea Surface Temperature data set (HadISST) (1973:2016)

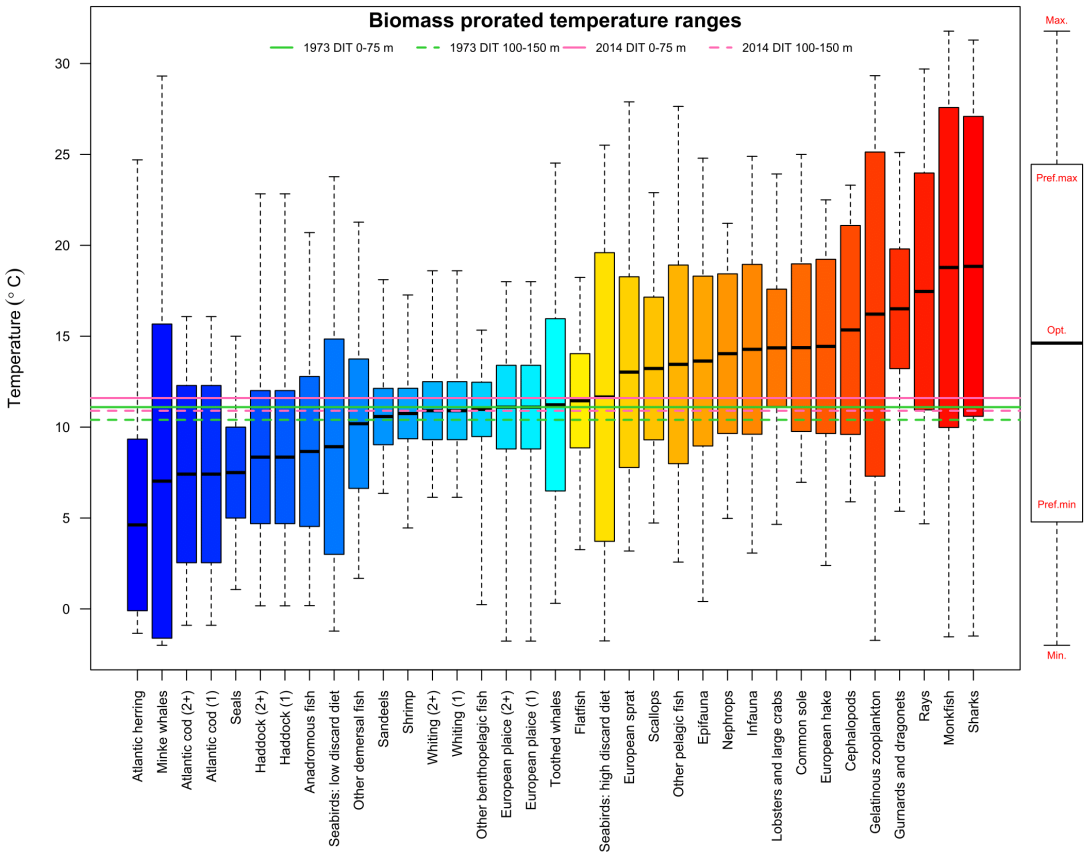


Figure 3.4. *Temperature ranges for functional groups in the Irish Sea Ecosim model. Data taken from AquaMaps (Kaschner et al., 2016).*

3.4Time series data

In order to accurately simulate biomass and catch trends, Ecosim requires the incorporation of time series data to both calibrate and drive simulations. The Irish Sea model uses 50 calibration time series (biomass and catch) for functional groups. Calibration time series are shown in Figure 3.5 and Figure 3.6 (see Annex 2.1 for data sources). Four types of forcing time series were used in the model: fishing effort, fishing mortality, forced catch, and abiotic forcing. Fishing fleet catch rates were driven by fishing effort time series. Effort trends for otter trawl, beam trawl, and *Nephrops* trawl were taken from scientific data, whereas trends for pelagic nets, gillnets, pots, dredge, and longline from 1973-2003 were reconstructed by fishers' during WKIrish workshops (Figure 3.7, see Annex 2.2 for data sources). From 2003-2016 STECF effort data was used for these fleets (STECF, 2018). The landings of sharks, whiting, anadromous fish, and infauna could not be simulated using the effort time series and were therefore driven by fishing mortality (sharks and whiting) or forced catch (anadromous fish, infauna). Abiotic forcing time series were assigned using Pearson's correlation analysis (section 3.2).

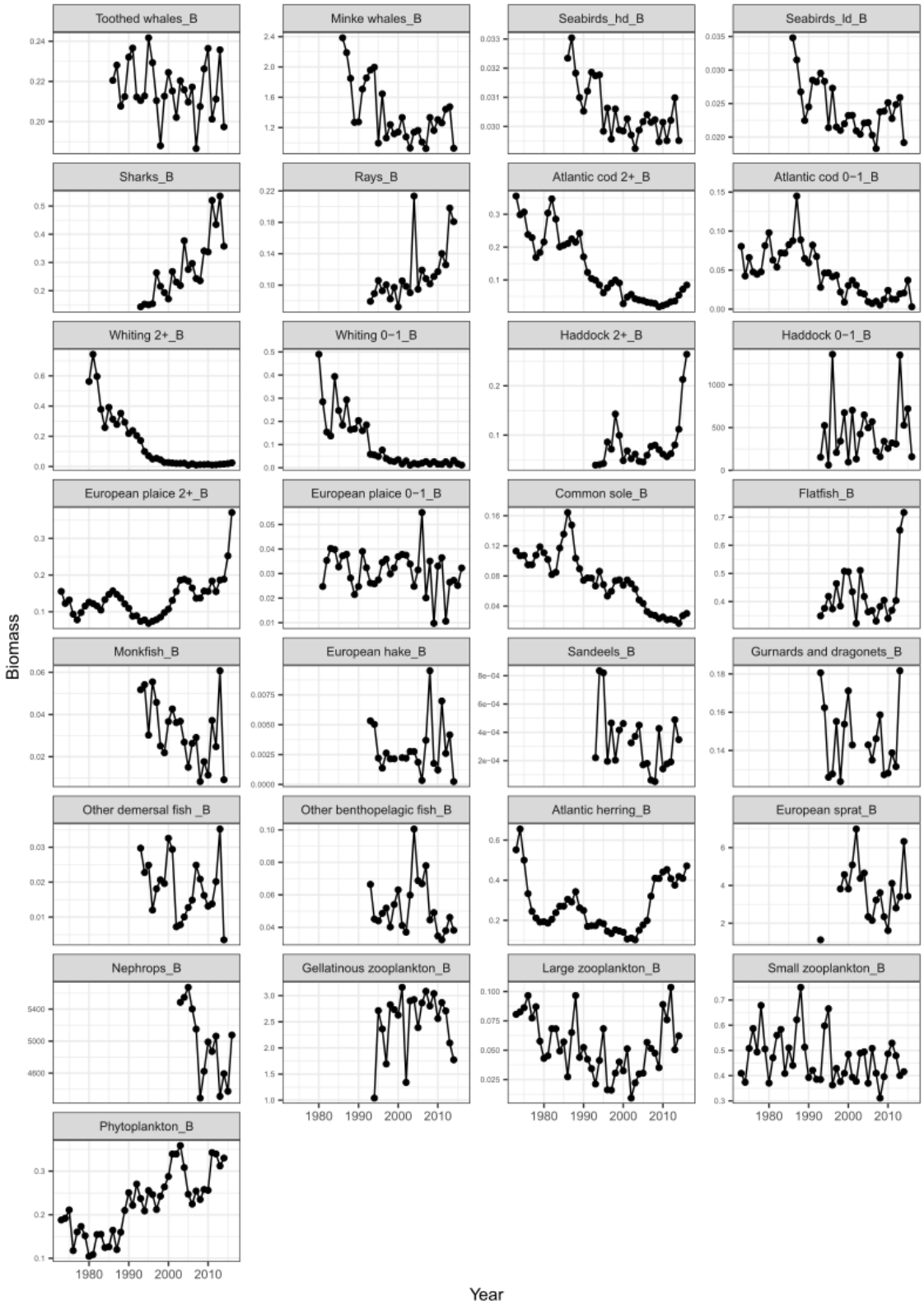


Figure 3.5. *Biomass calibration time series used for fitting the Irish Sea Ecosim model (1973-2016). For references see Annex 2.1.*

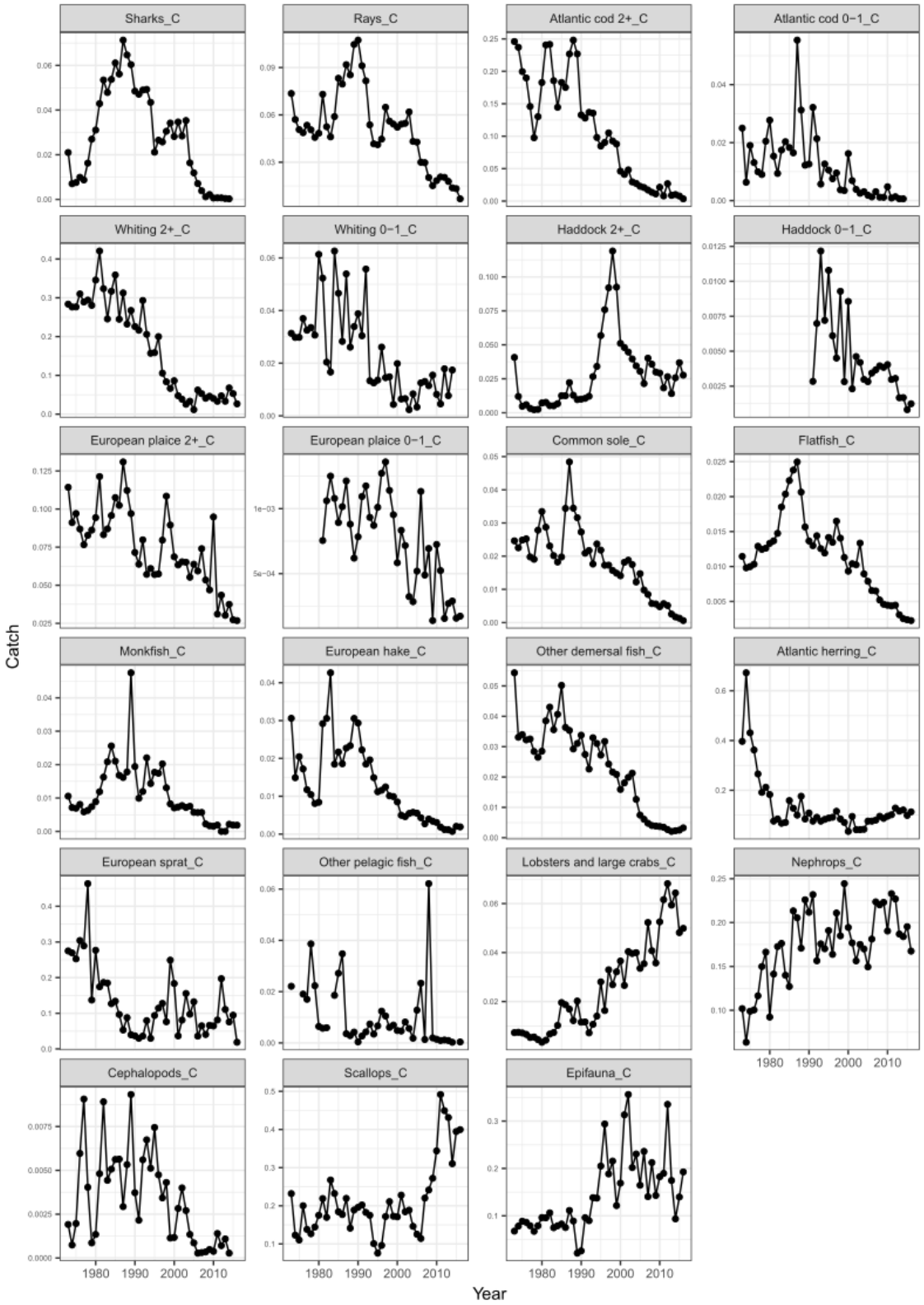


Figure 3.6. *Catch calibration time series used for fitting the Irish Sea Ecosim model (1973-2016). For references see Annex 2.1.*

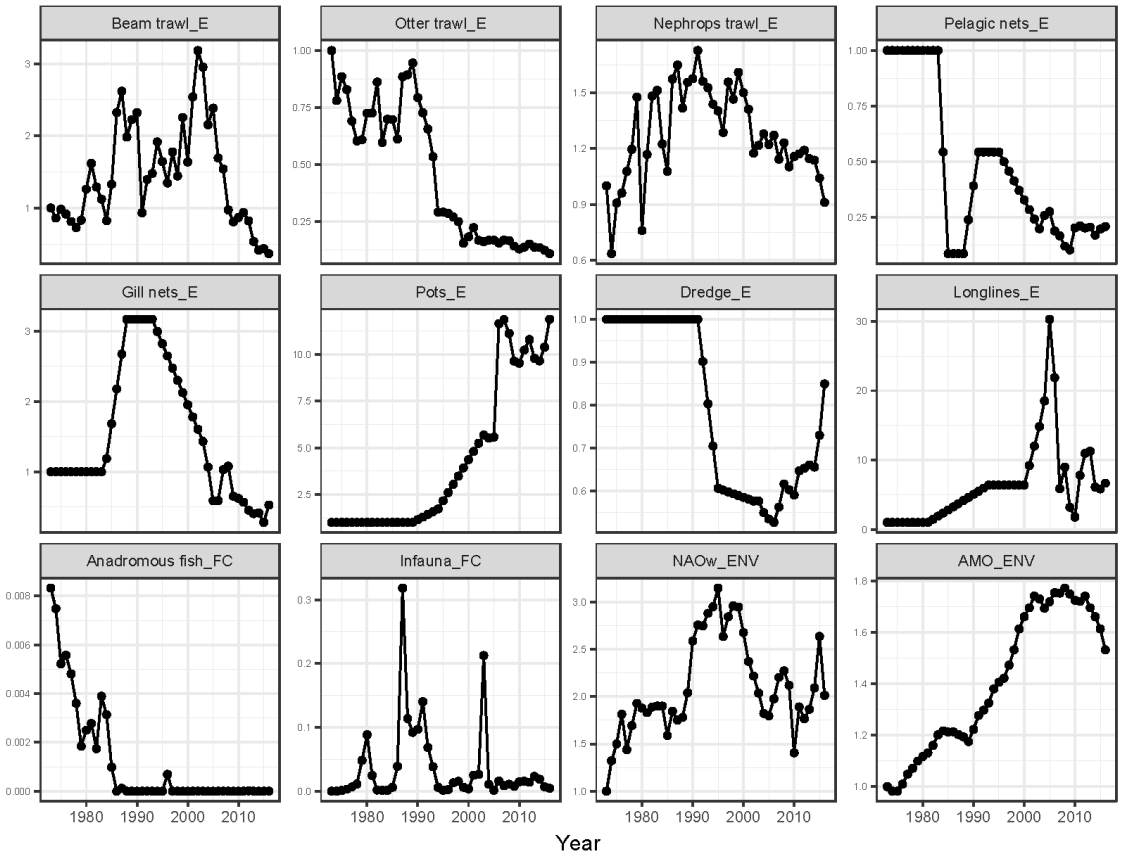


Figure 3.7. *Forcing time series used in the Irish Sea Ecosim model (1973-2016), including fishing fleet effort drivers (E), forced catch (FC), and environmental drivers (ENV). For references see Annex 2.2.*

3.5Time series weighting

Time-series are weighted differently when calculating sum of squares according to our judgement on data reliability and relevance. The choice of weightings are subjective and vulnerability estimates have been shown to not be very sensitive applied weightings (ICES 2017e). In the key run we therefore only alter the weightings for time series with low confidence based on data origin (i.e. mammal and bird biomass trends from a generalised additive model), trend noise (i.e. IBTS_VIIa biomass survey estimates for sandeels, monkfish, hake, demersal fish, benthopelagic fish), and group composition (i.e. trends for groups with multiple species such as demersal fish and pelagic fish catch). Weightings of 0.5 were applied to trends associated with low confidence as shown in Figure 3.11 whilst all others were left at the default setting of 1.

3.6Time series fitting

Ecosim uses the foraging arena theory (Ahrens et al., 2012) to quantify “vulnerabilities”, which represent the degree to which a change in predator biomass will impact predation mortality for a given prey. Vulnerabilities are adjusted to statistically fit model simulations to observed data. This is done by applying multipliers to the rate with which a prey moves between being vulnerable and not vulnerable

(Christensen and Walters, 2004). Multipliers can range from one to infinity with two as the default. Vulnerabilities with multipliers greater than two indicate top-down control, where predator biomass drives prey mortality, whereas vulnerabilities with multipliers between one and two suggest bottom-up control, where even large increases in predator biomass cause only a limited increase in the consumption rate of that predator on the given prey, therefore the biomass of the prey regulates predator consumption (Christensen et al., 2008).

Vulnerability multipliers can be estimated for predator-prey interactions or for whole predator diets. The automated fitting routine (Scott et al., 2016) facilitates the estimation of one or the other, however it is ecologically probable that, whilst species may exert a general top-down or bottom-up forcing on the majority of their prey due to their behaviour or morphology, some interactions may be unique as a result of prey behaviour, habitat, or morphology. Estimating predator-prey vulnerabilities alone accounts for these unique interactions, however the number of vulnerabilities that can be estimated this way is largely limited by the available degrees of freedom. As 52 calibration time series were used we were able to estimate a maximum of 51 predator/prey vulnerabilities before the model could be considered as 'overfitted', thus leaving the majority of the vulnerability matrix at its default setting of two. However, estimating predator vulnerabilities alone homogenises the impact of a group's biomass on its prey, meaning potentially important dynamics may be overlooked.

We therefore propose a new approach for the ecological optimisation of vulnerability multipliers which has the potential to improve model performance whilst remaining within the given degrees of freedom. Firstly, the automated stepwise fitting routine was used to find the model of best fit with predator vulnerabilities (first stage best fit, Figure 3.8). The best fit model was determined by the minimum difference between model simulations and time series observations using the weighted sum of squared differences (SS) and the Akaike Information Criterion for small sample sizes (AICc) (Akaike, 1974, Burnham and Anderson, 2003). Secondly, using the best fit model and the remaining degrees of freedom, we systematically searched for the most sensitive predator-prey vulnerabilities which would improve the statistical fit of model simulations to observed data (second stage best fit). The AICc was manually calculated for each iteration to take into account the predator vulnerabilities already estimated during the stepwise fitting procedure. The model iteration with the lowest AICc was used as the best fit model.

Once environmental drivers were applied to the model the automated stepwise fitting routine was used to generate a set of model iterations with altered predator vulnerability multiplies. Here we identified the first stage best fit model to include fishing and 26 estimated predator vulnerabilities (Figure 3.8, Table 3.1). At this stage the models SS was reduced from 1251 to 754 and the AICc was reduced from -

829 to -1753. As 26 parameters were estimated we were still capable to estimate a maximum of 25 additional predator/prey vulnerabilities to optimise model fit whilst retaining statistical integrity. Using a stepwise approach, the most sensitive predator/prey vulnerabilities were altered incrementally from the addition of 1 to 25 parameter changes. The second stage best fit model included the 26 predator vulnerabilities from the first stage best fit with the alteration of 14 predator/prey vulnerability multipliers (Figure 3.8, Table 3.2). The second stage best fit and final model had an improved SS of 656 and an AICc of -1984.

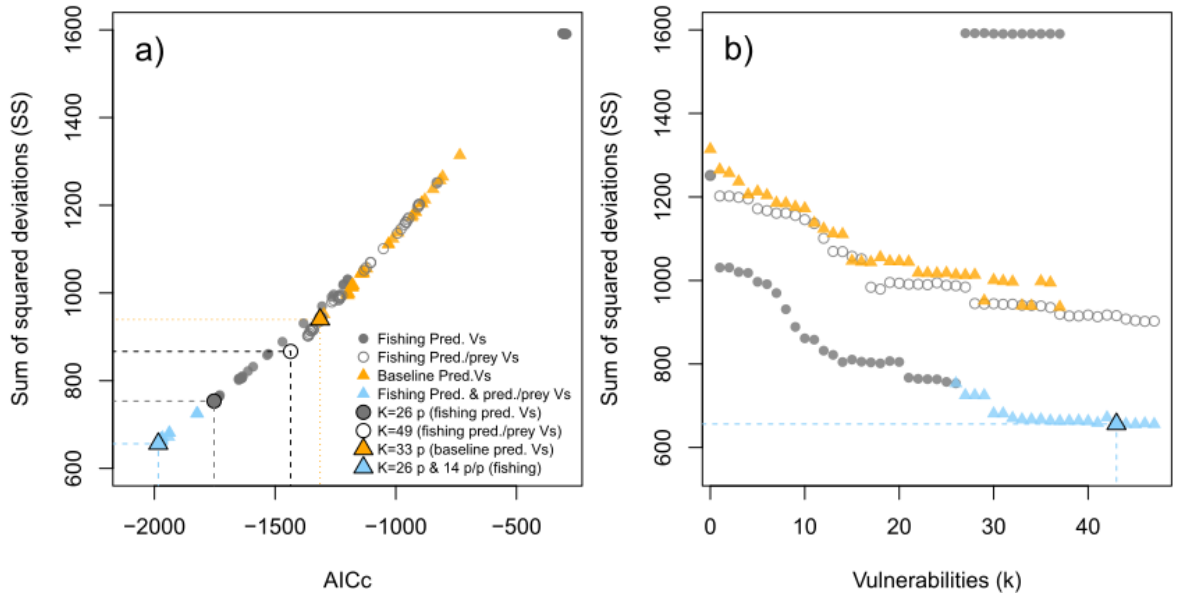


Figure 3.8 . *Ecosim fitting. (a) Sensitivity of the sum of squared deviations (SS) and Akaike's information criterion (AICc) to fitting predator (p) and predator/prey (p/p) vulnerabilities (Vs). Best fit models are highlighted by larger points and reference lines, with the legend indicating the number of parameters (K) estimated. (b) Change in SS as a function of vulnerability parameters estimated.*

Table 3.1. *Fitted predator vulnerabilities applied to specific groups in the key run during the first stage of model fitting.*

Functional group	Vulnerability
Sharks	2.17
Rays	2.42
Atlantic cod 2+	>100
Atlantic cod 0-1	>100
Whiting 2+	1.53
Whiting 0-1	>100
Haddock 2+	2.63
Haddock 0-1	1.00
European plaice 2+	1.10
European plaice 0-1	1.67
Common sole	4.31
Flatfish	1.00
Monkfish	1.00
European hake	1.00
Gurnards and dragonets	1.00
Atlantic herring	1.30
European sprat	2.54
Other pelagic fish	1.00
Nephrops	>100
Shrimp	1.00
Cephalopods	1.00
Scallops	16.10
Infauna	1.00
Gelatinous zooplankton	1.00
Large zooplankton	3.63
Small zooplankton	1.77

Table 3.2. *Fitted predator/prey vulnerabilities applied to predator/prey pairs in the key run during the second stage of model fitting. These 14 pairs were the most sensitive to altered vulnerabilites.*

Functional group		Vulnerability
Predator	Prey	
Whiting 2+	Atlantic cod 0-1	>100
	European hake	1.00
	Other benthopelagic fish	>100
	Other pelagic fish	1.10
	European sprat	>100
Haddock 2+	Nephrops	2.64
Common sole	Epifauna	1.59
	Infauna	>100
Atlantic herring	Large zooplankton	1.45
European sprat	Small zooplankton	1.00
Large zooplankton	Small zooplankton	1.85
	Phytoplankton	>100
	Detritus	>100
Small zooplankton	Phytoplankton	2.66

3.7 Fitting diagnostics and performance

Plots of sum of squares residuals of model predictions to observation data are given in Figures 3.9 and 3.10 for all functional groups. The rank order of SS values contributing to the total SS (Figure 3.11) shows that the largest contributions generally come from key run catch simulations, particularly for 'Other pelagic fish', which is most likely due to the models use of fishing effort to drive catch simulations rather than using group specific fishing mortality which is often capable of capturing the impact of additional variables such as changes in catchability over time. Biomass and catch simulations for all functional groups are shown in Figures 3.12 and 3.13 with 95% confidence intervals derived from 1,000 Monte Carlo model permutaions varying Ecopath basic input parameters using data pedigree information.

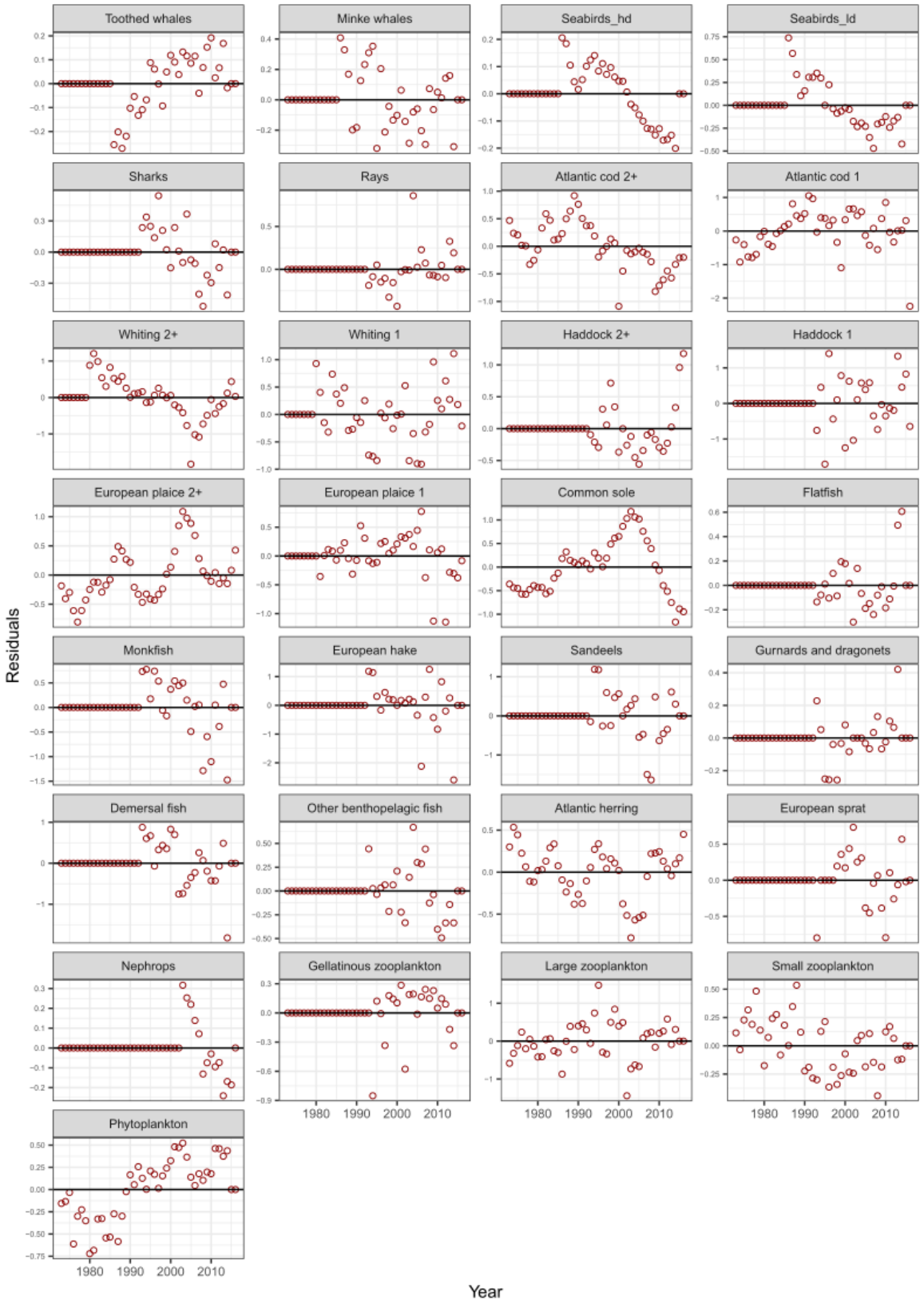


Figure 3.9. Residuals for relative biomass simulations vs observed data.

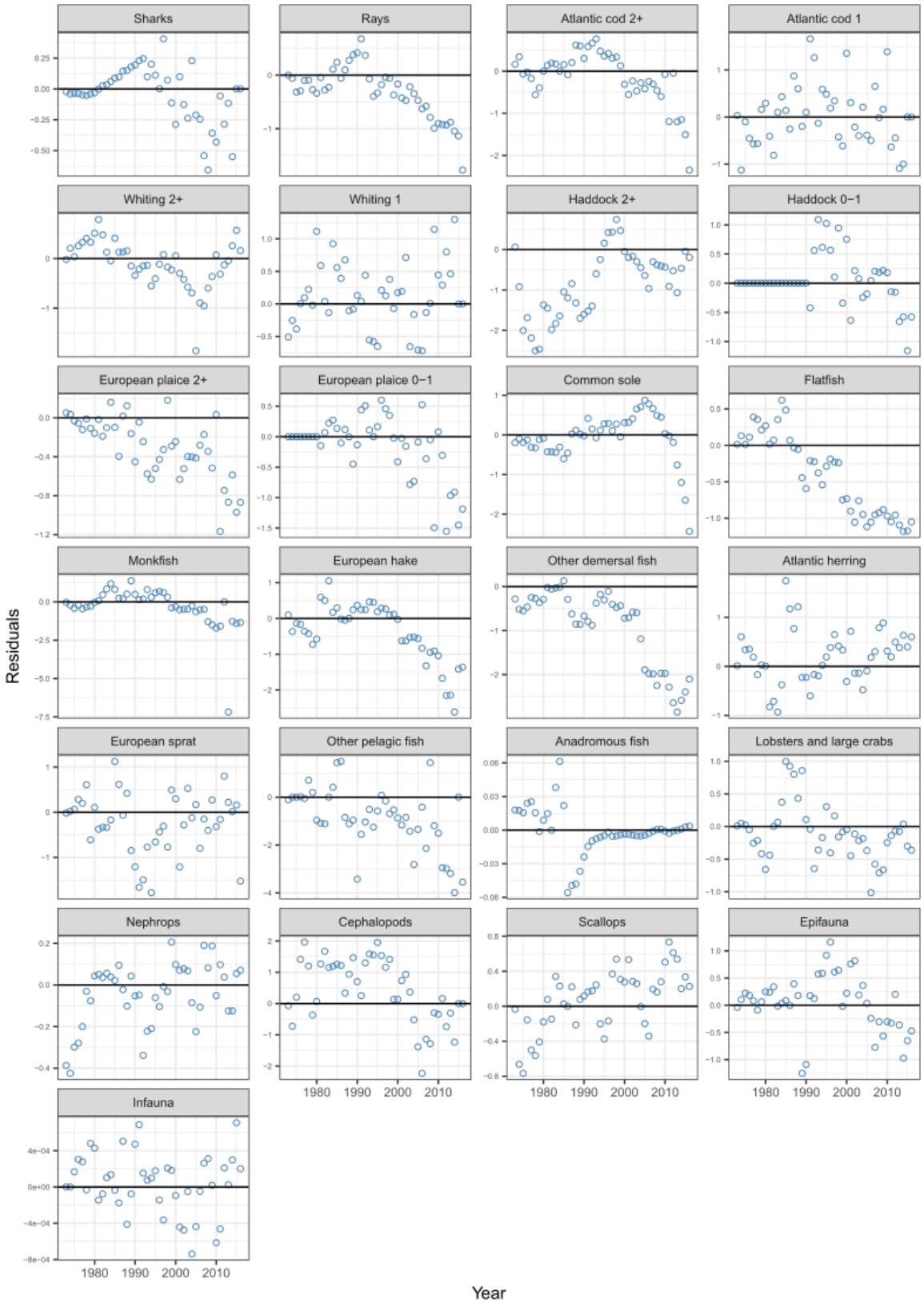


Figure 3.10. Residuals for catch simulations vs observed data.

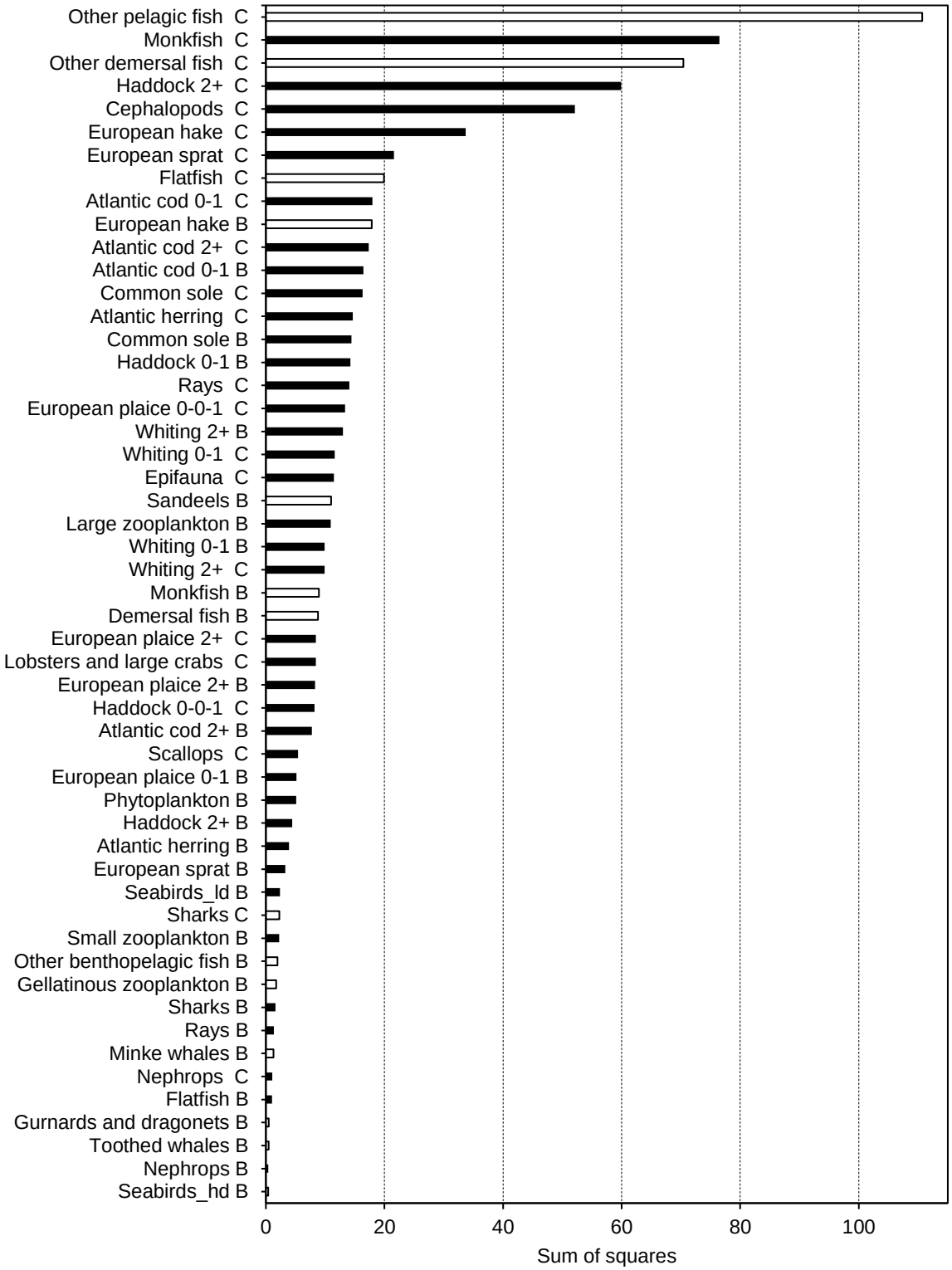


Figure 3.11. *Sum of squares contributions from all biomass (B) and catch (C) simulations in the fitted key run. Time series with low weightings (0.5) based on data reliability and relevance are hollow. All others use the default weighting of 1.*

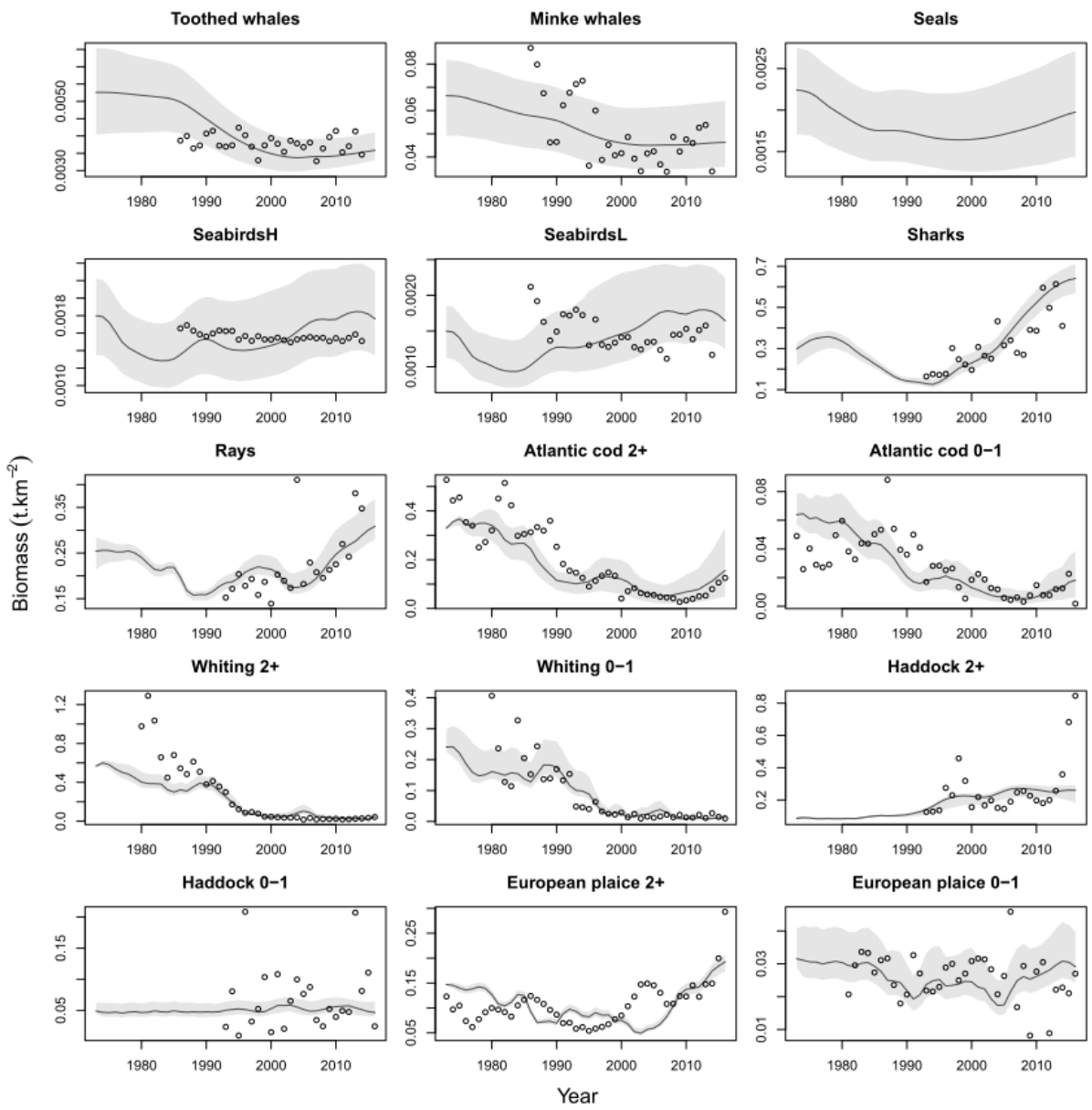


Figure 3.12. Ecosim predicted biomass trends (1/3) for functional groups in the Irish Sea Ecopath with Ecosim model (page 1 of 3). Black lines indicate model simulations against observed data (points). The shaded area indicates 95% confidence intervals based on Monte Carlo simulations varying Ecopath basic input parameters (B, PB, QB, diet).

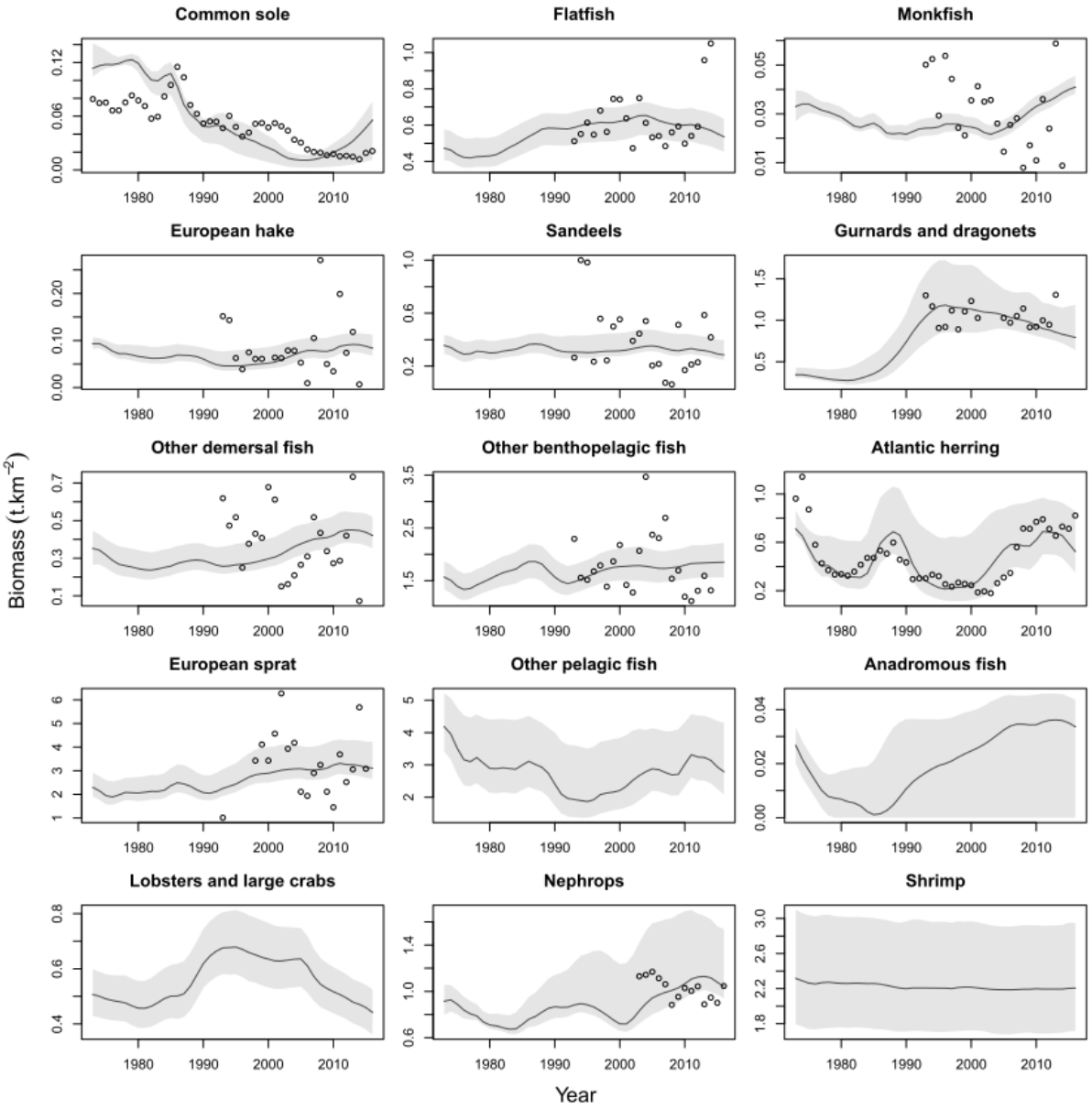


Figure 3.12 continued. *Ecosim predicted biomass trends for functional groups in the Irish Sea Ecopath with Ecosim model (page 2 of 3). Black lines indicate model simulations against observed data (points). The shaded area indicates 95% confidence intervals based on Monte Carlo simulations varying Ecopath basic input parameters (B, PB, QB, diet).*

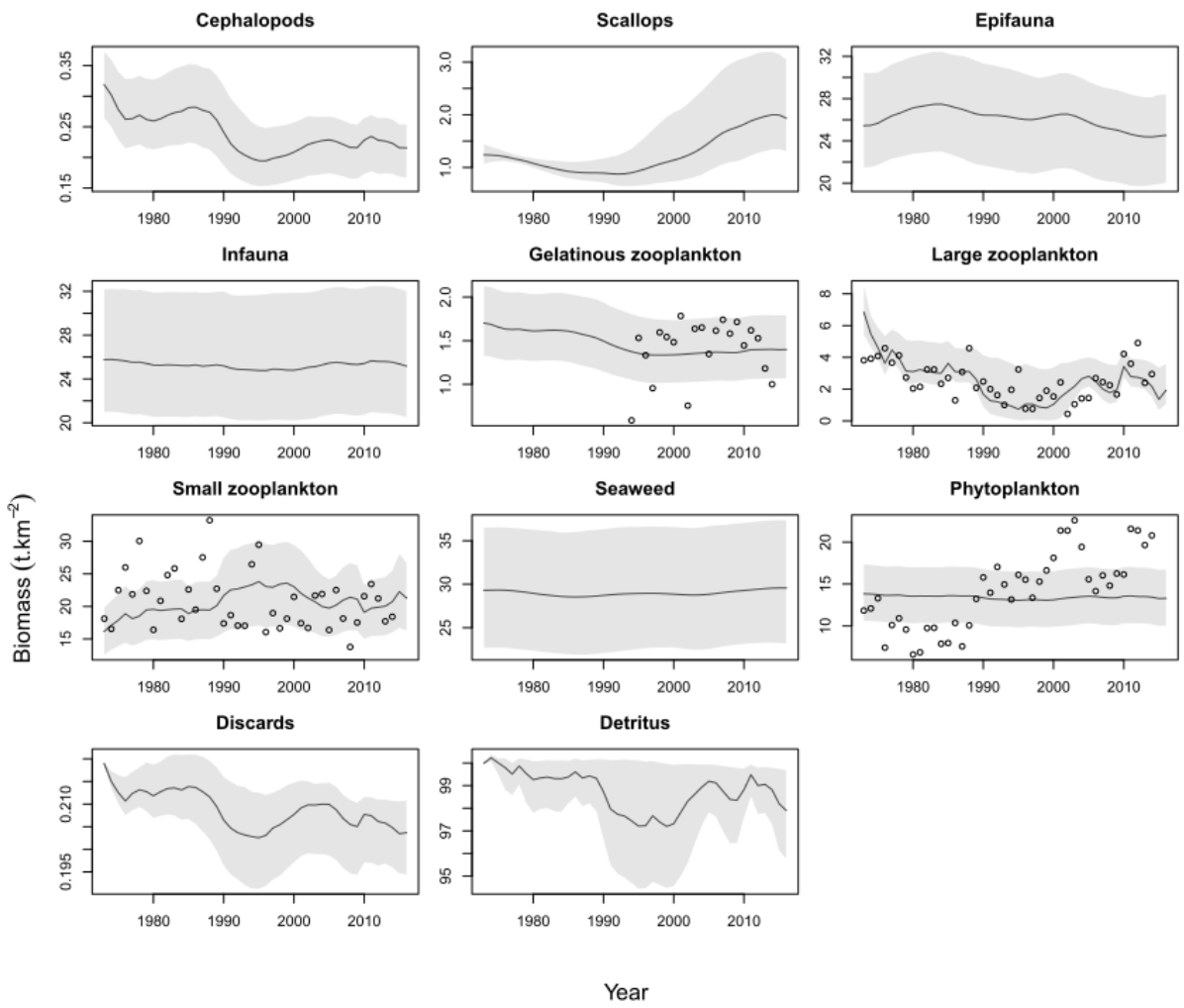


Figure 3.12 continued. Ecosim predicted biomass trends for functional groups in the Irish Sea Ecopath with Ecosim model (page 3 of 3). Black lines indicate model simulations against observed data (points). The shaded area indicates 95% confidence intervals based on Monte Carlo simulations varying Ecopath basic input parameters (B, PB, QB, diet).

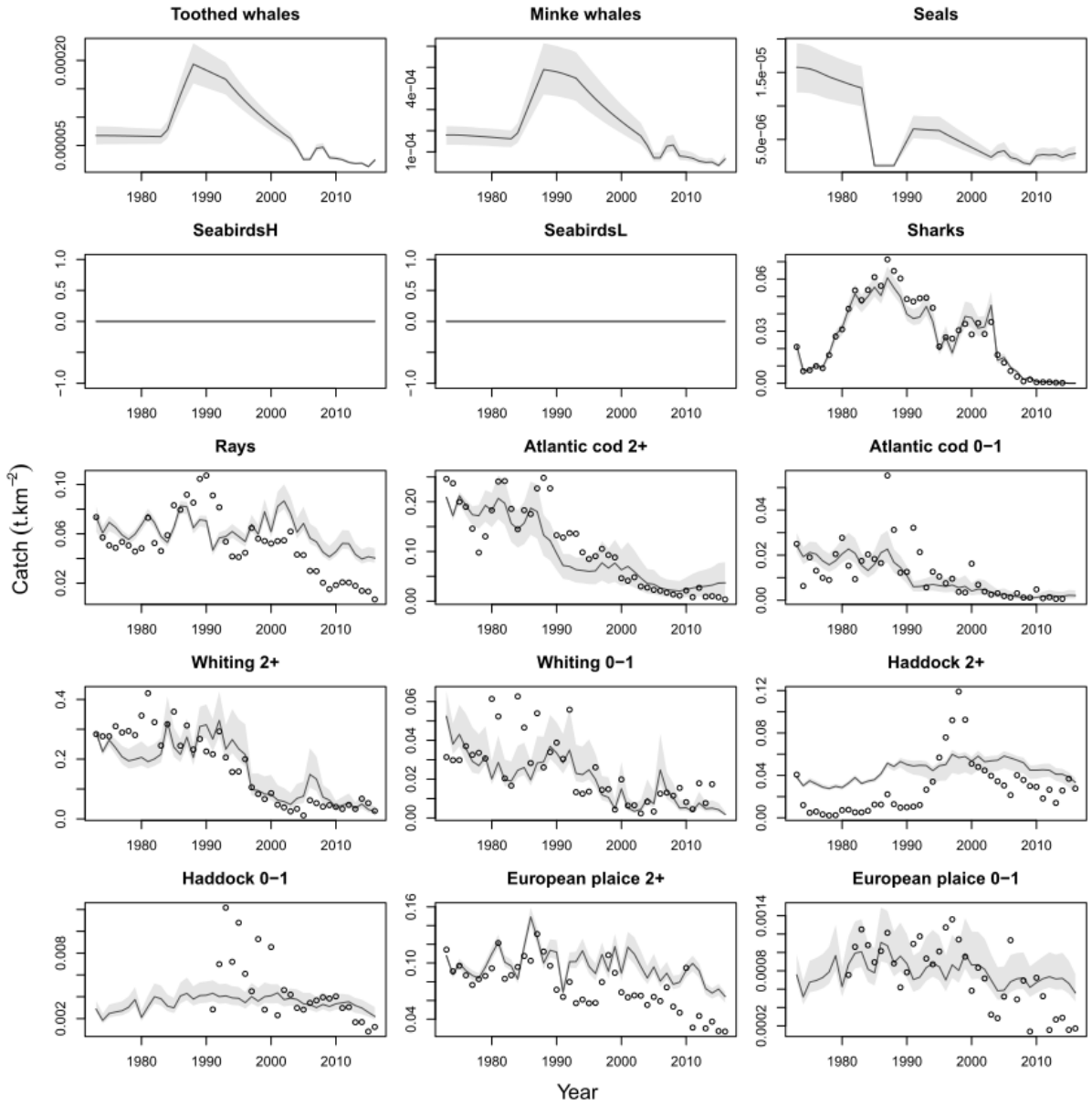


Figure 3.13. Ecosim predicted catch trends for functional groups in the Irish Sea Ecopath with Ecosim model (page 1 of 3). Black lines indicate model simulations against observed data (points). The shaded area indicates 95% confidence intervals based on Monte Carlo simulations varying Ecopath basic input parameters (B, PB, QB, diet).

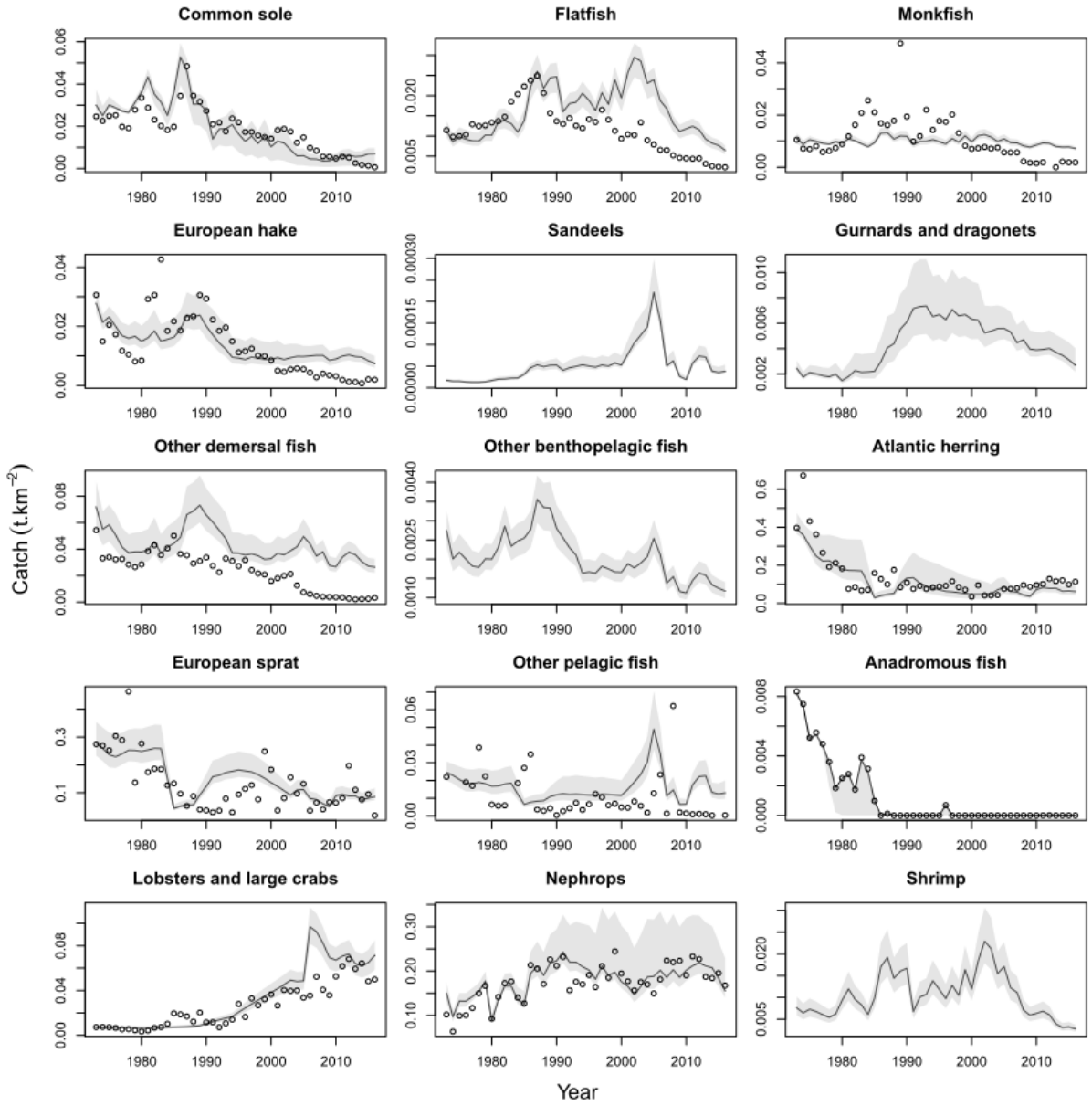


Figure 3.13 continued. Ecosim predicted catch trends for functional groups in the Irish Sea Ecopath with Ecosim model (page 2 of 3). Black lines indicate model simulations against observed data (points). The shaded area indicates 95% confidence intervals based on Monte Carlo simulations varying Ecopath basic input parameters (B, PB, QB, diet).

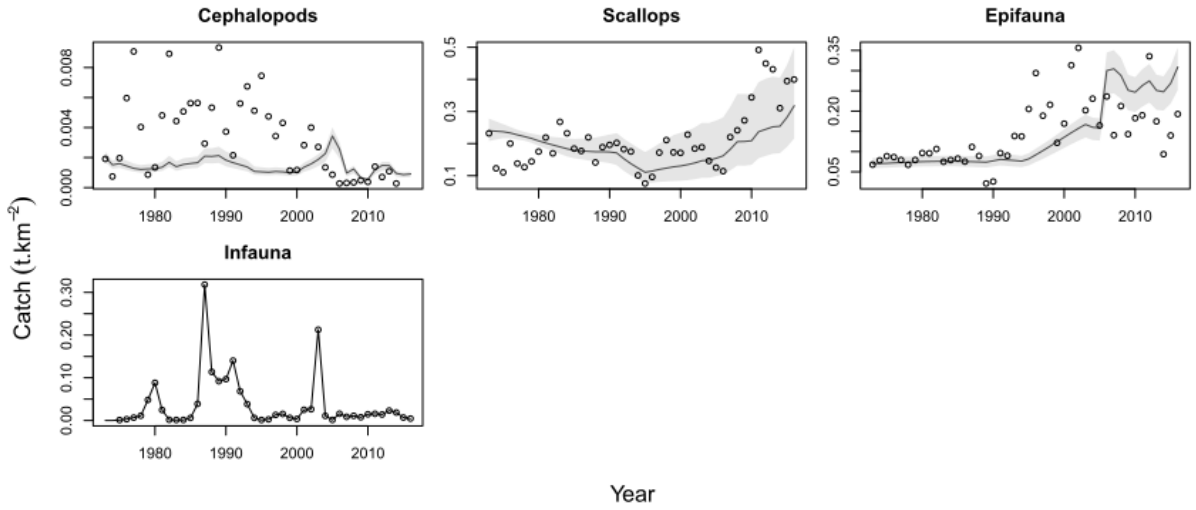


Figure 3.13 continued. *Ecosim predicted catch trends for functional groups in the Irish Sea Ecopath with Ecosim model (page 3 of 3). Black lines indicate model simulations against observed data (points). The shaded area indicates 95% confidence intervals based on Monte Carlo simulations varying Ecopath basic input parameters (B, PB, QB, diet).*

Chapter 4: Key Run Outputs

4.1 Model fits and mortality estimates

The fit of model biomass and catch predictions to observed data (relative biomass and catch) along with estimates of total, fishing, and predation mortality for selected key species of interest are shown in Figure 4.1.

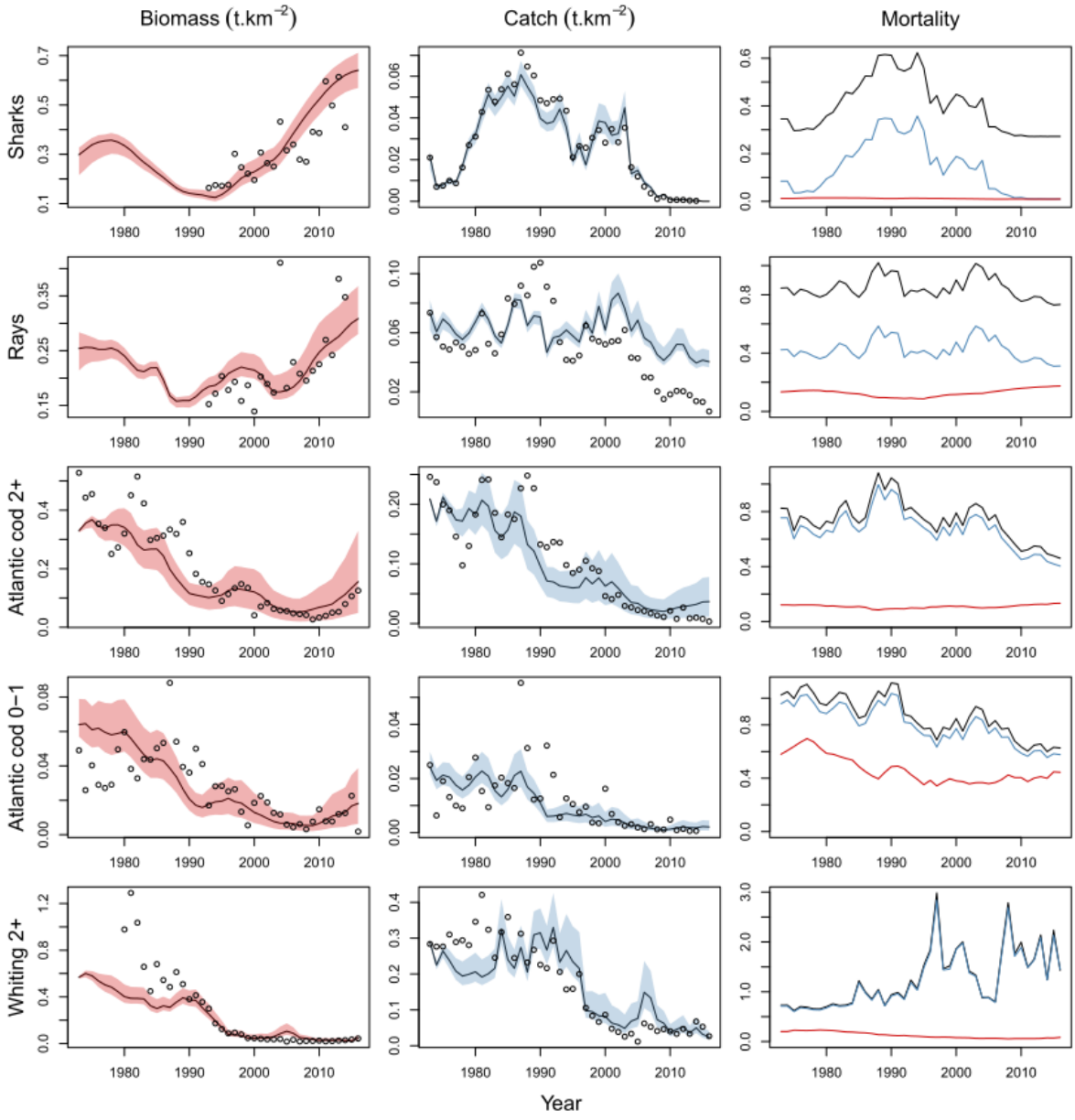


Figure 4.1. *Ecosim simulated biomasses, catches, and mortalities for key functional groups in the Irish Sea EwE key run (page 1 of 3). For biomass and catch simulations, solid lines represent key run simulations, dots represent observed data, and the shaded areas indicate 95% confidence intervals based on Monte Carlo simulations varying Ecopath input parameters using data pedigree confidence intervals. Mortalities are shown as total mortlaity (**black line**), fishing mortality (blue line) and predation mortality (red line).*

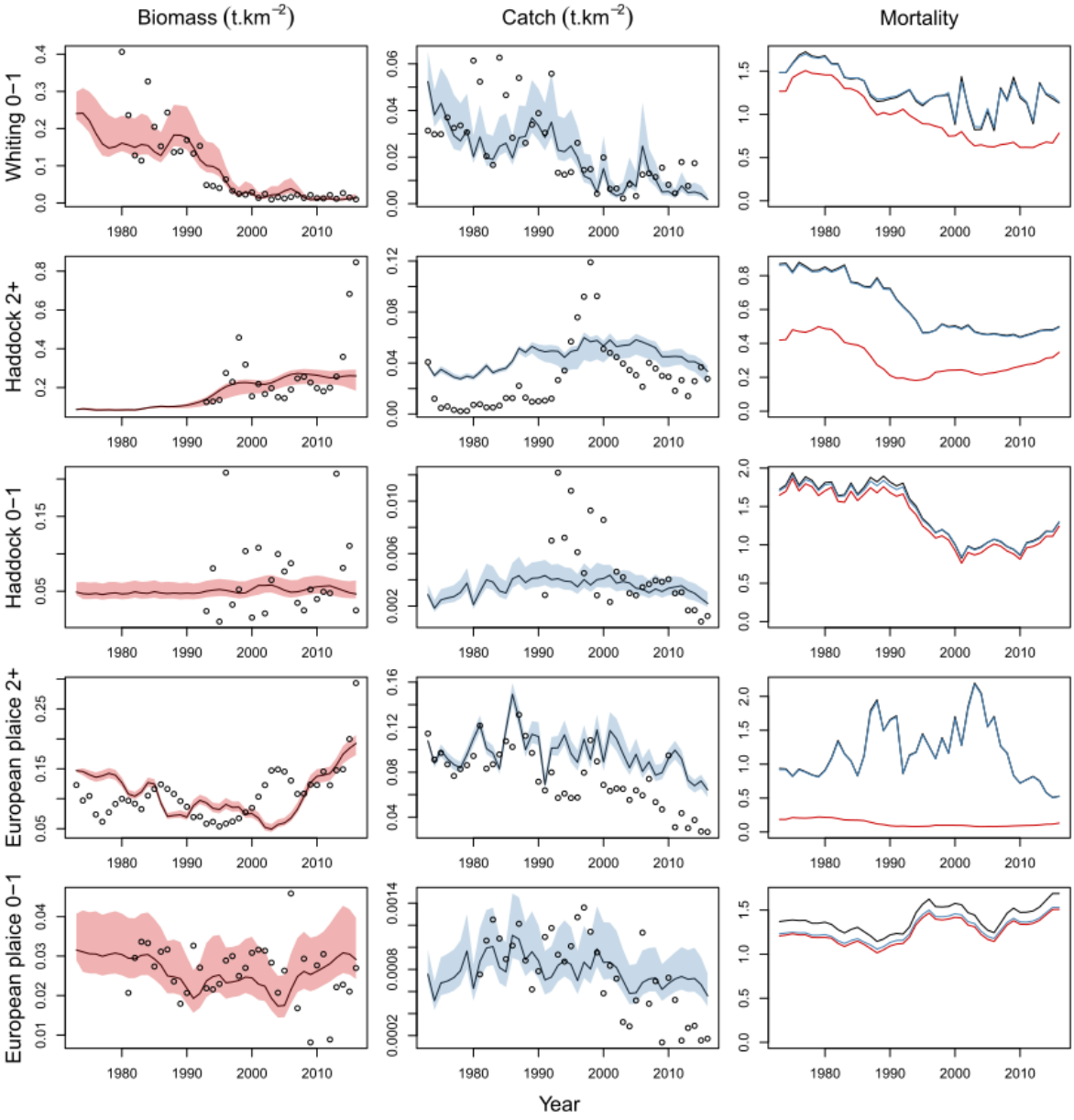


***Figure 4.1 continued**. Ecosim simulated biomasses, catches, and mortalities for key functional groups in the Irish Sea EwE key run (page 2 of 3). For biomass and catch simulations, solid lines represent key run simulations, dots represent observed data, and the shaded areas indicate 95% confidence intervals based on Monte Carlo simulations varying Ecopath input parameters using data pedigree confidence intervals. Mortalities are shown as total mortlaity (**black line**), fishing mortality (blue line) and predation mortality (red line).*

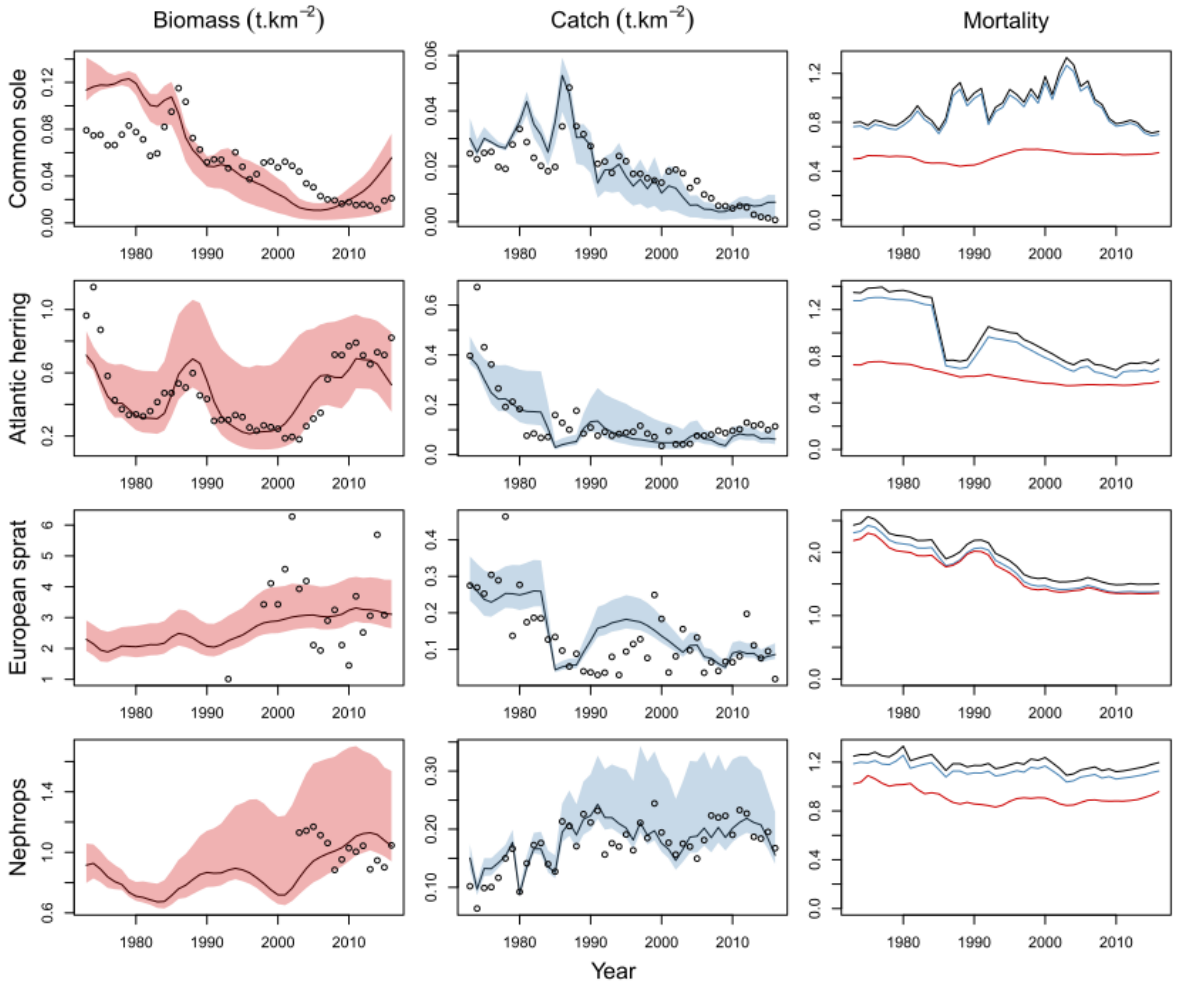


***Figure 4.1 continued**. Ecosim simulated biomasses, catches, and mortalities for key functional groups in the Irish Sea EwE key run (page 3 of 3). For biomass and catch simulations, solid lines represent key run simulations, dots represent observed data, and the shaded areas indicate 95% confidence intervals based on Monte Carlo simulations varying Ecopath input parameters using data pedigree confidence intervals. Mortalities are shown as total mortlaity (**black line**), fishing mortality (blue line) and predation mortality (red line).*

4.2 Estimates of Fmsy

The key run was used to estimate Fmsy ranges for each functional group using two approaches: "Full compensation" and "Stationary system". The full compensation assessment takes into account the indirect changes in biomass of other species caused through trophic linkages. During this assessment method, the F of the assessed species is incremented across all ranges of F over a long simulation period (100 years) until stock depletion whilst groups that are not being assessed are set to F=Fcurrent and allowed to interact. During the Stationary system assessment there are no interactions between species (the biomasses of the groups that are not assessed are kept constant through the simulations). Both methods are well explained by Walters et al. (2005). Key run Fmsy estimates for all species are shown

in Table 4.1, whilst MSY curves are plotted for key commercial species in Figure 4.2 and compared to ICES reference points in Figure 4.3.

Table 4.1. *Irish Sea EwE estimated Fmsy reference points compared with ICES single species reference points for VIIa stocks.*

Functional group	EwE-IS Key Run 2016: Full compensation Fmsy	EwE-IS Key Run 2016: Stationary system Fmsy	ICES				
			Fmsy	$Fmsy_{upper}$	$Fmsy_{lower}$	Flim	Fpa
Toothed whales	0.01	0.01	-	-	-	-	-
Minke whales	0.01	0.01	-	-	-	-	-
Seals	0.01	0.01	-	-	-	-	-
Rays	0.29	0.27	-	-	-	-	-
Atlantic cod 2+	0.24	0.24	0.44	0.46	0.34	0.81	0.58
Atlantic cod 0-1	0.36	0.30	-	-	-	-	-
Whiting 2+	0.46	0.38	0.22	0.29	0.16	0.37	0.22
Whiting 0-1	0.33	0.27	-	-	-	-	-
Haddock 2+	0.42	0.36	0.28	0.35	0.2	0.53	0.38
Haddock 0-1	0.00	0.00	-	-	-	-	-
European plaice 2+	0.28	0.28	0.2	0.29	0.13	0.48	0.35
European plaice 0-1	0.01	0.01	-	-	-	-	-
Common sole	0.22	0.20	0.2	0.24	0.16	0.29	0.21
Flatfish	0.06	0.06	-	-	-	-	-
Monkfish	0.18	0.18	-	-	-	-	-
European hake	0.25	0.24	-	-	-	-	-
Sandeels	0.00	0.00	-	-	-	-	-
Gurnards and dragonets	0.02	0.02	-	-	-	-	-
Other demersal fish	0.31	0.31	-	-	-	-	-
Other benthopelagic fish	0.00	0.00	-	-	-	-	-
Atlantic herring	0.26	0.23	0.27	0.35	0.2	0.4	0.29
European sprat	0.14	0.14	-	-	-	-	-
Other pelagic fish	0.02	0.02	-	-	-	-	-
Anadromous fish	0.00	0.00	-	-	-	-	-
Lobsters and large crabs	0.26	0.26	-	-	-	-	-
Nephrops	0.42	0.38	0.18	0.18	0.12		-
Shrimp	0.01	0.01	-	-	-	-	-
Cephalopods	0.02	0.02	-	-	-	-	-
Scallops	0.34	0.34	-	-	-	-	-
Epifauna	0.06	0.06	-	-	-	-	-
Infauna	0.00	0.00	-	-	-	-	-

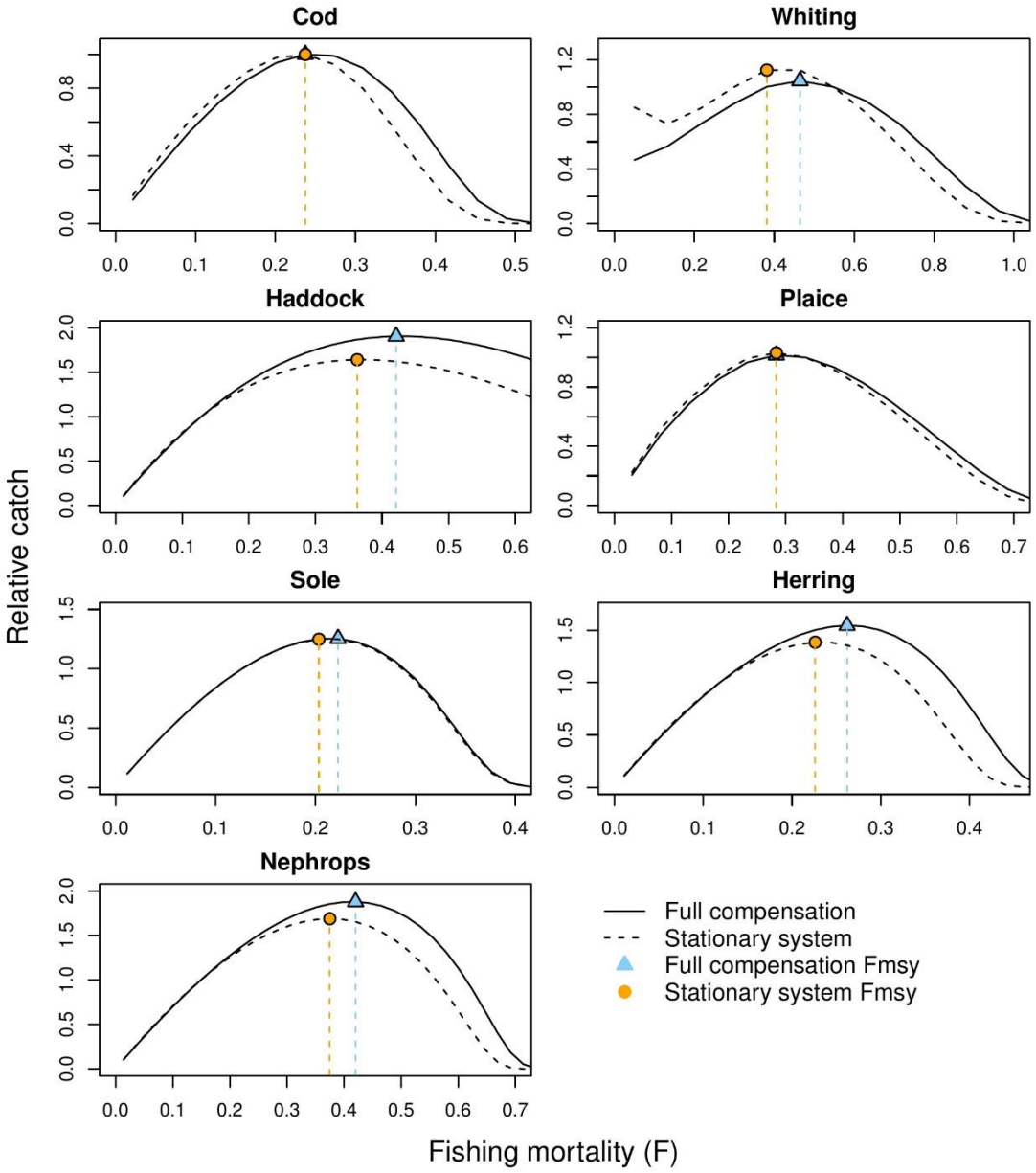


Figure 4.2. Relative catch as a function of F and Fmsy reference points for commercial important species as simulated under "stationary" and "full compensation" system dynamics.

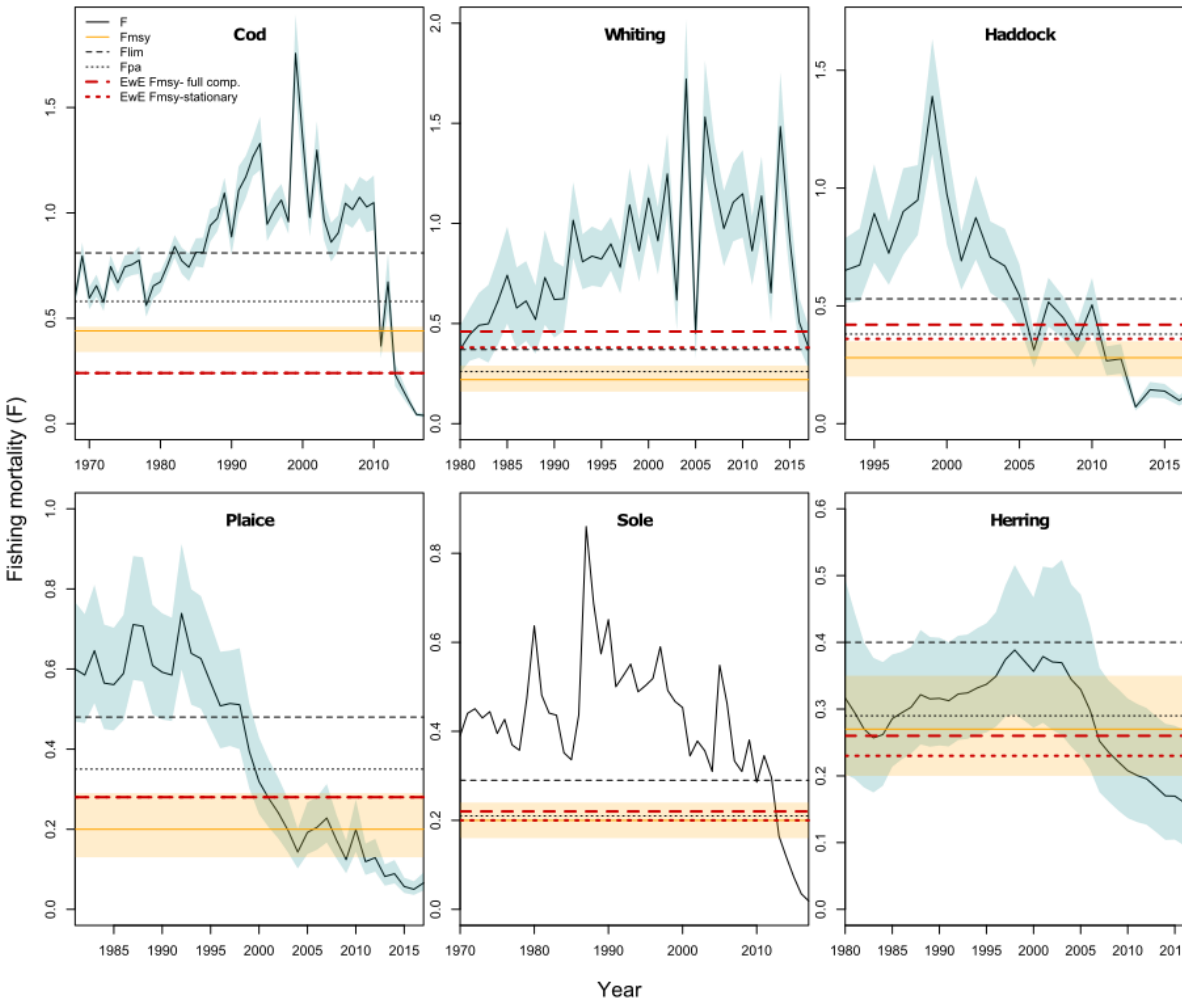


Figure 4.3. *Multi-species fishing mortality estimates from the Irish Sea EwE key run in comparison to ICES fishing mortality trends and reference points for assessed stocks. Blue shading: fishing mortality high and low limits; Orange shading: ICES Fmsy range ($Fmsy_{lower}$-$Fmsy_{upper}$).*

4.3 Ecosystem indicator trends

Changes in selected system and community indicators are shown in Figure 4.4. Referring to the figure, these include:

- Trends in total system biomass and catch and the biomass and catch of finfish, invertebrates, and mammals.
- Changes to the trophic level of the community and catch.
- Primary production and total system production. Due to the uncertainty in initial Ecopath inputs it is dificult to dicern a trend for production in Figure 4.4, therefore the anomaly of production (with uncertainty) is shown in Figure 4.5.
- Meaure of food web diversity (Shannon diversity index).

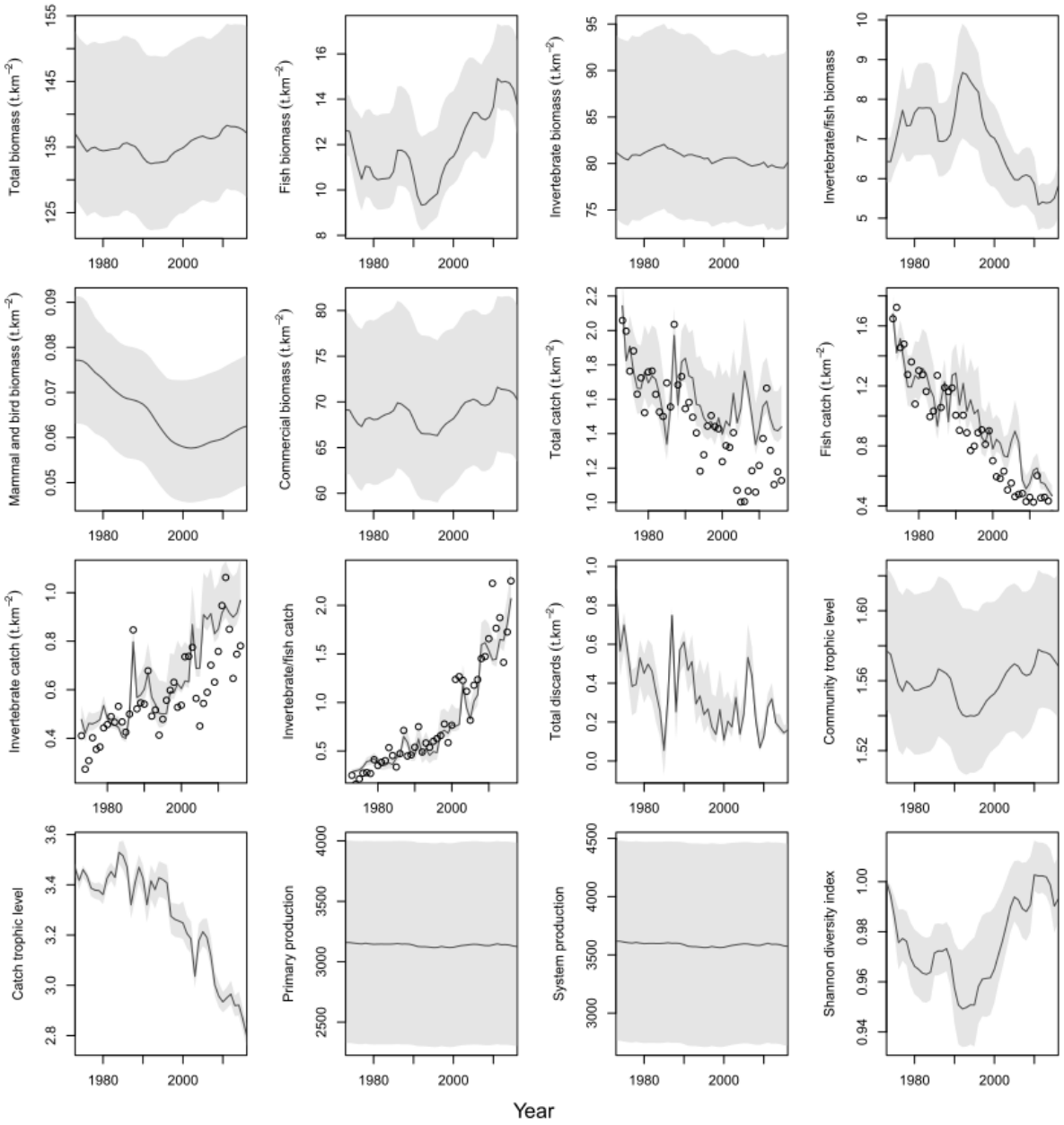


Figure 4.4. *Ecosystem indicators derived from the Irish Sea EwE key run. Shaded areas indicate 95% confidence intervals based on Monte Carlo simulations varying Ecopath input parameters using data pedigree confidence intervals. Model estimated indicators of catch are accompanied by observed data from ICES catch statistics (points).*

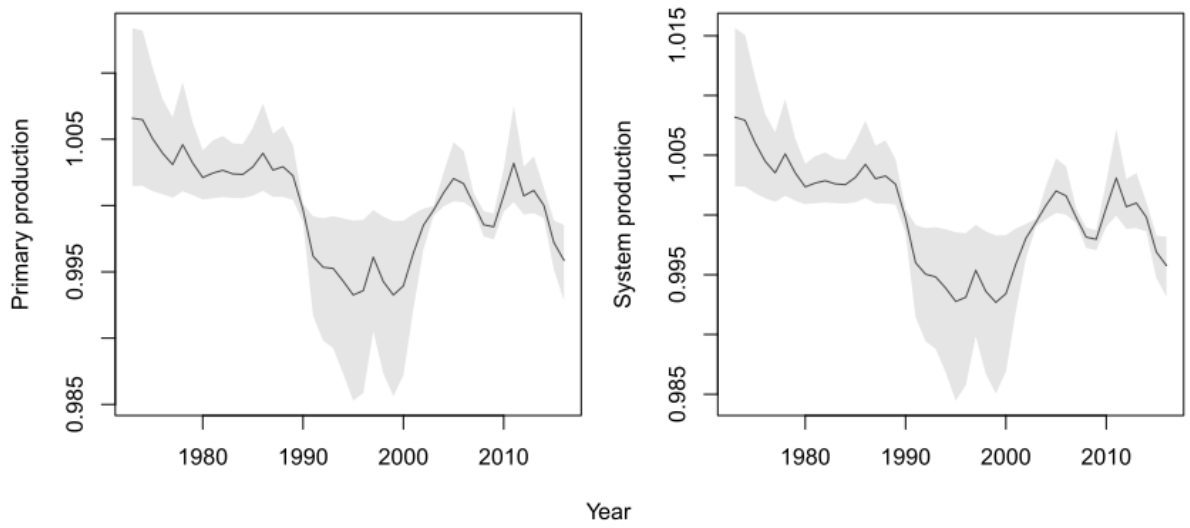


Figure 4.5. *Anomalies of primary production and total system production derived from the Irish Sea key run. Shaded areas indicate 95% confidence intervals based on Monte Carlo simulations varying Ecopath input parameters using data pedigree confidence intervals.*

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Annex

Annex 1.1

Ecopath input parameters, references and assumptions. All biomass units (**B**) are in units of $t.km^{-2}$ (area covered = 58,000 km^{-2}) whilst production/biomass (**PB**) and consumption/biomass (**QB**) are annual ratios ($year^{-1}$). Where necessary, assumptions were made regarding the ecotrophic efficiency (**EE**) or production/consumption (**PQ**; $year^{-1}$) parameters in order to estimate missing parameters (B, PB, and/or QB). We also provide references for diet composition (**DC**). A more detailed discussion of and justification of input parameters can be found in the Irish Sea Ecopath technical report (Bentley et al., 2018). Parameters which have been subsequently updated since the publication of the technical report have been denoted with '*'.

Functional group	Parameter	Value	Source	Comment
1: Toothed whales	B	0.00526	(Hammond et al., 2013, Bjorge and Tolley, 2009)	Estimate from the Celtic Sea population and knowledge from Dr Hernández-Milián.
	PB	0.02	(Trites et al., 1999, Mackinson and Daskalov, 2008)	Parameter reflects half of the maximum rate for annual whale population increase (4%)
	QB	15.52	(Trites et al., 1999)	Mean daily ration calculated as a function of individual weight.
	EE	0.642	Calculated by EwE	-
	PQ	0.001	Calculated by EwE	-
	DC	See Annex 1.2	(Santos et al., 2001, Hernandez-Milian et al., 2015, Rogan and Hernández-Milián, 2011)	The parameterisation of marine mammal diets were guided by marine mammal expert Dr Hernández-Milián.
2: Minke whales	B	0.0664	(Wall, 2013, Hammond et al., 2013)	Estimate from the Celtic Sea population and knowledge from Dr Hernández-Milián.
	PB	0.02	(Trites et al., 1999)	Parameter reflects half of the maximum rate for annual whale population increase (4%)
	QB	6.58	(Trites et al., 1999)	Mean daily ration calculated as a function of individual weight.
	EE	0.136	Calculated by EwE	-
	PQ	0.003	Calculated by EwE	-
	DC	See Annex 1.2	(Pierce et al., 2004, Ryan et al., 2013)	The parameterisation of marine mammal diets were guided by marine mammal expert Dr Hernández-Milián.
3: Seals	B	0.00225	(Bonner, 1981, Bonner, 1972, Summers et al., 1980, Lyons, 2004, Ó'Cadhla and Strong, 2007)	Biomass estimated with guidance from Dr Hernández-Milián.
	PB	0.06	(Trites et al., 1999, Small and DeMaster, 1995)	Parameter reflects half of the maximum rate for annual seal population increase (12%)
	QB	14.43	(Trites et al., 1999)	Mean daily ration calculated as a function of individual weight.
	EE	0.117	Calculated by EwE	-
	PQ	0.004	Calculated by EwE	-
	DC	See Annex 1.2	(Kierly et al., 2000, Philpott, 2001, Kavanagh et al., 2010, Gosch, 2017, Gosch et al., 2014)	The parameterisation of marine mammal diets were guided by marine mammal expert Dr Hernández-Milián.
4: Seabirds (high discard diets) *Includes species for which discards constitutes >10% of their diet.*	B	0.0018	(ICES, 2002)	Value based on report from the ICES Seabird Ecology Working Group. This estimate reflects population sizes in 2001 and was used in the absence of data for earlier years.
	PB	0.4	(Trites et al., 1999, Serpetti et al., 2017)	Based on the estimate of Trites et al. (1999) for the mortality rate of piscivorous birds as adopted by the

				West Coast of Scotland EwE model (Serpetti et al., 2017)
	QB	66.02	(Nilsson and Nilsson, 1976)	Calculated from daily ration and body weight (equation from Nilsson and Nilsson (1976)).
	EE	0	Calculated by EwE	Group neither consumed by other groups nor caught by fisheries in the model.
	PQ	0.006	Calculated by EwE	-
	DC	See Annex 1.2	See Table 5 in Bentley et al. (2018) for an extensive list of diet references.	-
5: Seabirds (low discard diets) *Includes species for which discards constitutes <10% of their diet.*	B	0.0015	(ICES, 2002)	Value based on report from the ICES Seabird Ecology Working Group. This estimate reflects population sizes in 2001 and was used in the absence of data for earlier years.
	PB	0.4	(Trites et al., 1999, Serpetti et al., 2017)	Based on the estimate of Trites et al. (1999) for the mortality rate of piscivorous birds as adopted by the West Coast of Scotland EwE model (Serpetti et al., 2017)
	QB	69.21	(Nilsson and Nilsson, 1976)	Calculated from daily ration and body weight (equation from Nilsson and Nilsson (1976)).
	EE	0	Calculated by EwE	Group neither consumed by other groups nor caught by fisheries in the model.
	PQ	0.006	Calculated by EwE	-
	DC	See Annex 1.2	See Table 5 in Bentley et al. (2018) for an extensive list of diet references.	-
6: Sharks	B	0.29	(ICES, 2017a, ICES, 2017b)	Estimate based on the BTS-VIIa and NIGFS surveys for the Irish Sea (VIIa).
	PB	0.4	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	4.01	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.344	Calculated by EwE	-
	PQ	0.1	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014, ICES, 2018e, Bentley et al., In press)	Diet designed using CEFAS fish stomach records (DAPSTOM) and fishers' knowledge.
7: Rays	B	0.254	Calculated by EwE	Survey estimates (1993-2016) were too low to balance production with mortality (*F* and *M*).
	PB	0.845	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	3.28	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.5	-	Assumption applied to estimate biomass and provide realistic responses to changes in *F*.
	PQ	0.258	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014, ICES, 2018e, Bentley et al., In press)	Diet designed using CEFAS fish stomach records (DAPSTOM) and fishers' knowledge.
8: Adult cod (2+) *Multi-stanza group; linked to Juvenile cod (0-1)*	B	0.339	(ICES, 2017c, ICES, 2018c)	Estimated Age 2+ biomass for the year 1973 from ASAP model output for the Irish Sea (VIIa).
	PB	0.82	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*;

				catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	3.12	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.92	Calculated by EwE	-
	PQ	0.263	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM) from cod with length greater than 57.8 cm (*Lmat*).
9: Juvenile cod (0-1) *Multi-stanza group; linked to Adult cod (2+)*	B	0.0662	Calculated by EwE	Estimate based on adult cod parameters (leading group)
	PB	1.01	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	7.641	Calculated by EwE	Estimate based on adult cod parameters (leading group)
	EE	0.92	Calculated by EwE	-
	PQ	0.263	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM) from cod with length less than 57.8 cm (*Lmat*).
10: Adult whiting (2+) *Multi-stanza group; linked to Juvenile whiting (0-1)*	B	0.56	(ICES, 2017c, ICES, 2018c)	Estimated Age 2+ biomass for the year 1980 from ASAP model output for the Irish Sea (VIIa).
	PB	0.762	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	5.39	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.948	Calculated by EwE	-
	PQ	0.141	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014, ICES, 2018e, Bentley et al., In press)	Diet designed using CEFAS fish stomach records (DAPSTOM) from whiting with length greater than 25.1 cm (*Lmat*) and fishers' knowledge.
11: Juvenile whiting (0-1) *Multi-stanza group; linked to Adult whiting (2+)*	B	0.236	Calculated by EwE	Estimate based on adult whiting parameters (leading group)
	PB	1.524	(Lees and Mackinson, 2007, Mackinson and Daskalov, 2008, Serpetti et al., 2017)	Assumed to be 2 times adult PB
	QB	11.88	Calculated by EwE	Estimate based on adult whiting parameters (leading group)
	EE	0.994	Calculated by EwE	-
	PQ	0.128	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM) from whiting with length less than 25.1 cm (*Lmat*).
12: Adult haddock (2+) *Multi-stanza group; linked to Juvenile haddock (0-1)*	B	0.086	(ICES, 2017c, ICES, 2018c)	Assessment data prior to 1993 is unavailable, therefore an initial biomass was estimated as the average biomass from the period 1993-2000 from ASAP model output for the Irish Sea (VIIa).
	PB	0.89	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	4.42	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.962	Calculated by EwE	-
	PQ	0.201	Calculated by EwE	-

	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM) from haddock with length greater than 37.1 cm (*Lmat*).
13: Juvenile haddock (0-1) *Multi-stanza group; linked to Adult haddock (2+)*	B	0.0484	Calculated by EwE	Estimate based on adult whiting parameters (leading group)
	PB	1.789	(Lees and Mackinson, 2007, Mackinson and Daskalov, 2008, Serpetti et al., 2017)	Assumed to be 2 times adult PB
	QB	10.28	Calculated by EwE	Estimate based on adult whiting parameters (leading group)
	EE	0.999	Calculated by EwE	-
	PQ	0.128	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM) from haddock with length less than 37.1 cm (*Lmat*).
14: Adult plaice (2+) *Multi-stanza group; linked to Juvenile plaice (0-1)*	B	0.155	(ICES, 2017c, ICES, 2018c)	Estimated Age 2+ biomass for the year 1973 from SAM model output for the Irish Sea (VIIa).
	PB	0.79	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	3.75	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.997	Calculated by EwE	-
	PQ	0.211	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014, ICES, 2018e, Bentley et al., In press)	Diet designed using CEFAS fish stomach records (DAPSTOM) from plaice with length greater than 36.1 cm (*Lmat*) and fishers' knowledge.
15: Juvenile haddock (0-1) *Multi-stanza group; linked to Adult haddock (2+)*	B	0.0315	Calculated by EwE	Estimate based on adult plaice parameters (leading group)
	PB	1.38	(Lees and Mackinson, 2007, Mackinson and Daskalov, 2008, Serpetti et al., 2017)	Assumed to be 2 times adult PB
	QB	10.3	Calculated by EwE	Estimate based on adult plaice parameters (leading group)
	EE	0.934	Calculated by EwE	-
	PQ	0.134	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014, ICES, 2018e, Bentley et al., In press)	Diet designed using CEFAS fish stomach records (DAPSTOM) from plaice with length less than 36.1 cm (*Lmat*) and fishers' knowledge.
16: Common sole	B	0.113	(ICES, 2018c)	Estimated biomass for the year 1973 from XSA model output for the Irish Sea (VIIa).
	PB	0.81	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	5.61	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.958	Calculated by EwE	-
	PQ	0.144	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
17: Flatfish	B	0.47	(ICES, 2017a, ICES, 2017b)	Estimate based on the BTS-VIIa and NIGFS surveys for the Irish Sea (VIIa).
	PB	1.518	Calculated by EwE	
	QB	6.07	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.956	Calculated by EwE	-
	PQ	0.25	-	We estimate a PQ of 0.25 to facilitate the calculation of a PB parameter
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).

18: Monkfish	B	0.033	(ICES, 2017a, ICES, 2017b)	Estimate based on the BTS-VIIa and NIGFS surveys for the Irish Sea (VIIa).
	PB	0.49	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	2.746	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.687	Calculated by EwE	-
	PQ	0.178	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014, ICES, 2018e, Bentley et al., In press)	Diet designed using CEFAS fish stomach records (DAPSTOM) and fishers' knowledge.
19: European hake	B	0.093	Calculated by EwE	Biomass from trawl surveys suggest hake may be under-represented (potentially due to catchability).
	PB	0.981	Calculated by EwE	-
	QB	3.924	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.25	-	We estimate a PQ of 0.25 to facilitate the calculation of a PB parameter
	DC	See Annex 1.2	(Pinnegar, 2014, ICES, 2018e, Bentley et al., In press)	Diet designed using CEFAS fish stomach records (DAPSTOM) and fishers' knowledge.
20: Sandeels	B*	0.354	Calculated by EwE	Biomass from trawl surveys suggest sandeels may be under-represented (potentially due to catchability).
	PB	2.053	Calculated by EwE	-
	QB	8.21	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.25	-	We estimate a PQ of 0.25 to facilitate the calculation of a PB parameter
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
21: Gurnards and dragonets	B	0.34	(ICES, 2017a, ICES, 2017b)	Estimate based on the BTS-VIIa and NIGFS surveys for the Irish Sea (VIIa).
	PB	1.403	Calculated by EwE	When compared to other models through PREBAL analysis, the initial PB was very low. We therefore estimate PQ in order to balance the group and achieve a PB more consistent with other models (West Coast of Scotland, North Sea).
	QB	5.61	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.976	Calculated by EwE	
	PQ	0.25	-	We estimate a PQ of 0.25 to facilitate the calculation of a PB parameter
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
22: Other demersal fish	B	0.359	Calculated by EwE	Diet structures placed too much predation pressure on the survey biomass estimate, therefor a biomass was estimated for the 'other demersal fish' group.
	PB	0.96	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).

	QB	4.32	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.22	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
23: Other benthopelagic fish	B	1.435	Calculated by EwE	-
	PB	1.845	Calculated by EwE	-
	QB	7.38	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.25	-	We estimate a PQ of 0.25 to facilitate the calculation of a PB parameter
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
24: Atlantic herring	B*	0.72	(Molloy, 2006, ICES, 2016a)	TSB for 1973 from 2016 ICES age based analytical assessment (ICES, 2016a). Subsequent assessments only hindcast to 1980.
	PB*	1.35	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	6.59	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.95	Calculated by EwE	-
	PQ	0.205	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
25: European sprat	B	2.294	Calculated by EwE	-
	PB	2.425	Calculated by EwE	-
	QB	9.7	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.25	-	We estimate a PQ of 0.25 to facilitate the calculation of a PB parameter
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
26: Other pelagic fish	B	4.147	Calculated by EwE	-
	PB	1.278	Calculated by EwE	-
	QB	5.11	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.25	-	We estimate a PQ of 0.25 to facilitate the calculation of a PB parameter
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
27: Anadromous fish	B	0.03	(Lees and Mackinson, 2007)	Estimate from Lees and Mackinson (2007) Irish Sea model which derived its biomass estimate from expert advice.
	PB	0.644	(Allen, 1971, Pauly et al., 1990)	Equivalent to the instantaneous rate of total mortality (Allen, 1971) which is calculated as fishing mortality (*F*; catch/biomass) plus natural mortality (*M*; Pauly et al. (1990)).
	QB	3.17	(Pauly et al., 1990, Christensen and Pauly, 1992)	Calculated using an empirical model which incorporates water temperature and feeding characteristics.

	EE	0.336	Calculated by EwE	-
	PQ	0.203	Calculated by EwE	-
	DC	See Annex 1.2	(Pinnegar, 2014)	Diet designed using CEFAS fish stomach records (DAPSTOM).
28: Lobsters and large crabs	B	0.5	Calculated by EwE	-
	PB	0.62	(Tumbiolo and Downing, 1994)	Empirical model for the production rate of marine invertebrates
	QB	4.133	Calculated by EwE	-
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.15	(Christensen, 1995)	We estimate a PQ of 0.15 to facilitate the calculation of a QB parameter
	DC	See Annex 1.2	(Barker and Gibson, 1977, Bernárdez et al., 2000, Brey, 2001)	-
29: *Nephrops*	B	0.916	Calculated by EwE	-
	PB	1.27	(Tumbiolo and Downing, 1994)	Empirical model for the production rate of marine invertebrates
	QB	8.467	Calculated by EwE	-
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.15	(Christensen, 1995)	We estimate a PQ of 0.15 to facilitate the calculation of a QB parameter
	DC	See Annex 1.2	(Cristo and Cartes, 1998)	-
30: Shrimp	B	2.249	Calculated by EwE	-
	PB	2.67	(Tumbiolo and Downing, 1994)	Empirical model for the production rate of marine invertebrates
	QB	17.8	Calculated by EwE	-
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.15	(Christensen, 1995)	We estimate a PQ of 0.15 to facilitate the calculation of a QB parameter
	DC	See Annex 1.2	(Simpson et al., 1970, Oh et al., 2001)	-
31: Cephalopods	B	0.315	Calculated by EwE	-
	PB	1.981	(Tumbiolo and Downing, 1994)	Empirical model for the production rate of marine invertebrates
	QB	15	(Lees and Mackinson, 2007, Mackinson and Daskalov, 2008, Serpetti et al., 2017)	Little information was available regarding cephalopod consumption rates in the Irish Sea, therefore an estimate was taken from other local models.
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.132	Calculated by EwE	-
	DC	See Annex 1.2	(Pierce et al., 1994, Collins and Pierce, 1996)	-
32: Scallops	B	1.214	Calculated by EwE	-
	PB	1.146	(Tumbiolo and Downing, 1994)	Empirical model for the production rate of marine invertebrates
	QB	7.643	Calculated by EwE	-
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.15	(Christensen, 1995)	We estimate a PQ of 0.15 to facilitate the calculation of a QB parameter
	DC	See Annex 1.2	(Serpetti et al., 2017)	-
33: Epifauna	B*	24.648	Calculated by EwE	-
	PB*	1.7	(Tumbiolo and Downing, 1994)	Empirical model for the production rate of marine invertebrates
	QB*	4.028	Calculated by EwE	-
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.25	(Mackinson and Daskalov, 2008, Serpetti et al., 2017)	We estimate a PQ of 0.25 to facilitate the calculation of a QB parameter
	DC	See Annex 1.2	(Serpetti et al., 2017)	-
34: Infauna	B*	25.005	Calculated by EwE	-

	PB*	1.7	(Tumbiolo and Downing, 1994)	Empirical model for the production rate of marine invertebrates
	QB*	6.8	Calculated by EwE	-
	EE	0.95	-	We estimate an EE of 0.95 to facilitate the calculation of a B parameter
	PQ	0.25	(Mackinson and Daskalov, 2008, Serpetti et al., 2017)	We estimate a PQ of 0.25 to facilitate the calculation of a QB parameter
	DC	See Annex 1.2	(Serpetti et al., 2017)	-
35: Gelatinous zooplankton	B	1.7	(Stephen Beggs, per.comms)	Estimate based on AFBI jellyfish surveys between May and June. Value taken from earliest survey date (1994).
	PB	0.762	Calculated by EwE	-
	QB	2.54	(Martinussen and Båmstedt, 1995, Brey, 2001)	Calculated using carbon food rations (Martinussen and Båmstedt, 1995) converted into wet weight (Brey, 2001).
	EE	0.938	Calculated by EwE	-
	PQ	0.3	(Lees and Mackinson, 2007, Mackinson and Daskalov, 2008)	We estimate a PQ of 0.3 to facilitate the calculation of a PB parameter
	DC	See Annex 1.2	(Martinussen and Båmstedt, 1995)	-
36: Large zooplankton	B*	6.81	(Pitois and Fox, 2006, Richardson et al., 2006)	Plankton abundances from the SAHFOS continuous plankton recorder (Richardson et al., 2006) were converted into biomass (Pitois and Fox, 2006).
	PB	10	(Lees and Mackinson, 2007, Serpetti et al., 2017)	-
	QB	33.3	Calculated by EwE	-
	EE	0.682	Calculated by EwE	-
	PQ	0.3	(Lees and Mackinson, 2007, Mackinson and Daskalov, 2008)	We estimate a PQ of 0.3 to facilitate the calculation of a QB parameter
	DC	See Annex 1.2	(Lees and Mackinson, 2007, Serpetti et al., 2017)	-
37: Small zooplankton	B*	16.18	(Pitois and Fox, 2006, Richardson et al., 2006)	Plankton abundances from the SAHFOS continuous plankton recorder (Richardson et al., 2006) were converted into biomass (Pitois and Fox, 2006).
	PB	18	(Serpetti et al., 2017)	-
	QB	60	Calculated by EwE	-
	EE	0.521	Calculated by EwE	-
	PQ	0.3	(Lees and Mackinson, 2007, Mackinson and Daskalov, 2008)	We estimate a PQ of 0.3 to facilitate the calculation of a QB parameter
	DC	See Annex 1.2	(Lees and Mackinson, 2007, Serpetti et al., 2017)	-
38: Seaweed	B	29.3	(Burrows et al., 2018)	Estimated biomass of seaweed in the Irish Sea extracted from a UK wide model of seaweed biomass.
	PB	1	(Mike Burrows, per.comms)	-
	EE	0.393	Calculated by EwE	-
39: Phytoplankton	B	13.83	(Gowen et al., 2000)	Gowen et al. (2000) provide a production estimate of 97 $g.C.m^{-2}.yr$ for the Irish Sea,
	PB*	226.3	(Heymans, 2001)	The conversion ratio for C to wet weight was taken from Heymans (2001), bringing the PB close to those in the North Sea and Baltic Sea models.
	EE	0.327	Calculated by EwE	-
40: Discards	B*	0.219	(STECF, 2018, ICES, 2019)	Total discards were estimated using discard data from the STECF database and ICES stock assessments.
	EE	0.127	Calculated by EwE	-
41: Detritus	B	100	(Lees and Mackinson, 2007, Mackinson and Daskalov, 2008, Serpetti et al., 2017)	Includes dissolved and particulate organic matter.
	EE	0.115	Calculated by EwE	-

Annex 1.2

Ecopath diet matrix showing the proportions of each prey in the diets of predators.

	Predator/Prey	1	2	3	4	5	6	7	8	9	10
1	Toothed whales										
2	Minke whales										
3	Seals										
4	Seabirds (high discard diet)										
5	Seabirds (low discard diet)										
6	Sharks	0.040		0.009							
7	Rays			0.045			0.020	0.010			
8	Atlantic cod 2+	0.005		0.040		0.015	0.001	0.040			
9	Atlantic cod 0-1	0.020		0.009	0.012		0.005	0.010			0.005
10	Whiting 2+	0.094		0.031			0.004		0.060	0.005	0.010
11	Whiting 0-1	0.160			0.037		0.008	0.020	0.110	0.015	0.020
12	Haddock 2+						0.012		0.020		
13	Haddock 0-1						0.008	0.005	0.035		0.010
14	European plaice 2+	0.001		0.002			0.005		0.002		
15	European plaice 0-1					0.010	0.005		0.004		
16	Common sole	0.001		0.020			0.008	0.004	0.010	0.017	
17	Flatfish	0.100		0.057	0.037	0.010	0.151	0.050	0.104	0.150	0.042
18	Monkfish	0.002		0.001							
19	European hake	0.060							0.004		0.016
20	Sandeels	0.050	0.005	0.060	0.091	0.348	0.002	0.076		0.105	0.005
21	Gurnards and dragonets			0.023	0.036	0.075	0.070	0.003	0.070	0.048	
22	Other demersal fish	0.143		0.195	0.083	0.021	0.007		0.074	0.001	
23	Other benthopelagic fish	0.200		0.195	0.040	0.013	0.185	0.003	0.069	0.072	0.201
24	Atlantic herring	0.050	0.195	0.045	0.087	0.128	0.011		0.050	0.050	0.020
25	European sprat	0.050	0.195	0.045	0.087	0.128	0.002		0.002	0.002	0.280
26	Other pelagic fish	0.001	0.400	0.075	0.118	0.097	0.196	0.005	0.037	0.033	0.302
27	Anadromous fish	0.001		0.090		0.018					
28	Lobsters and large crabs			0.008		0.014	0.014	0.013	0.036	0.109	
29	Nephrops			0.016			0.075	0.116	0.184	0.101	0.007
30	Shrimp		0.100				0.002	0.020	0.003	0.006	0.002
31	Cephalopods	0.024	0.090	0.035	0.002		0.037	0.007	0.043	0.077	0.015
32	Scallops						0.007	0.050		0.005	
33	Epifauna				0.107	0.001	0.159	0.423	0.028	0.113	0.041
34	Infauna				0.050	0.084	0.005	0.029	0.001	0.003	0.001
35	Gelatinous zooplankton						0.001		0.054		0.001
36	Large zooplankton		0.009		0.013	0.018	0.001	0.002	0.001	0.088	0.002
37	Small zooplankton		0.006				0.001	0.003	0.001	0.001	0.001
38	Seaweed						0.001				
39	Phytoplankton										
40	Discards				0.201	0.020		0.112			0.020
41	Detritus										

	Predator/Prey	11	12	13	14	15	16	17	18	19	20
1	Toothed whales										
2	Minke whales										
3	Seals										
4	Seabirds (high discard diet)										
5	Seabirds (low discard diet)										
6	Sharks										
7	Rays								0.010		
8	Atlantic cod 2+								0.035		
9	Atlantic cod 0-1								0.020	0.010	
10	Whiting 2+							0.001	0.010		
11	Whiting 0-1							0.004	0.118	0.008	
12	Haddock 2+								0.005		
13	Haddock 0-1								0.005	0.009	
14	European plaice 2+								0.005		
15	European plaice 0-1							0.005	0.065		
16	Common sole							0.004		0.012	
17	Flatfish	0.006	0.046					0.018	0.175		
18	Monkfish										
19	European hake								0.011		
20	Sandeels	0.026	0.032		0.068	0.049		0.012			0.050
21	Gurnards and dragonets	0.002				0.037		0.020	0.075		
22	Other demersal fish								0.032		
23	Other benthopelagic fish	0.064	0.008	0.027	0.027	0.001		0.063	0.098	0.100	
24	Atlantic herring								0.104	0.150	
25	European sprat	0.411	0.029	0.024		0.001		0.018	0.001	0.019	
26	Other pelagic fish	0.190				0.068		0.127	0.190	0.449	
27	Anadromous fish										
28	Lobsters and large crabs		0.020	0.036	0.032			0.001			
29	Nephrops	0.065	0.318	0.100				0.001	0.001		
30	Shrimp	0.034	0.035	0.022	0.001	0.003		0.001	0.000	0.003	
31	Cephalopods		0.006			0.008		0.010	0.040		
32	Scallops				0.049	0.049					
33	Epifauna	0.092	0.449	0.546	0.324	0.279	0.225	0.492	0.001		0.078
34	Infauna	0.021	0.045	0.167	0.203	0.366	0.775	0.062			
35	Gelatinous zooplankton	0.003						0.070			
36	Large zooplankton	0.087	0.001	0.066	0.037	0.001		0.074	0.001	0.240	0.473
37	Small zooplankton	0.001	0.001	0.012		0.059		0.001			0.400
38	Seaweed		0.011			0.002		0.018			
39	Phytoplankton										
40	Discards				0.259	0.078					
41	Detritus										

	Predator/Prey	21	22	23	24	25	26	27	28	29	30
1	Toothed whales										
2	Minke whales										
3	Seals										
4	Seabirds (high discard diet)										
5	Seabirds (low discard diet)										
6	Sharks										
7	Rays										
8	Atlantic cod 2+										
9	Atlantic cod 0-1										
10	Whiting 2+										
11	Whiting 0-1		0.029								
12	Haddock 2+										
13	Haddock 0-1										
14	European plaice 2+										
15	European plaice 0-1	0.005									
16	Common sole	0.005									
17	Flatfish	0.010	0.003		0.001						
18	Monkfish										
19	European hake										
20	Sandeels	0.001	0.017					0.600			
21	Gurnards and dragonets	0.030	0.004	0.010	0.001						
22	Other demersal fish		0.100								
23	Other benthopelagic fish	0.115	0.100	0.005	0.001		0.010	0.076	0.011		
24	Atlantic herring		0.058					0.133			
25	European sprat	0.006	0.023				0.130	0.130			
26	Other pelagic fish	0.175	0.427	0.030	0.001		0.050				
27	Anadromous fish										
28	Lobsters and large crabs	0.001		0.003					0.019		
29	Nephrops	0.020	0.003	0.010							
30	Shrimp	0.026	0.050	0.016	0.014		0.223	0.003	0.095	0.014	
31	Cephalopods	0.038	0.003	0.005						0.020	
32	Scallops										
33	Epifauna	0.105	0.093	0.137	0.050	0.017	0.003		0.222	0.400	0.111
34	Infauna	0.038	0.001	0.088	0.083	0.002		0.033	0.139	0.069	0.404
35	Gelatinous zooplankton	0.050		0.020	0.020	0.005					
36	Large zooplankton	0.026	0.046	0.248	0.772	0.476	0.584	0.026		0.120	0.185
37	Small zooplankton	0.348	0.045	0.350	0.060	0.500				0.092	0.300
38	Seaweed			0.078					0.122		
39	Phytoplankton										
40	Discards								0.126	0.286	0.001
41	Detritus								0.266		

	Predator/Prey	31	32	33	34	35	36	37
1	Toothed whales							
2	Minke whales							
3	Seals							
4	Seabirds (high discard diet)							
5	Seabirds (low discard diet)							
6	Sharks							
7	Rays							
8	Atlantic cod 2+							
9	Atlantic cod 0-1							
10	Whiting 2+	0.001						
11	Whiting 0-1	0.002						
12	Haddock 2+							
13	Haddock 0-1							
14	European plaice 2+							
15	European plaice 0-1							
16	Common sole							
17	Flatfish							
18	Monkfish							
19	European hake							
20	Sandeels	0.020						
21	Gurnards and dragonets	0.004						
22	Other demersal fish							
23	Other benthopelagic fish	0.050				0.100		
24	Atlantic herring	0.020						
25	European sprat							
26	Other pelagic fish	0.029						
27	Anadromous fish							
28	Lobsters and large crabs	0.010						
29	Nephrops							
30	Shrimp					0.050		
31	Cephalopods	0.010						
32	Scallops			0.010				
33	Epifauna	0.080		0.090		0.100		
34	Infauna	0.002		0.150	0.035			
35	Gelatinous zooplankton	0.020						
36	Large zooplankton	0.471				0.250	0.012	
37	Small zooplankton	0.281		0.100		0.250	0.350	0.030
38	Seaweed			0.100				
39	Phytoplankton		0.500	0.375	0.500	0.250	0.500	0.800
40	Discards							
41	Detritus		0.500	0.175	0.465		0.138	0.170

Annex 1.3

Landings and discards by fleet (t.km^{-2}) for the Irish Sea Ecopath model.

Landings:

Functional group	Beam trawls	Otter trawls	Nephrops trawl	Pelagic nets	Gill nets	Pots	Dredge	Longline	Total
1 Toothed whales	-	-	-	-	-	-	-	-	-
2 Minke whales	-	-	-	-	-	-	-	-	-
3 Seals	-	-	-	-	-	-	-	-	-
4 Seabirds (high discard diet)	-	-	-	-	-	-	-	-	-
5 Seabirds (low discard diet)	-	-	-	-	-	-	-	-	-
6 Sharks	3.36x10-6	5.00x10-3	1.66x10-3	1.00x10-6	2.00x10-3	1.44x10-4	5.19x10-7	3.08x10-3	0.012
7 Rays	2.15x10-2	2.49x10-2	3.04x10-3	3.10x10-5	5.11x10-4	4.51x10-4	1.42x10-4	2.49x10-4	0.051
8 Atlantic cod 2+	4.21x10-2	1.15x10-1	3.61x10-2	2.50x10-6	1.25x10-6	1.25x10-6	1.25x10-6	4.44x10-4	0.194
9 Atlantic cod 0-1	8.15x10-3	1.36x10-2	2.01x10-3	2.06x10-6	1.03x10-6	1.03x10-6	1.03x10-6	4.58x10-5	0.024
10 Whiting 2+	4.11x10-6	1.95x10-1	4.42x10-2	4.11x10-6	2.06x10-6	2.06x10-6	2.06x10-6	4.36x10-4	0.240
11 Whiting 0-1	2.00x10-6	3.09x10-2	2.35x10-4	2.00x10-6	1.00x10-6	1.00x10-6	1.00x10-6	4.53x10-5	0.031
12 Haddock 2+	2.03x10-3	2.72x10-2	2.50x10-3	2.00x10-6	1.00x10-6	1.00x10-6	1.00x10-6	1.18x10-6	0.032
13 Haddock 0-1	1.48x10-5	6.86x10-5	9.63x10-6	1.48x10-5	7.38x10-6	7.38x10-6	7.38x10-6	2.14x10-9	1.30x10-4
14 European plaice 2+	6.25x10-2	4.15x10-6	7.48x10-3	2.76x10-6	1.38x10-6	1.38x10-6	1.38x10-6	1.84x10-6	0.070
15 European plaice 0-1	1.95x10-5	5.31x10-6	8.85x10-7	3.54x10-6	1.77x10-6	1.77x10-6	1.77x10-6	2.70x1010	1.70x10-5
16 Common sole	2.30x10-2	4.33x10-4	6.00x10-3	1.00x10-6	3.54x10-6	3.64x10-6	9.39x10-5	1.12x10-6	0.030
17 Flatfish	5.06x10-3	2.08x10-3	1.97x10-3	9.57x10-6	1.76x10-5	5.06x10-6	7.74x10-6	5.33x10-7	0.009
18 Monkfish	4.00x10-3	2.09x10-3	2.17x10-3	9.72x10-6	8.46x10-5	8.49x10-6	1.22x10-4	-	0.008
19 European hake	4.63x10-4	1.96x10-2	2.71x10-3	1.66x10-5	1.61x10-3	6.81x10-7	1.05x10-5	-	0.024
20 Sandeels	1.00x10-5	-	5.00x10-7	-	-	-	-	6.52x10-6	1.70E-05
21 Gurnards and dragonets	2.00x10-6	1.19x10-3	2.79x10-7	1.00x10-6	1.00x10-6	-	1.00x10-6	7.96x10-6	0.001
22 Other demersal fish	1.65x10-4	3.97x10-2	4.08x10-3	2.03x10-5	4.12x10-3	7.11x10-5	4.26x10-5	5.23x10-4	0.049
23 Other benthopelagic fish	2.00x10-6	1.89x10-3	5.65x10-6	1.00x10-6	1.00x10-6	1.00x10-6	1.00x10-6	3.51x10-5	0.002
24 Atlantic herring	-	1.14x10-3	5.25x10-4	3.88x10-1	5.91x10-3	-	-	2.03x10-5	0.396
25 European sprat	7.73x10-6	3.25x10-3	3.66x10-3	2.60x10-1	7.57x10-3	1.10x10-5	1.00x10-6	-	0.274
26 Other pelagic fish	6.80x10-6	3.96x10-4	1.35x10-5	2.00x10-2	4.25x10-6	1.09x10-5	1.00x10-6	2.00x10-3	0.022
27 Anadromous fish	-	-	5.00x10-7	-	1.00x10-6	1.00x10-6	-	1.67x10-3	0.002
28 Lobsters and large crabs	2.88x10-6	3.82x10-6	1.44x10-5	1.82x10-6	8.95x10-5	7.20x10-3	3.11x10-6	3.94x10-7	0.007
29 Nephrops	9.51x10-5	1.46x10-3	1.20x10-1	2.00x10-6	1.00x10-6	9.43x10-5	1.00x10-6	-	0.122
30 Shrimp	7.07x10-3	-	5.00x10-7	-	1.00x10-6	1.00x10-5	1.00x10-6	2.78x10-7	0.007
31 Cephalopods	1.92x10-5	1.63x10-3	1.34x10-4	1.00x10-6	1.00x10-6	1.00x10-6	1.00x10-6	1.37x10-4	0.002
32 Scallops	4.43x10-4	1.57x10-5	7.15x10-4	-	6.75x10-7	1.00x10-6	2.30x10-1	-	0.231
33 Epifauna	4.11x10-6	1.00x10-6	5.00x10-7	-	1.66x10-4	2.27x10-2	4.49x10-2	1.19x10-7	0.068
34 Infauna	-	-	-	-	-	3.71x10-8	7.23x10-6	1.48x1011	7.27x10-6
35 Gelatinous zooplankton	-	-	-	-	-	-	-	-	-
36 Large zooplankton	-	-	-	-	-	-	-	-	-
37 Small zooplankton	-	-	-	-	-	-	-	-	-
38 Seaweed	-	-	-	-	-	-	-	-	-
39 Phytoplankton	-	-	-	-	-	-	-	-	-
40 Discards	-	-	-	-	-	-	-	-	-
41 Detritus	-	-	-	-	-	-	-	-	-
Sum	0.177	0.487	0.239	0.669	0.022	0.031	0.275	0.009	1.908

Discards:

Functional group	Beam trawls	Otter trawls	Nephrops trawl	Pelagic nets	Gill nets	Pots	Dredge	Longline	Total
1 Toothed whales	-	-	-	-	6.75x10-5	-	-	-	6.75x10-5
2 Minke whales	-	-	-	-	1.80x10-4	-	-	-	1.80x10-4
3 Seals	-	-	-	1.58x10-5	-	-	-	-	1.58x10-5
4 Seabirds (high discard diet)	-	-	-	-	-	-	-	-	-
5 Seabirds (low discard diet)	-	-	-	-	-	-	-	-	-
6 Sharks	-	9.36x10-3	1.51x10-2	-	-	-	-	-	2.45x10-2
7 Rays	7.94x10-3	4.52x10-3	4.68x10-3	9.23x10-8	1.36x10-6	4.61x10-6	5.60x10-3	-	2.27x10-2
8 Atlantic cod 2+	2.70x10-3	1.53x10-3	1.65x10-2	-	-	6.00x10-6	3.31x10-4	-	2.11x10-2
9 Atlantic cod 0-1	1.59x10-4	9.04x10-5	9.76x10-4	-	-	3.61x10-7	1.96x10-5	-	1.25x10-3
10 Whiting 2+	7.20x10-5	5.04x10-4	5.66x10-2	-	-	2.00x10-6	3.00x10-6	-	5.71x10-2
11 Whiting 0-1	1.22x10-5	8.57x10-5	2.00x10-2	2.99x1010	6.08x10-9	3.24x10-7	4.55x10-7	1.73x10-8	2.01x10-2
12 Haddock 2+	1.65x10-4	7.46x10-4	5.37x10-3	-	-	-	1.30x10-5	-	6.29x10-3
13 Haddock 0-1	7.10x10-5	3.21x10-4	2.31x10-3	3.43x10-9	-	8.11x10-8	5.43x10-6	-	2.71x10-3
14 European plaice 2+	1.00x10-2	-	1.10x10-2	-	-	-	-	-	2.10x10-2
15 European plaice 0-1	1.26x10-4	1.92x10-5	5.62x10-4	-	2.14x10-9	1.80x10-7	1.31x10-5	-	7.20x10-4
16 Common sole	1.67x10-4	9.82x10-6	1.74x10-4	-	-	-	9.45x10-6	-	3.60x10-4
17 Flatfish	2.50x10-4	6.24x10-5	1.68x10-3	-	1.03x10-6	1.45x10-6	7.11x10-5	-	2.07x10-3
18 Monkfish	2.53x10-4	2.59x10-4	1.11x10-3	-	-	5.15x10-6	8.12x10-4	-	2.44x10-3
19 European hake	1.25x10-5	6.95x10-4	2.64x10-3	-	-	7.54x10-6	5.24x10-7	-	3.35x10-3
20 Sandeels	-	-	-	-	-	-	-	-	-
21 Gurnards and dragonets	2.29x10-5	1.63x10-4	1.03x10-3	-	-	-	-	-	1.22x10-3
22 Other demersal fish	4.68x10-4	1.93x10-3	1.22x10-3	-	1.53x10-6	-	-	-	3.62x10-3
23 Other benthopelagic fish	3.38x10-5	3.49x10-5	4.37x10-4	-	-	2.63x10-7	1.21x10-5	-	5.19x10-4
24 Atlantic herring	1.00x10-6	4.54x10-6	3.26x10-5	4.84x1011	-	1.14x10-9	7.66x10-8	-	3.83x10-5
25 European sprat	-	9.43x10-8	8.72x10-7	-	-	2.97x10-9	2.99x10-9	-	9.72x10-7
26 Other pelagic fish	6.50x10-7	2.84x10-5	7.91x10-4	-	-	-	4.41x10-5	1.94x10-7	8.64x10-4
27 Anadromous fish	-	-	-	-	-	-	-	-	-
28 Lobsters and large crabs	3.40x10-5	3.50x10-5	2.12x10-4	-	-	-	-	-	2.81x10-4
29 Nephrops	-	9.10x10-5	2.89x10-2	-	-	-	-	-	2.90x10-2
30 Shrimp	-	-	-	-	-	-	-	-	-
31 Cephalopods	-	-	9.29x10-5	-	-	-	1.86x10-5	-	1.11x10-4
32 Scallops	3.40x10-4	1.51x10-4	-	-	-	-	1.72x10-3	-	2.21x10-3
33 Epifauna	-	-	-	-	-	-	-	-	-
34 Infauna	-	-	-	-	-	-	-	-	-
35 Gelatinous zooplankton	-	-	-	-	-	-	-	-	-
36 Large zooplankton	-	-	-	-	-	-	-	-	-
37 Small zooplankton	-	-	-	-	-	-	-	-	-
38 Seaweed	-	-	-	-	-	-	-	-	-
39 Phytoplankton	-	-	-	-	-	-	-	-	-
40 Discards	-	-	-	-	-	-	-	-	-
41 Detritus	-	-	-	-	-	-	-	-	-
Sum	0.023	0.021	0.171	1.59x10-5	2.51x10-4	2.80x10-5	0.009	2.11x10-7	0.224

Annex 2.1

Calibration time series

Functional group	Time series type	Source
Toothed whales	Biomass-model	GAM model, VIIa, (Waggitt et al., In prep.)
Minke whales	Biomass-model	GAM model, VIIa, (Waggitt et al., In prep.)
Seabirds (high discard diet)	Biomass-model	GAM model, VIIa, (Waggitt et al., In prep.)
Seabirds (low discard diet)	Biomass-model	GAM model, VIIa, (Waggitt et al., In prep.)
Sharks	Biomass-survey	BTS-VIIa (ICES, 2017b)
	Catches	ICES catch statistics (ICES, 2018a)
Rays	Biomass-survey	BTS-VIIa (ICES, 2017b)
	Catches	ICES catch statistics (ICES, 2018a)
Atlantic cod 2+	Biomass-assessment	SAM Age2+, WKIrish (ICES, 2017c, ICES, 2018c)
	Catches	WKIrish/WGCSE, catch in tonnes (landings + discards) (ICES, 2017c, ICES, 2018c)
Atlantic cod 0-1	Biomass-assessment	SAM Age 0-1, WKIrish (ICES, 2017c, ICES, 2018c)
	Catches	WKIrish/WGCSE, catch in tonnes (landings + discards) (ICES, 2017c, ICES, 2018c)
Whiting 2+	Biomass-assessment	ASAP Age2+, WKIrish (ICES, 2017c, ICES, 2018c)
	Catches	WKIrish/WGCSE, catch in tonnes (landings + discards) (ICES, 2017c, ICES, 2018c)
Whiting 0-1	Biomass-assessment	ASAP Age 0-1, WKIrish (ICES, 2017c, ICES, 2018c)
	Catches	WKIrish/WGCSE, catch in tonnes (landings + discards) (ICES, 2017c, ICES, 2018c)
Haddock 2+	Biomass-assessment	ASAP Age2+, WKIrish (ICES, 2017c, ICES, 2018c)
	Catches	WKIrish/WGCSE, catch in tonnes (landings + discards) (ICES, 2017c, ICES, 2018c)
Haddock 0-1	Biomass-assessment	ASAP Age 0-1, WKIrish (ICES, 2017c, ICES, 2018c)
	Catches	WKIrish/WGCSE, catch in tonnes (landings + discards) (ICES, 2017c, ICES, 2018c)
European plaice 2+	Biomass-assessment	SAM Age2+, WKIrish (ICES, 2017c, ICES, 2018c)
	Catches	WKIrish/WGCSE, catch in tonnes (landings + discards) (ICES, 2017c, ICES, 2018c)
European plaice 0-1	Biomass-assessment	SAM Age 0-1, WKIrish (ICES, 2017c, ICES, 2018c)
	Catches	WKIrish/WGCSE, catch in tonnes (landings + discards) (ICES, 2017c, ICES, 2018c)
Common sole	Biomass-assessment	XSA (ICES, 2018c)
	Catches	Landings data from WGCSE (ICES, 2018c)
Flatfish	Biomass-survey	BTS-VIIa (ICES, 2017b)
	Catches	ICES catch statistics (ICES, 2018a)
Monkfish	Biomass-survey	BTS-VIIa (ICES, 2017b)
	Catches	ICES catch statistics (ICES, 2018a)
European hake	Biomass-survey	BTS-VIIa (ICES, 2017b)
	Catches	ICES catch statistics (ICES, 2018a)
Sandeels	Biomass-survey	BTS-VIIa (ICES, 2017b)
Gurnards and dragonets	Biomass-survey	BTS-VIIa (ICES, 2017b)
Other demersal fish	Biomass-survey	BTS-VIIa (ICES, 2017b)
	Catches	ICES catch statistics (ICES, 2018a)
Other benthopelagic fish	Biomass-survey	BTS-VIIa (ICES, 2017b)
Atlantic herring	Biomass-assessment	SAM, HAWG, (ICES, 2018b)
	Catches	Landings data from HAWG (ICES, 2018b)
European sprat	Biomass-survey	AFBI annual herring acoustic survey, HAWG, (ICES, 2018b)
	Catches	ICES catch statistics (ICES, 2018a)
Other pelagic fish	Catches	ICES catch statistics (ICES, 2018a)
Lobsters and large crabs	Catches	ICES catch statistics (ICES, 2018a)
Nephrops	Biomass-assessment	UWTV index for Irish Sea West (FU15) (ICES, 2018c)
	Catches	WGCSE, catch in tonnes (landings + discards) (ICES, 2018c) and info from ICES catch statistics from 1973-2003 (ICES, 2018a)
Cephalopods	Catches	ICES catch statistics (ICES, 2018a)
Epifauna	Catches	ICES catch statistics (ICES, 2018a)
Scallops	Catches	ICES catch statistics (ICES, 2018a)
Gelatinous zooplankton	Biomass-survey	Annual average weight from 1994-2016 from S. Beggs, as calculated in Lynam et al. (2011).
Large zooplankton	Biomass-survey	SAHFOS CPR, > 2 mm, VIIa
Small zooplankton	Biomass-survey	SAHFOS CPR, < 2 mm, VIIa
Phytoplankton	Biomass-survey	SAHFOS CPR, PCI, VIIa

Annex 2.2

Forcing time series

Forcing Series	Group(s)	Target variable	Source
Fishing effort	Beam trawl, otter trawl, Nephrops trawl	Effort	Beam and otter trawl effort from WGCSE (ICES, 2018c). Nephrops trawl effort taken from Coughlan et al. (2015) who reconstructed the trend based on catch per unit effort (CPUE).
Fishing effort	Pelagic nets, gill nets, pots, dredge, longline	Effort	Fishers' knowledge effort trends (ICES, 2018e, Bentley et al., 2019c)
Fishing mortalities (yield/biomass)	Sharks, whiting 2+, whiting 0-1	Fishing mortality (F)	Group catch/biomass
Forced catch	Anadromous fish, infauna	Catch	ICES catch statistics (ICES, 2018a). EwE fleet structure could not simulate the landings of these groups.
Winter North Atlantic Oscillation index (NAOw)	Large zooplankton	Other mortality	Winter (December through March) index of the NAO is based on the difference of normalised sea level pressure between Lisbon, Portugal and Reykjavik, Iceland (NCAR, 2019b). Assigned to large zooplankton mortality based on Pearson's correlation analysis.
Atlantic Multidecadal Oscillation (AMO)	Cod, whiting	Larval recruitment success	The AMO is defined here as the annual mean area averaged sea surface temperature (SST) anomalies over the North Atlantic basin (NCAR, 2019a). Assigned to cod and whiting based on Pearson's correlation analysis.

NOTES

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